

Introduction to Physiology

Dr. Doc

[24/09/2026 7:39 AM] Imaanaa Harmee:

💧 Introduction to Human Physiology

✦ Definition and Nature of Physiology

▽ Physiology is the branch of biomedical science that studies the normal functions of the human body. It is also the science of regulation, explaining how different organs and body systems work together to maintain normal physiological parameters.

✦ Importance of Learning Physiology

▽ Understanding normal body functions helps in identifying, diagnosing, and treating abnormal conditions (pathologies). Knowledge of normal physiology forms the foundation of medicine and clinical practice.

✦ Fields of Physiology

▽ Physiology covers a wide range of topics, from the study of viral and bacterial physiology to cellular physiology, organ physiology, and the complex functions of the human body.

✦ Historical Background

🎯 William Harvey

▽ Described the circulation of blood and is widely known as the "Father of Physiology."

🎯 Claude Bernard

▽ Introduced the concept of the extracellular fluid (ECF) as the "internal environment" (milieu intérieur), emphasizing its importance in maintaining normal body function.

🎯 Walter Cannon

▽ Coined the term homeostasis, describing the body's ability to maintain a stable internal environment despite external changes.

✦ Relationship with Other Sciences

▽ Physiology is closely related to anatomy, biochemistry, pathology, pharmacology, and physics, as these sciences help explain the structure, function, and regulation of the human body.

✦ Physiology as a Quantitative Science

▽ Physiology is a quantitative science, meaning that physiological parameters are measured and expressed using numerical values and standard units for accurate assessment.

✦ Units of Measurement

▽ Standard units commonly used in physiology include length (meter, centimeter), volume (liter, milliliter), weight (kilogram, gram), and temperature (°C or Kelvin).

★ Key Point

▽ Physiology is the study of normal body functions and their regulation. It provides the scientific basis for understanding health, diagnosing diseases, and applying medical treatments.

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💧 Homeostasis and Regulatory Systems

✦ Concept of Homeostasis

▽ Homeostasis is the maintenance of a stable internal environment (extracellular fluid, ECF) despite changes in the external or internal environment. It ensures that body functions remain within normal physiological limits for optimal cell function.

✦ Organ Contributions to Homeostasis

▽ Different organs work together to maintain homeostasis. The lungs regulate oxygen and carbon dioxide concentrations, the cardiovascular system transports oxygen, nutrients, hormones, and wastes, the kidneys regulate water, electrolytes, and ionic concentration, while the gastrointestinal tract (GIT) absorbs nutrients, water, and electrolytes needed by the body.

✦ Regulatory Systems

🧠 The Nervous Regulatory Mechanism

▽ The nervous system regulates body functions rapidly through action potentials and neurotransmitters, producing immediate responses to maintain homeostasis.

✓ **The Reflex Arc** consists of receptors that detect stimuli, a sensory (afferent) pathway that carries information to the integrating center (brain and spinal cord), a motor (efferent) pathway, and effectors (muscles or glands) that produce the response.

🧠 The Hormonal Regulatory Mechanism

▽ The endocrine system regulates body functions through hormones, which are chemical

messengers released into the bloodstream to act on distant target organs.

✓ **Examples:** Parathyroid hormone (PTH) regulates calcium levels, Aldosterone regulates sodium balance, and Antidiuretic hormone (ADH) promotes water retention.

✦ Comparison of Hormones and Neurotransmitters

▽ Both hormones and neurotransmitters act by binding to specific receptors and help maintain homeostasis.

✓ **Neurotransmitters** act rapidly, have short-lasting, and localized effects.

✓ **Hormones** act more slowly, have longer-lasting, and often widespread effects.

✦ Homeostatic Values and Parameters

▽ Homeostasis maintains normal physiological parameters such as body fluid volume, osmolality, body temperature, blood pH, blood gases, and electrolyte concentrations.

▽ It also regulates waste products including bilirubin, creatinine, blood urea nitrogen (BUN), and uric acid, as well as cardiac output, blood flow, RBC count, WBC count, hemoglobin concentration, and hematocrit.

✦ Feedback Control Mechanisms

🌀 Negative Feedback Mechanism (NFM)

▽ Negative feedback opposes the initial stimulus, restoring the body to its normal state and maintaining homeostasis.

✓ **Examples:** Regulation of arterial blood pressure (ABP), blood glucose level (BGL), and body temperature (BT).

🌀 Positive Feedback Mechanism (PFM)

▽ Positive feedback amplifies the initial stimulus, causing the response to continue until a specific event is completed. Although it can create a vicious cycle, it is essential in certain physiological processes.

✓ **Examples:** Blood clotting, generation of action potentials, labor (childbirth), and the LH surge during ovulation.

★ Key Point

▽ **Homeostasis is maintained through the coordinated actions of the nervous and endocrine systems, with negative feedback serving as the primary mechanism for stability, while positive feedback amplifies specific physiological events when necessary.**

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💧 Structural Levels of Organization

✦ Levels of Organization

▽ The human body is organized into six structural levels, with each level building upon the previous one to form a complete living organism.

✓ **Chemical Level:** The simplest level, consisting of atoms and molecules that combine to form the basic building blocks of cells.

✓ **Cellular Level:** Cells are the basic structural and functional units of the body. Different cell types perform specialized functions.

✓ **Tissue Level:** Similar cells group together to form tissues. The four basic tissue types are epithelial, connective, muscle, and nervous tissue.

✓ **Organ Level:** Two or more tissue types combine to form an organ, such as the heart, lungs, kidneys, or liver, each performing specific functions.

✓ **Organ System Level:** Related organs work together as an organ system to carry out major physiological functions, such as the digestive, cardiovascular, respiratory, and nervous systems.

✓ **Organism Level:** All organ systems function together in a coordinated manner to form the human organism, maintaining life and homeostasis.

✦ Key Point

▽ The levels of organization progress from **Chemical → Cellular → Tissue → Organ → Organ System → Organism**, with each level becoming more complex and contributing to the normal functioning of the human body.

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💧 Cell Physiology

✦ Cell Types

▽ Cells are classified into Prokaryotic and Eukaryotic cells based on their structure and complexity.

✓ **Prokaryotic cells:** Are small, lack a true nucleus and membrane-bound organelles, and have their genetic material freely suspended in the cytoplasm.

✓ **Eukaryotic cells:** Are larger and more complex, containing a true nucleus and several membrane-bound organelles that perform specialized cellular functions.

✦ Cell Structures and Components

🌀 Protoplasm

▽ Protoplasm is the living substance of the cell and consists mainly of water, electrolytes, proteins, lipids, and carbohydrates. It includes both the cytoplasm and the nucleus, where most cellular activities occur.

🌀 Plasma Membrane

▽ The plasma membrane forms the outer boundary of the cell and is composed of approximately 55% proteins, 42% lipids, and 3% carbohydrates.

✓ **Functions:** It separates the cell from its environment, regulates the passage of substances, contains receptors for cell signaling, enables cell-to-cell communication, and conducts electrical impulses in excitable cells.

✓ **Structure:** The membrane follows the Fluid Mosaic Model, consisting of a phospholipid bilayer with cholesterol that maintains membrane stability and fluidity.

✓ **Membrane Proteins:** Integral (transmembrane) proteins function as channels, carriers, pumps, receptors, and enzymes, whereas peripheral proteins provide structural support and are associated with the cytoskeleton and glycocalyx.

✓ **Membrane Carbohydrates:** Form part of the glycocalyx and are involved in immune recognition, cell adhesion, and receptor functions.

🌀 Cytoplasm

▽ The cytoplasm is the region between the plasma membrane and nucleus. It consists of the cytosol and various cell organelles, where most metabolic reactions take place.

🌀 The Nucleus

▽ The nucleus is the control center of the cell, containing DNA (genes) that regulate cellular activities. It is the site of synthesis of mRNA, rRNA, and tRNA, while the nucleolus is responsible for producing ribosomes.

🌀 Functions of Nucleotides

▽ Nucleotides are the building blocks of DNA and RNA, serve as energy molecules such as ATP and GTP, and act as biosynthetic mediators in many cellular reactions.

✦ Cellular Organelles

🌀 Ribosomes

▽ Ribosomes are the sites of protein synthesis, translating genetic information from mRNA into proteins.

🌀 Endoplasmic Reticulum (ER)

▽ **Rough ER** contains ribosomes and is responsible for the synthesis and transport of proteins.

▽ **Smooth ER** lacks ribosomes and functions in lipid synthesis, glycogen and calcium storage, and drug metabolism (detoxification).

🌀 Golgi Complex

▽ The Golgi complex modifies, sorts, packages, and secretes proteins and lipids, forming secretory granules and lysosomes.

🌀 Mitochondria

▽ Mitochondria are known as the "power plants" of the cell because they produce ATP through aerobic respiration using oxygen, supplying energy for cellular activities.

🌀 Lysosomes

▽ Lysosomes contain hydrolytic enzymes that digest foreign particles, worn-out organelles, and cellular debris, acting as the cell's intracellular digestive system.

🌀 Peroxisomes

▽ Peroxisomes are small organelles containing enzymes such as catalase, which detoxify harmful substances by converting hydrogen peroxide into water and oxygen.

🌀 Cytoskeletal System

▽ The cytoskeleton is composed mainly of microfilaments and microtubules, which maintain cell shape, provide mechanical support, facilitate intracellular transport, and enable cell movement and division.

★ Key Point

▽ **Eukaryotic cells possess specialized organelles that work together to regulate protein synthesis, energy production, transport, digestion, communication, and structural support, ensuring normal cellular function and survival.**

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💧 Membrane Transport and Signaling

✦ Transport Mechanisms

🌀 Simple Diffusion

▽ Simple diffusion is the passive movement of molecules from an area of higher concentration to lower concentration without the use of energy. The rate of diffusion depends on the concentration gradient, lipid solubility, membrane permeability, and temperature.

🌀 Osmosis

▽ Osmosis is the movement of water molecules across a selectively permeable membrane through aquaporin channels. Water moves from an area of low solute concentration to high solute concentration until equilibrium is established.

🌀 Facilitated Diffusion

▽ Facilitated diffusion is a carrier-mediated passive transport process in which substances move down their concentration gradient without using ATP. This mechanism is specific for particular molecules and becomes saturated when all carrier proteins are occupied.

🌀 Active Transport

▽ Active transport moves substances against their concentration or electrochemical gradient, requiring cellular energy to maintain normal intracellular conditions.

✓ **Primary Active Transport:** Uses ATP directly to move substances across the membrane. Important examples include the $\text{Na}^+\text{--K}^+$ ATPase pump, which maintains sodium and potassium gradients, and the Ca^{2+} ATPase pump, which regulates intracellular calcium levels.

✓ **Secondary Active Transport:** Uses the energy stored in ion gradients created by primary active transport. It occurs through symporters, where two substances move in the same direction, or antiporters, where substances move in opposite directions.

🌀 Vesicular Transport

▽ Vesicular transport allows the movement of large molecules and particles that cannot cross the plasma membrane directly.

✓ **Endocytosis** transports materials into the cell by pinocytosis (cell drinking), phagocytosis (cell eating), or receptor-mediated endocytosis, which is highly specific.

✓ **Exocytosis** transports substances out of the cell, releasing hormones, enzymes, neurotransmitters, and other secretory products into the extracellular space.

✦ Cell Junctions (Intercellular Connections)

▽ Cells are connected by Cell Adhesion Molecules (CAMs) such as cadherins, integrins, and selectins, which maintain tissue organization, mechanical stability, and communication.

✓ **Tight Junctions:** Form a sealed barrier between adjacent cells, controlling paracellular permeability and preventing leakage.

✓ **Desmosomes and Adherens Junctions:** Provide strong mechanical attachment between cells, maintaining tissue integrity during stretching and mechanical stress.

✓ **Gap Junctions:** Create direct communication channels between neighboring cells, allowing the movement of ions and small molecules for coordinated cellular activities, especially in cardiac and smooth muscle.

✦ Intercellular Signaling

▽ Cells communicate through different forms of intercellular signaling, enabling coordination of physiological processes and maintenance of homeostasis.

✓ **Contact-dependent signaling:** Requires direct contact between neighboring cells.

✓ **Paracrine signaling:** Local chemical messengers act on nearby target cells.

✓ **Synaptic signaling:** Neurons release neurotransmitters to communicate rapidly with target cells across synapses.

✓ **Endocrine signaling:** Hormones are released into the bloodstream and act on distant target organs.

✓ **Autocrine signaling:** Cells release signaling molecules that act on the same cell that produced them.

▽ Cellular responses occur through receptors, which are classified as cell-surface receptors for water-soluble signaling molecules and intracellular receptors for lipid-soluble molecules such as steroid hormones.

★ Key Point

▽ **Membrane transport, cell junctions, and intercellular signaling work together to regulate the movement of substances, maintain cell and tissue integrity, and coordinate communication between cells, ensuring the body's normal function and homeostasis.**

[24/09/2026 7:39 AM] Imaanaa Harmee:

💧 Body Fluid Environment

✦ Fluid Distribution

▽ Body fluids make up about 60% of total body weight. They are divided into Intracellular Fluid (ICF), which accounts for about 2/3 of total body fluid, and Extracellular Fluid (ECF), which makes up the remaining 1/3. The ICF is the fluid inside cells, while the ECF surrounds the cells and provides their immediate environment.

✦ ECF Subdivisions

▽ The Extracellular Fluid (ECF) is further divided into interstitial fluid (lymph), which surrounds tissue cells, and blood plasma, the liquid portion of blood that transports nutrients, hormones, gases, and waste products throughout the body.

✦ Transcellular Fluid

▽ Transcellular fluid is a specialized portion of the ECF found within body cavities. It includes cerebrospinal fluid (CSF), aqueous humor of the eye, digestive secretions, and other specialized fluids that perform specific physiological functions.

✦ Constituents of Body Fluids

▽ The composition of body fluids differs between compartments. The ICF contains a high concentration of potassium (K^+) and phosphate ions, whereas the ECF is rich in sodium (Na^+) and chloride (Cl^-). These electrolyte differences are essential for normal cell function, nerve impulse transmission, and muscle contraction.

✦ Water Balance

▽ The body maintains water balance by matching daily water intake with daily water loss. Water is gained through drinking fluids, food intake, and metabolic reactions, while it is lost through insensible evaporation (skin and lungs), sweating, feces, and urine.

✦ Osmotic Dynamics

🌀 Osmotic Pressure

▽ Osmotic pressure is the force produced by dissolved particles (osmoles) that draws water across a selectively permeable membrane. It depends on the number of solute particles, not their size.

🌀 Tonicity

▽ Tonicity describes the effect of a solution on cell volume.

✓ **Isotonic solution:** Has the same osmotic concentration as body fluids, so cell size remains unchanged.

✓ **Hypotonic solution:** Has a lower solute concentration than the cell, causing water to

enter the cell, making it swell.

✓ **Hypertonic solution:** Has a higher solute concentration than the cell, causing water to leave the cell, making it shrink.

✦ Overall Body Composition

▽ The average human body is composed of approximately 60% water, 18% proteins, 15% fats, and 7% minerals. These components are essential for maintaining the body's structure, metabolism, and physiological functions.

★ Key Point

▽ **Body fluids are distributed between the ICF and ECF, with each compartment having a unique electrolyte composition. Proper water balance, osmotic pressure, and tonicity are essential for maintaining cell volume, normal physiological function, and homeostasis.**

[24/09/2026 7:39 AM] Imaanaa Harmee:

Respiratory System Physiology

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[24/09/2026 7:39 AM] Imaanaa Harmee: ✦ ▽ **I. Introduction to the Respiratory System**

✦ 1. General Considerations

▽ The respiratory system is essential for life because body cells require a continuous supply of oxygen and must remove carbon dioxide, which is produced during metabolism. Survival also depends on the proper functioning of body systems and an adequate supply of food, water, and air.

▽ In unicellular organisms, gas exchange occurs by simple diffusion because the cells are directly exposed to their surroundings. In large multicellular animals, simple diffusion is not sufficient, so specialized respiratory organs such as the lungs are required for efficient gas exchange.

✦ 2. Prime Functions of the Lung

▽ The primary function of the lungs is gas exchange. Oxygen from inhaled air enters the alveoli and diffuses into the blood, while carbon dioxide moves from the blood into the alveoli and is removed during expiration.

▽ The lungs also perform other important functions, including the metabolism of certain compounds, filtration of unwanted materials from the blood, and serving as a reservoir for blood.

✦ 3. Definition and Phases of Respiration

▽ **External respiration** → exchange of O_2 and CO_2 between the alveoli and blood. Oxygen enters the blood while carbon dioxide leaves the blood and enters the alveoli.

▽ **Internal or cellular respiration** → occurs at the tissue and cellular level. Cells use oxygen during metabolic reactions to break down nutrients and produce energy, while carbon dioxide is produced as a waste product.

✦ 4. Stages of Respiration

▽ **Pulmonary ventilation** → movement of air into and out of the lungs through inspiration and expiration.

▽ **External respiration** → oxygen is loaded into the blood and carbon dioxide is unloaded into the alveoli.

▽ **Transport of respiratory gases** → the blood carries oxygen from the lungs to the tissues and transports carbon dioxide from the tissues back to the lungs.

▽ **Internal respiration** → oxygen is unloaded from the blood to the tissues, while carbon dioxide is loaded into the blood.

✦ 5. Non Respiratory Functions

▽ The respiratory system helps with water and heat elimination and provides olfaction, allowing the detection of different odors.

▽ It helps maintain acid base balance by controlling the amount of carbon dioxide removed from the body and contributes to vocalization or phonation through the movement of air across the vocal cords.

▽ The respiratory tract provides defense through mucus, cilia, macrophages, and the cough reflex, which help trap, remove, and destroy harmful particles and microorganisms.

▽ The lungs also participate in metabolism, including the conversion of Angiotensin I to Angiotensin II, and help enhance venous return by changes in pressure within the thoracic cavity during breathing.

[24/09/2026 7:39 AM] Imaanaa Harmee: ✦ ▽ **II. Functional Structures of the Respiratory System**

✦ 1. Anatomical Structural Parts

▽ The respiratory system is divided anatomically into the upper respiratory system and the lower respiratory system. These two divisions work together to receive air from the environment, conduct it through the respiratory passages, and finally deliver it to the areas where gas exchange occurs.

▽ **The upper respiratory system** includes the nose and pharynx. The nose provides the main entrance for inspired air and helps filter, warm, and moisten it. The pharynx serves as a passageway through which air moves toward the lower respiratory tract.

▽ **The lower respiratory system** consists of the larynx, trachea, bronchi, and lungs. These structures provide a continuous pathway for air movement and contain the respiratory structures responsible for ventilation and exchange of oxygen and carbon dioxide.

✦ 2. Physiological Functional Zones

▽ The respiratory system can also be divided according to function into the conducting zone and the respiratory zone. Each zone has a different role in the movement and exchange of respiratory gases.

▽ **The conducting zone** includes the nose, pharynx, larynx, trachea, bronchi, and terminal bronchioles. Its main function is to transport air toward the respiratory zone. As air passes through these structures, it is filtered, warmed, and moistened, helping protect the delicate tissues deeper in the lungs.

▽ **The respiratory zone** consists of the respiratory bronchioles, alveolar ducts, alveolar sacs, and alveoli. These structures form the functional region of the lungs where O_2 moves into the blood and CO_2 moves from the blood into the alveolar air.

✦ 3. Specific Conducting Structures

▽ The nasal cavity is an important part of the conducting pathway. It contains the nasal septum, which divides the cavity into right and left portions, and the nostrils or anterior nares, which provide an entrance for air.

▽ The **nasal cavity** also contains the vestibules, which are located near the nostrils and contain keratinized epithelium that helps protect the entrance of the respiratory tract. The nasal conchae or turbinates increase the surface area of the nasal cavity and create air currents that improve the warming, moistening, and filtering of inspired air.

▽ The larynx connects the pharynx to the trachea and serves several important functions. It contains the vocal folds, which vibrate as air passes through them and produce sound for speech. The larynx also acts as a protective switching mechanism, helping direct air toward the respiratory tract and food toward the digestive tract during swallowing.

▽ The trachea and bronchial tree form the major conducting pathway into the lungs. The trachea is a musculo cartilaginous tube that maintains an open passage for air. It divides into progressively smaller bronchi and bronchioles, forming approximately 23 generations of branching airways.

▽ The ciliary system works together with mucus as an important defense mechanism. Mucus traps dust, particles, microorganisms, and other inhaled materials, while coordinated movement of the cilia carries this material upward toward the pharynx. This process is known

as the mucus escalator.

✦ 4. The Lungs and Alveoli

▽ Each lung is surrounded by a double layered pleural membrane. The outer layer is called the parietal pleura, while the inner layer covering the lung surface is called the visceral pleura.

▽ Between the two pleural layers is the pleural cavity, which contains a small amount of lubricating fluid. This fluid reduces friction between the pleural surfaces and allows the lungs to move smoothly during inspiration and expiration.

▽ The lungs contain approximately 300 million alveoli, which are microscopic air sacs located at the ends of the respiratory passages. Their walls are extremely thin and are surrounded by numerous blood capillaries, providing a very large surface area for the exchange of respiratory gases.

[24/09/2026 7:39 AM] Imaanaa Harmee: ✓ **Type I pneumocytes** → form the major thin lining of the alveoli and provide the main surface through which O₂ and CO₂ are exchanged between alveolar air and pulmonary blood.

✓ **Type II pneumocytes** → produce surfactant, which reduces the surface tension of the fluid lining the alveoli and helps prevent the alveoli from collapsing, particularly during expiration.

✓ **Type III cells or alveolar macrophages** → provide an important defense mechanism within the alveoli by engulfing and removing inhaled particles, microorganisms, and cellular debris, helping maintain clean and functional alveolar surfaces.

[24/09/2026 7:39 AM] Imaanaa Harmee: ✦ ▽ **III. Pulmonary Ventilation (Mechanics of Breathing)**

✦ 1. Principles of Airflow

▽ **Pulmonary ventilation** refers to the movement of air into and out of the lungs. Airflow depends mainly on differences in pressure between the atmosphere and the lungs. When the pressure inside the lungs changes, air moves in or out according to the pressure gradient.

▽ **Boyle's Law** states that pressure and volume are inversely proportional when temperature remains constant. As the volume of a gas increases, its pressure decreases, and when volume decreases, pressure increases. This relationship can be expressed as $V \propto 1/P$.

▽ Air always moves from an area of higher pressure to an area of lower pressure. Therefore, air enters the lungs when intrapulmonary pressure becomes lower than atmospheric pressure and leaves the lungs when intrapulmonary pressure becomes higher than atmospheric pressure.

✦ 2. Mechanics of Inspiration and Expiration

▽ **Expansion and contraction of the lungs** occur mainly through movements of the

diaphragm and ribs. The diaphragm is the major muscle of respiration, while the external intercostal muscles help elevate the ribs and expand the thoracic cavity.

▽ **During inspiration**, the diaphragm contracts and moves downward, while the external intercostal muscles elevate the ribs. This increases the volume of the thoracic cavity and lungs, causing intrapulmonary pressure to fall below atmospheric pressure. Air therefore flows into the lungs.

▽ **During normal expiration**, the diaphragm and inspiratory muscles relax and the lungs return toward their resting size because of their natural elastic recoil. Lung volume decreases, intrapulmonary pressure rises, and air flows out of the lungs.

▽ **During forced expiration**, additional muscles are recruited to actively decrease thoracic volume. The abdominal muscles and internal intercostal muscles contract to push air out more forcefully, such as during coughing or vigorous exercise.

✦ 3. Pressures and Forces

▽ **Intrapleural pressure** is the pressure within the pleural cavity between the lungs and chest wall. It is normally slightly negative, approximately -5 to -7.5 cm H₂O, which helps keep the lungs expanded and prevents them from collapsing.

▽ **Surface tension** is produced by the attractive forces between molecules of the fluid lining the alveoli. According to the Law of Laplace, alveolar pressure can be expressed as $P = 2T/r$, where P is pressure, T is surface tension, and r is the radius of the alveolus.

▽ Because of surface tension, smaller alveoli have a greater tendency to collapse than larger alveoli. Without a mechanism to reduce this force, maintaining open alveoli would require greater pressure and work.

▽ **Surfactant** is a phospholipid substance, mainly dipalmitoylphosphatidylcholine (DPPC), produced by Type II pneumocytes. It reduces alveolar surface tension, decreases the work required for breathing, and helps prevent alveolar collapse.

✦ 4. Compliance and Work of Breathing

▽ **Lung compliance** refers to the ability of the lungs to expand in response to a change in pressure. It is defined as the change in lung volume produced by a unit change in pressure. The average lung compliance is approximately 200 mL/cm H₂O.

▽ High compliance means the lungs expand relatively easily with a small change in pressure, whereas low compliance means greater pressure is required to produce the same amount of lung expansion.

▽ **The work of breathing** is the energy required to move air into and out of the lungs. It can be divided into **compliance or elastic work**, which overcomes elastic forces of the lungs and chest wall, **structural or tissue resistance work**, which overcomes tissue viscosity, and **airway resistance work**, which overcomes resistance to airflow through the respiratory

passages.

[24/09/2026 7:39 AM] Imaanaa Harmee: ✦ ▽ **IV. Pulmonary Volumes, Capacities, and Ventilation Rates**

✦ 1. Pulmonary Volumes

▽ **Tidal Volume (VT)** → the amount of air inhaled or exhaled during a normal, quiet breath. It represents the normal movement of air during resting breathing and is approximately 500 mL in an average adult.

▽ **Inspiratory Reserve Volume (IRV)** → the additional amount of air that can be forcefully inhaled after a normal inspiration. It represents the extra inspiratory capacity available when deeper breathing is required and is approximately 3000 mL.

▽ **Expiratory Reserve Volume (ERV)** → the additional amount of air that can be forcefully exhaled after a normal expiration. It represents the extra volume that can be expelled beyond a normal breath and is approximately 1100 mL.

▽ **Residual Volume (RV)** → the amount of air that remains in the lungs after a maximum forced expiration. This air cannot normally be voluntarily expelled and helps keep the lungs and alveoli partially expanded. It is approximately 1200 mL.

✦ 2. Pulmonary Capacities

▽ **Inspiratory Capacity (IC)** → the maximum amount of air that can be inspired after a normal expiration. It combines the normal tidal volume with the additional inspiratory reserve volume. $IC = VT + IRV$

▽ **Functional Residual Capacity (FRC)** → the amount of air that remains in the lungs at the end of a normal, quiet expiration. It represents the combination of the expiratory reserve volume and the residual volume. $FRC = ERV + RV$

▽ **Vital Capacity (VC)** → the maximum amount of air that can be exhaled after taking a maximum inspiration. It includes the inspiratory reserve volume, tidal volume, and expiratory reserve volume. $VC = IRV + VT + ERV$

▽ **Total Lung Capacity (TLC)** → the total amount of air contained in the lungs after a maximum inspiration. It represents the sum of the vital capacity and residual volume. $TLC = VC + RV$

▽ **Note** → because RV cannot be directly measured by ordinary spirometry, indirect methods such as helium dilution are required to determine RV and capacities that include it, including FRC and TLC.

✦ 3. Ventilation Rates

▽ **Minute Ventilation** → the total volume of air moved into or out of the respiratory

system in one minute. It is calculated by multiplying tidal volume by respiratory rate. Minute Ventilation = $VT \times \text{Respiratory Rate}$, with an average value of approximately 6 L/min at rest.

▽ **Alveolar Ventilation (VA)** → the amount of fresh air that actually reaches the respiratory exchange areas of the lungs each minute. Unlike minute ventilation, it takes into account the air that remains in the conducting passages and does not participate directly in gas exchange.

▽ **Dead Space** → the portion of inspired air that does not participate in gas exchange. **Anatomic dead space** refers to air within the conducting airways, while **physiologic dead space** includes anatomic dead space plus any alveolar spaces that receive ventilation but have little or no effective gas exchange.

[24/09/2026 7:39 AM] Imaanaa Harmee: ✦ ▽ **V. Pulmonary Circulation and Capillary Dynamics**

✦ 1. Dual Circulation

▽ The lungs have a dual blood supply consisting of the pulmonary circulation and bronchial circulation. These two circulations have different pressures, flow rates, and functions within the respiratory system.

▽ **Pulmonary circulation** → a low pressure, high flow circulation that carries deoxygenated blood from the right side of the heart to the lungs. Its main purpose is to bring blood into close contact with the alveoli for efficient gas exchange.

▽ **Bronchial circulation** → a high pressure, low flow circulation that supplies oxygenated blood to the supporting tissues of the lungs, including the bronchi and other structures of the respiratory system.

✦ 2. Regional Blood Flow Distribution

▽ Pulmonary blood flow is not distributed equally throughout the lungs. It is strongly influenced by hydrostatic pressure and gravity, especially when a person is standing upright. Blood flow is generally greater toward the base of the lungs and lower toward the apex.

▽ In an upright person, the base of the lung receives greater blood flow and is associated with Zone 3, while regions toward the upper part of the lung have lower blood flow and are associated with Zone 2.

▽ **Exercise Effect** → during exercise, pulmonary blood flow increases significantly. Additional pulmonary capillaries are recruited and distended, including vessels in the upper regions of the lungs. This helps increase the available area for blood flow and reduces regional differences in pulmonary perfusion.

✦ 3. Fluid Dynamics and Edema

▽ **Starling Forces** → movement of fluid across pulmonary capillary walls is determined by

the balance between forces pushing fluid outward and forces pulling fluid inward. The total outward force is approximately 29 mm Hg, while the inward force is approximately 28 mm Hg, producing a small net filtration pressure of about 1 mm Hg.

▽ This slight net filtration normally allows a small amount of fluid to move from the pulmonary capillaries into the surrounding tissues. The lymphatic system helps remove excess fluid and maintain normal lung function.

▽ **Pulmonary Edema** → occurs when excessive fluid accumulates in the lung tissues or alveolar spaces. It may result from heart failure, which increases pulmonary capillary pressure, or from damage to the pulmonary membrane, which increases fluid movement out of the capillaries.

▽ Pulmonary edema develops when capillary hydrostatic pressure becomes greater than plasma colloid osmotic pressure, causing excessive filtration of fluid into the pulmonary tissues and potentially interfering with normal gas exchange.

[24/09/2026 7:39 AM] Imaanaa Harmee: ✦ ▽ **VI. Pulmonary Gas Exchange**

✦ 1. Gas Pressures

▽ **Partial Pressure** → the pressure exerted by an individual gas within a mixture of gases. The partial pressure of a gas depends on its proportion or concentration in the mixture. For example, PO_2 is proportional to the concentration of oxygen present.

▽ **Henry's Law** → the amount of a gas that dissolves in a liquid depends on its partial pressure and solubility. Carbon dioxide is approximately 20 times more soluble than oxygen, allowing it to dissolve and diffuse more readily in body fluids.

✦ 2. The Respiratory Membrane

▽ The respiratory membrane is the thin barrier separating the air within the alveoli from the blood in pulmonary capillaries. It consists of approximately six distinct layers and is extremely thin, measuring about 0.6 micrometers.

▽ The respiratory membrane has a very large surface area of approximately 70 m^2 , providing extensive space for the rapid diffusion of O_2 and CO_2 between the alveoli and pulmonary capillary blood.

✓ **Solubility** → gases with greater solubility diffuse more easily through the respiratory membrane.

✓ **Surface Area** → a larger respiratory surface provides more area for gas exchange and increases the rate of diffusion.

✓ **Membrane Thickness** → a thinner respiratory membrane allows gases to diffuse more rapidly, while increased thickness slows diffusion.

✓ **Molecular Weight** → gases with lower molecular weight generally diffuse more rapidly than gases with higher molecular weight.

✓ **Temperature** → changes in temperature can influence the rate of gas diffusion.

✦ 3. Ventilation Perfusion Ratio (VA/Q)

▽ The ventilation perfusion ratio (VA/Q) compares the amount of air reaching the alveoli with the amount of blood flowing through the pulmonary capillaries. Efficient gas exchange requires appropriate matching between ventilation and perfusion.

▽ **VA/Q = 0** → there is perfusion without ventilation. Blood reaches the alveoli, but no air reaches them, producing a physiologic shunt.

▽ **VA/Q = ∞** → there is ventilation without perfusion. Air reaches the alveoli, but there is no blood flow available for gas exchange, producing physiologic dead space or wasted ventilation.

▽ **Normal regional differences** → in an upright person, the apex of the lung has a relatively high VA/Q ratio because ventilation is reduced less than blood flow, making it more similar to dead space.

▽ The base of the lung has a relatively low VA/Q ratio because blood flow increases more than ventilation, making it more similar to a shunt.

[24/09/2026 7:39 AM] Imaanaa Harmee: ✦ ▽ **VII. Gas Transport in the Blood**

✦ 1. Oxygen Transport

▽ Oxygen is transported in the blood in two main forms. Approximately 3% is dissolved directly in plasma, while about 97% is transported bound to hemoglobin within red blood cells.

▽ **Hemoglobin Saturation** → arterial blood is normally about 97% saturated with oxygen at a PO₂ of approximately 95 mm Hg. Venous blood is approximately 75% saturated at a PO₂ of about 40 mm Hg, reflecting oxygen removal by the tissues.

▽ **Oxygen Hemoglobin Dissociation Curve** → the relationship between PO₂ and hemoglobin saturation produces an S shaped curve. The plateau region favors oxygen loading in the lungs, while the steep region promotes oxygen unloading in the tissues.

✦ 2. Curve Shifting Factors

✓ **Right Shift** → indicates a reduced affinity of hemoglobin for oxygen, allowing more oxygen to be released to the tissues. It occurs with an increase in H⁺ concentration, decreased pH, increased PCO₂, increased temperature, or increased 2,3 DPG.

✓ **Left Shift** → indicates an increased affinity of hemoglobin for oxygen, causing

hemoglobin to hold oxygen more strongly. It occurs when H^+ , PCO_2 , temperature, or 2,3 DPG decreases.

✓ **Bohr Effect** → increased CO_2 and H^+ concentration in metabolically active tissues reduces hemoglobin's affinity for oxygen and promotes greater oxygen unloading to those tissues.

✦ 3. Carbon Monoxide (CO) Poisoning

✓ **Carbon Monoxide Affinity** → carbon monoxide has approximately 210 times greater affinity for hemoglobin than oxygen. It therefore binds strongly to hemoglobin and reduces the amount of oxygen that can be transported

✓ **Effect on Oxygen Dissociation Curve** → CO binding shifts the oxygen dissociation curve toward the left, increasing the remaining hemoglobin's affinity for oxygen and making oxygen unloading to the tissues more difficult.

✓ **Blood Appearance** → in carbon monoxide poisoning, the blood may appear bright red because of the formation of carboxyhemoglobin.

✦ 4. Carbon Dioxide Transport

▽ Carbon dioxide is transported in the blood in three major forms: approximately 7% dissolved in plasma, 70% as bicarbonate ions (HCO_3^-), and 23% bound to proteins as carbamino compounds, mainly carbaminohemoglobin.

✓ **Chloride Shift** → in systemic capillaries, bicarbonate ions leave red blood cells and enter the plasma. To maintain electrical neutrality, Cl^- enters the red blood cells to replace the exiting bicarbonate ions.

✓ **Haldane Effect** → deoxygenated hemoglobin has a greater ability to bind and carry CO_2 and H^+ than oxygenated hemoglobin. This facilitates CO_2 uptake at the tissues and promotes CO_2 release when blood reaches the lungs and becomes oxygenated.

[24/09/2026 7:39 AM] Imaanaa Harmee: ✦ ▽ **VIII. Respiratory Regulation**

✦ 1. Respiratory Centers (Medulla and Pons)

▽ The respiratory centers are located mainly in the medulla and pons of the brainstem. They control the rhythm and depth of breathing by coordinating the activity of the respiratory muscles.

✓ **Dorsal Respiratory Group (DRG)** → primarily controls inspiration and helps establish the basic rhythm of breathing. It generates a gradual “ramp” signal that progressively increases inspiratory muscle activity.

✓ **Pneumotaxic Center** → controls the “switch-off” point of inspiration. By regulating the

duration of inspiration, it influences the rate and depth of breathing.

✓ **Ventral Respiratory Group (VRG)** → becomes active mainly during heavy exercise or increased respiratory demand, providing additional respiratory drive for forceful breathing.

✦ 2. Chemical Regulation

✓ **Central Chemoreceptors** → are located in the brainstem and are sensitive mainly to changes in H^+ concentration in the brain and surrounding fluid. They are primarily stimulated indirectly when CO_2 crosses the blood-brain barrier and produces H^+ .

✓ **Peripheral Chemoreceptors** → are located mainly in the carotid bodies and aortic bodies. They are primarily sensitive to low arterial O_2 , while also responding to changes in CO_2 and H^+ .

✓ **Acclimatization** → during prolonged exposure to low oxygen levels, the respiratory system gradually adapts. Over approximately 2–3 days, the respiratory center becomes less sensitive to CO_2 , allowing ventilation to remain increased and helping improve oxygen uptake.

[24/09/2026 7:39 AM] Imaanaa Harmee: ✦ ▽ **IX. Respiratory Insufficiency and Pathophysiology**

✦ 1. Basic Pathological States

▽ **Hypoxia** → refers to an inadequate supply of oxygen to the tissues. It occurs when the tissues do not receive enough oxygen to meet their metabolic needs.

▽ **Hypoxemia** → refers to a decreased level of oxygen in arterial blood, usually indicated by an arterial PO_2 of less than 60 mmHg.

▽ **Cyanosis** → refers to a bluish discoloration of the skin and mucous membranes caused by an increased amount of deoxygenated hemoglobin in the blood, generally more than 5 g/dL.

▽ **Hypercapnia** → refers to an excessive amount of CO_2 in the body fluids or blood, usually resulting from inadequate removal of carbon dioxide through ventilation.

▽ **Dyspnea** → refers to the uncomfortable sensation of difficulty in breathing or “air hunger.” It may involve a feeling of not being able to ventilate adequately and can produce significant mental distress.

✦ 2. Lung Defects

▽ **Restrictive Disorders** → conditions in which lung expansion is restricted, making it difficult for the lungs to expand normally. Examples include pulmonary fibrosis and scoliosis.

▽ **Obstructive Disorders** → conditions in which airflow through the respiratory passages

is limited, particularly during expiration. Examples include asthma and emphysema.

✦ 3. Specific Abnormalities

▽ **Chronic Pulmonary Emphysema** → a condition characterized by airflow obstruction and destruction of alveolar walls, commonly associated with long-term cigarette smoking. Destruction of alveolar tissue reduces the surface area available for gas exchange and may lead to pulmonary hypertension and heart failure.

▽ **Pneumonia** → an inflammatory condition of the lungs in which the alveoli become filled with fluid and inflammatory blood cells. This process, known as consolidation, interferes with normal gas exchange.

▽ **Asthma** → a condition involving spastic contraction of bronchiolar smooth muscle, which narrows the airways and obstructs airflow. It is often associated with IgE-mediated allergic reactions and may be triggered by various allergens or other stimuli.

▽ **Atelectasis** → refers to the collapse of alveoli, which may occur because of airway obstruction or insufficient surfactant. Collapsed alveoli reduce the effective surface area available for gas exchange.

[24/09/2026 7:39 AM] Imaanaa Harmee:

Renal System Physiology

Dr. Doc

[24/09/2026 7:39 AM] Imaanaa Harmee: ☯ 1. Body Composition & Dietary Constituents

✦ A. Overall Body Composition

▽ The human body is composed mainly of water, together with proteins, fats, minerals, carbohydrates, and other substances. These components are essential for maintaining body structure, energy production, metabolism, and normal physiological functions.

▽ **Water — 60%:** Water is the largest component of the body and provides the environment in which most biochemical reactions occur. It also helps transport substances and regulate temperature.

▽ **Protein — 17%:** Proteins form important cellular and tissue structures and are needed for growth, repair, enzymes, hormones, and immune functions.

▽ **Fats — 13%:** Fats provide concentrated energy storage, form part of cell membranes, provide essential fatty acids, and help protect and insulate the body.

▽ **Minerals — 8%:** Minerals contribute to bones and teeth and are involved in fluid balance, nerve activity, muscle contraction, and metabolic reactions.

▽ **Carbohydrates — 1.5%:** Although present in a relatively small amount, carbohydrates

are an important source of readily available energy.

✦ B. Major Dietary Constituents

▽ **Water:** Maintains body fluids, supports chemical reactions, transports nutrients and wastes, and assists in temperature regulation.

▽ **Carbohydrates:** Mainly provide immediate energy for cellular activity, physical work, and heat production.

▽ **Fats:** Provide stored energy and essential fatty acids and contribute to cell structure and other physiological functions.

▽ **Proteins:** Essential for growth and tissue repair and serve as components of muscles, enzymes, hormones, antibodies, and other cellular structures.

▽ **Minerals:** Required for tissue development, bone formation, enzyme activity, fluid balance, and normal cellular functions.

▽ **Vitamins:** Needed in small amounts to support normal metabolic reactions, growth, and protection of cells.

▽ **Fiber and other indigestible materials:** Add bulk to food, support normal intestinal movement, provide a favorable environment for beneficial intestinal bacteria, and assist in the elimination of gastrointestinal waste.

✦ C. Water Content and Fluid Loss

▽ Water is the most important component for maintaining life because cells and tissues depend on an adequate fluid environment for normal function.

▽ In adults, water generally represents about 55–60% of total body weight.

▽ Body water is continuously lost through urine, sweat, feces, and respiration, so it must be replaced regularly.

▽ A 10% loss of total body fluid may cause approximately an 8% loss of body weight and represents a serious disturbance of fluid balance.

▽ A 20% loss of total body fluid may cause approximately a 15% loss of body weight and can be fatal.

▽ Maintaining homeostasis requires that fluid intake and fluid loss remain approximately balanced over time.

✦ D. Six Vital Functions of Body Fluids

- ▽ **1. Transport:** Body fluids transport nutrients, respiratory gases, hormones, and metabolic wastes between different organs and tissues.
- ▽ **2. Cellular metabolism:** Water provides the environment required for numerous chemical reactions that occur inside cells.
- ▽ **3. Solvent function:** It dissolves electrolytes and other substances, allowing them to circulate and participate in physiological reactions.
- ▽ **4. Temperature regulation:** Water helps distribute heat throughout the body and supports cooling mechanisms such as sweating and evaporation.
- ▽ **5. Digestion and waste removal:** Water assists digestion and helps move and eliminate wastes through the gastrointestinal and urinary systems.
- ▽ **6. Lubrication:** Body fluids reduce friction between moving structures, including joints and visceral organs.

✓ **Key idea:** The body depends heavily on water, while carbohydrates, fats, proteins, minerals, vitamins, and fiber work together to provide energy, structure, regulation, protection, digestion, and waste elimination.

[24/09/2026 7:39 AM] Imaanaa Harmee: ☞ **2. Fluid Compartments, Osmolarity, and Tonicity**

✦ A. The Two Major Body Fluid Compartments

- ▽ **Total body water (TBW)** in a standard adult is approximately **40 liters**. Body water is divided into two major compartments: **intracellular fluid (ICF)** and **extracellular fluid (ECF)**.
 - ▽ **Intracellular Fluid (ICF):** Approximately **25 liters**, representing about **62.5% of total body water**. It is the fluid located **inside the body's cells** and provides the environment required for cellular metabolism and normal cell function.
 - ▽ **Extracellular Fluid (ECF):** Approximately **15 liters**, representing about **37.5% of total body water**. It is the fluid located **outside the cells** and provides the environment surrounding cells and transporting substances between tissues.
- ✓ **ICF = fluid inside cells, while ECF = fluid outside cells.**

✦ B. ECF Sub-Compartments

- ▽ **1. Interstitial Fluid (ISF):** This is the fluid that surrounds and bathes the cells outside

the blood vessels. It provides a pathway through which nutrients, oxygen, wastes, and other substances move between the blood and cells.

▽ **2. Intravascular Fluid (Plasma):** This is the liquid portion of blood contained within the **blood vessels**. Plasma transports nutrients, hormones, gases, electrolytes, proteins, and metabolic wastes throughout the body.

▽ **3. Transcellular Fluid:** These are specialized, relatively small fluid compartments separated from the main ECF by epithelial or other cellular barriers.

▽ Important examples include **cerebrospinal fluid (CSF), intraocular fluid, pleural fluid, peritoneal fluid, synovial fluid, and digestive tract secretions**.

▽ Although these compartments are physically separate, they are all part of the broader **extracellular fluid system**.

▽ The **volume and composition** of body-fluid compartments must be regulated within narrow limits because changes in fluid balance can interfere with cell function, circulation, and normal physiological processes.

✦ C. Osmolarity and Osmotic Pulling Forces

▽ **Osmolarity** refers to the concentration of **osmotically active solute particles** in a solution. In simple terms, it describes how many dissolved particles are present in a given volume of solution.

▽ When a solution contains a **greater concentration of active particles**, it has a greater tendency to draw water toward it through a selectively permeable membrane.

▽ This water-attracting force is related to **osmotic pressure**. Therefore, differences in solute concentration between fluid compartments can cause water to move from one compartment to another.

▽ Normal blood serum osmolarity is tightly regulated at approximately **280–300 mOsm/L**.

✓ **Water moves in response to differences in effective solute concentration, helping maintain balance between fluid compartments.**

✦ D. Tonicity Classifications

▽ **Isotonic solution:** Has an effective osmolarity approximately equal to that of normal plasma. It produces little or no net movement of water into or out of cells, so **cell size remains relatively unchanged**.

▽ **Hypertonic solution:** Has a higher effective osmolarity than the cell interior. Water moves **out of cells into the extracellular fluid**, causing cells to **shrink**.

▽ **Hypotonic solution:** Has a lower effective osmolarity than the cell interior. Water moves **into cells**, causing them to **swell**.

✔ **Hypertonic → cells shrink | Isotonic → no major volume change | Hypotonic → cells swell.**

[24/09/2026 7:39 AM] Imaanaa Harmee: ✦ **E. Clinical Intravenous (IV) Fluid Dynamics**

✦ 1. Hypertonic IV Fluids

▽ Hypertonic solutions have a higher effective concentration of osmotically active particles than the intracellular environment.

▽ They increase the osmotic concentration of the ECF and cause **water to move from the intracellular compartment into the extracellular compartment**.

▽ As water leaves the cells, the cells **shrink**, while extracellular and vascular volume can increase.

▽ **Clinical uses:** Hypertonic fluids may be used in selected clinical situations to increase extracellular or intravascular volume and to help correct certain severe electrolyte or fluid abnormalities.

▽ Examples from the notes include **D5% NS and D5% LR**. The effect of a dextrose-containing solution can change after glucose is metabolized, so IV fluids should be classified according to their **clinical effect and composition**, not simply by their label.

✦ 2. Hypotonic IV Fluids

▽ Hypotonic solutions have a lower effective osmolarity than the intracellular environment.

▽ Water moves from the **ECF into the cells**, increasing intracellular water and causing cells to **swell**.

▽ **Clinical uses:** They may be used when the goal is to provide free water and correct certain conditions involving cellular dehydration, such as selected cases of **hyponatremia**.

▽ **Clinical danger:** Hypotonic fluids can increase cellular swelling and therefore may worsen **cerebral edema**. They are generally avoided in patients with **increased intracranial pressure (ICP)** unless specifically indicated and carefully managed.

▽ Examples include **0.45% normal saline ($\frac{1}{2}$ NS)** and **0.33% normal saline ($\frac{1}{3}$ NS)**.

✦ 3. Isotonic IV Fluids

▽ Isotonic fluids have an effective osmolarity close to that of plasma, so they do not produce a major osmotic shift of water into or out of cells.

▽ Their primary effect is to increase the **extracellular fluid volume**, including the intravascular space.

▽ **Clinical uses:** They are commonly used for **volume replacement**, such as in fluid losses from vomiting, diarrhea, bleeding, or other causes of extracellular fluid depletion.

▽ Examples include **0.9% normal saline, Lactated Ringer's solution, and other balanced isotonic crystalloids**.

▽ **Important correction:** **5% dextrose in water (D5W)** is initially approximately isotonic in the IV bag, but after the glucose is metabolized, the remaining free water distributes throughout total body water. Therefore, physiologically it behaves more like a **free-water/hypotonic solution**, rather than remaining isotonic.

✓ **The clinical effect of an IV fluid depends on its effective osmoles and how its components are distributed or metabolized after administration.**

✦ F. Albumin and Colloid Osmotic Pressure

▽ **Albumin** is the most abundant plasma protein and plays an important role in maintaining the movement of water between the blood vessels and interstitial spaces.

▽ Because albumin remains largely within the vascular compartment, it produces **colloid osmotic pressure**, also called **oncotic pressure**.

▽ This oncotic pressure helps draw water **from the interstitial space into the intravascular space**, helping maintain an appropriate blood volume.

▽ When serum albumin becomes significantly reduced, the plasma loses part of its normal oncotic pulling force.

▽ As a result, more fluid can remain in or move into the **interstitial spaces**, contributing to the development of **edema**.

▽ Low albumin can occur in conditions such as severe malnutrition, certain liver diseases, kidney diseases involving protein loss, and other disorders that reduce circulating albumin.

✓ **Key idea: Fluid balance depends on the controlled distribution of water between ICF and ECF, while osmolarity and tonicity determine how water shifts across cell membranes. Albumin and other plasma proteins help keep fluid within the vascular space through oncotic pressure.**

[24/09/2026 7:39 AM] Imaanaa Harmee: ☞ **3. Water Balance, Exchange, and Osmotic Equilibrium**

✦ A. The Rule of Fluid Balance

▽ **Fluid balance** is essential for maintaining **homeostasis**. The body must continuously regulate water intake and water loss so that the volume of body fluids remains relatively stable.

▽ Daily water intake can vary considerably depending on **climate, diet, physical activity, illness, and individual habits**.

▽ To maintain a stable body-fluid volume over time, **average water intake must approximately equal average water loss**.

▽ If water loss becomes greater than intake, the body develops a **negative water balance**, leading to dehydration and reduced body-fluid volume.

▽ If water intake is greater than loss for a prolonged period, excess water may accumulate and disturb normal fluid and electrolyte balance.

✓ **Fluid balance = water gained $\approx$ water lost over time.**

✦ B. Daily Intake of Water

▽ Under normal conditions, total daily water intake is approximately **2,300 mL/day**, although the actual amount varies from person to person.

▽ **1. Ingested Water — approximately 2,100 mL/day:** Most water enters the body through **drinks and water contained in food**. The amount consumed changes according to thirst, diet, environmental temperature, and activity level.

▽ **2. Metabolic Water — approximately 200 mL/day:** A smaller amount of water is produced **inside the body during cellular metabolism**, particularly during the oxidation of nutrients such as carbohydrates, fats, and proteins.

▽ Water produced metabolically contributes to total body water but is usually much smaller than the amount obtained from drinking and food.

✦ C. Daily Loss of Body Water

▽ Water is continuously lost from the body through several routes, including the **kidneys, skin, lungs, and gastrointestinal tract**.

▽ **1. Insensible Water Loss:** This is water loss that occurs continuously without conscious awareness. It mainly occurs through **evaporation from the respiratory tract and diffusion/evaporation through the skin**.

▽ Under ordinary conditions, insensible water loss is approximately **700 mL/day**, although it increases with **hot temperatures, fever, rapid breathing, and increased physical activity**.

▽ **2. Sensible Water Loss:** These are measurable forms of water loss, especially through **urine and sweat**. Fecal water loss is another measurable route.

▽ The **kidneys** are especially important because urinary water loss can be adjusted according to the body's hydration status.

✓ **Insensible loss occurs continuously and is not easily noticed, while sensible loss can be measured more directly.**

✦ D. Factors Influencing Total Body Water

▽ **Total body water (TBW)** is not the same percentage of body weight in every person. It varies mainly with **age, sex, and body composition**.

▽ **Infants — approximately 73%:** Infants have a relatively high proportion of body water compared with adults.

▽ **Adult males — approximately 60%:** Adult males generally have a higher percentage of body water because they tend to have a greater proportion of **lean tissue**, which contains more water.

▽ **Adult females — approximately 40–50%:** The percentage is generally lower because females, on average, have a greater proportion of **adipose tissue**, which contains considerably less water than lean tissue.

▽ **Obesity:** As the proportion of body fat increases, the percentage of body weight represented by water generally **decreases**, because adipose tissue contains relatively little water.

▽ **Old age:** Total body water tends to decrease with aging because of changes in body composition, particularly a reduction in lean tissue and an increase in the relative proportion of fat.

✓ **The more body fat a person has, the lower the percentage of body weight that is usually water.**

[24/09/2026 7:45 AM] Imaanaa Harmee: ✦ E. Forces Controlling Fluid Distribution

▽ Water is not distributed randomly throughout the body. Its movement between fluid compartments is controlled by **osmotic forces, hydrostatic pressure, colloid osmotic pressure, and membrane permeability**.

✦ 1. Distribution Within the ECF

- ▽ The extracellular fluid is divided mainly between the **plasma** and **interstitial fluid**.
- ▽ Movement of water between these two compartments occurs across **capillary walls**.
- ▽ The major forces controlling this exchange are **hydrostatic pressure** and **colloid osmotic pressure**.
- ▽ **Hydrostatic pressure** tends to push fluid out of blood vessels and toward the interstitial space.
- ▽ **Colloid osmotic pressure**, produced mainly by plasma proteins such as **albumin**, tends to pull water toward the vascular compartment.
- ▽ The balance between these forces helps determine how much fluid remains in the blood vessels and how much enters the interstitial tissues.

✦ 2. Distribution Between ECF and ICF

- ▽ Water movement between the **extracellular fluid and intracellular fluid** is controlled mainly by differences in the concentration of **osmotically active solutes** on either side of the cell membrane.
- ▽ **Sodium and its associated anions** are the major effective osmoles in the ECF, while **potassium and other intracellular solutes** are important inside cells.
- ▽ Cell membranes are highly permeable to **water**, allowing water to move rapidly between the ICF and ECF.
- ▽ In contrast, the cell membrane is much less freely permeable to many ions, including **Na⁺ and Cl⁻**. Therefore, these solutes contribute strongly to osmotic gradients across the membrane.
- ▽ Water shifts between the ICF and ECF until the effective osmotic forces on both sides are brought toward equilibrium.

✓ **Capillary exchange mainly regulates plasma ↔ interstitial fluid, while osmotic forces mainly regulate ECF ↔ ICF.**

✦ F. Basic Principles of Osmosis

- ▽ **Osmosis** is the net movement of water across a **selectively permeable membrane** in response to a difference in effective solute concentration.
- ▽ In simple terms, water tends to move from the side with **lower effective solute concentration** toward the side with **higher effective solute concentration**.

- ▽ When a solute such as **sodium chloride (NaCl)** is added to the ECF, the effective concentration of solute outside the cells increases.
- ▽ This creates an osmotic gradient that causes water to move **from the ICF toward the ECF**, resulting in a decrease in cell volume.
- ▽ Conversely, if the effective solute concentration of the ECF decreases, water tends to move **from the ECF into the ICF**, causing cells to increase in volume.
- ▽ Water continues to shift until the osmotic difference is reduced and an **osmotic equilibrium** is approached.

✦ G. Rate of Osmosis

- ▽ The **rate of osmosis** refers to how rapidly water moves across a selectively permeable membrane in response to an osmotic gradient.
- ▽ The greater the difference in effective solute concentration across the membrane, the stronger the driving force for water movement.
- ▽ The rate is also influenced by factors such as **membrane permeability, surface area, and the magnitude of the concentration gradient**.
- ✓ **A stronger osmotic gradient generally produces a greater driving force for water movement.**

✦ H. Osmotic Equilibrium

- ▽ The body's cells constantly exchange water with the extracellular environment. For normal cell function, the osmotic conditions of the ICF and ECF must remain closely balanced.
- ▽ If the ECF becomes excessively concentrated, water leaves cells and they **shrink**.
- ▽ If the ECF becomes excessively dilute, water enters cells and they **swell**.
- ▽ Therefore, maintaining the proper concentration of electrolytes and other effective solutes in the ECF is essential for maintaining **normal cell volume and function**.

[24/09/2026 7:45 AM] Imaanaa Harmee: ☞ **4. Fluid Volume Disturbances & Dehydration Categories**

✦ A. Major Fluid Volume Disturbances

- ▽ The body normally maintains a relatively stable volume of water and electrolytes within

the **intracellular and extracellular compartments**.

▽ Disturbances occur when the amount of body fluid becomes either **too high or too low**.

▽ The two major volume disturbances are:

▽ **Hypervolemia:** An excessive increase in body-fluid volume, particularly in the **extracellular and intravascular compartments**.

▽ **Hypovolemia:** A decrease in body-fluid volume caused by excessive loss of water and/or electrolytes.

✓ **Hypervolemia = too much body fluid | Hypovolemia = too little body fluid.**

✦ B. Hypervolemia — Fluid Excess

▽ **Hypervolemia** occurs when the body retains more fluid than it can effectively remove or when excessive fluid is introduced into the body.

▽ Excess fluid particularly affects the **extracellular compartment** and may increase the volume of blood circulating within the vessels.

✦ 1. Causes of Hypervolemia

▽ **Heart disease:** Impaired cardiac function can reduce effective circulation and promote retention of sodium and water by the kidneys.

▽ **Renal disease:** Diseased kidneys may be unable to excrete adequate amounts of sodium and water, resulting in fluid accumulation.

▽ **Liver disease:** Severe liver dysfunction can contribute to fluid retention and changes in plasma protein levels, particularly albumin, which can promote **edema**.

▽ **Protein malnutrition:** Severe protein deficiency can reduce plasma protein levels and therefore decrease **colloid osmotic pressure**, allowing more fluid to move into interstitial tissues.

▽ **Mismanagement of intravenous fluids:** Administration of excessive IV fluid can increase extracellular and intravascular volume beyond what the body can safely handle.

▽ Hypervolemia may manifest as **edema, increased body weight, elevated blood pressure, and fluid accumulation in tissues**, depending on the underlying cause and severity.

✓ **Hypervolemia is commonly associated with excessive retention or administration of sodium and water.**

✦ C. Hypovolemia — Fluid Deficit / Dehydration

▽ **Hypovolemia** occurs when fluid loss exceeds fluid replacement, resulting in a reduction of circulating blood volume and extracellular fluid volume.

▽ Severe hypovolemia can reduce **venous return, cardiac output, tissue perfusion, and blood pressure**.

✦ 1. Causes of Hypovolemia

▽ **Hemorrhage**: Loss of blood causes direct loss of water, electrolytes, plasma proteins, and blood cells.

▽ **Excessive diarrhea**: Large amounts of water and electrolytes can be lost through the gastrointestinal tract.

▽ **Excessive vomiting**: Repeated vomiting causes loss of water and electrolytes and can also disturb **acid–base balance**.

▽ **Excessive sweating**: Heavy sweating causes loss of water and electrolytes, particularly when fluid replacement is inadequate.

▽ **Mismanagement of intravenous fluids**: Inadequate or inappropriate fluid replacement may fail to correct ongoing fluid losses and allow hypovolemia to develop or worsen.

▽ Clinical manifestations can include **thirst, weakness, dizziness, reduced urine output, rapid heart rate, and low blood pressure**, particularly when fluid loss becomes significant.

✓ **Hypovolemia occurs when fluid loss is greater than fluid replacement.**

[24/09/2026 7:45 AM] Imaanaa Harmee: ✦ D. Dehydration Classifications

▽ **Dehydration** refers to a disturbance in body-fluid balance in which the body loses water and/or electrolytes.

▽ The classification depends on **which component of body fluid is lost and how the loss changes the concentration of the remaining body fluids**.

✦ 1. Isovolemic Dehydration

▽ **Isovolemic dehydration** occurs when **water and electrolytes are lost in approximately proportional amounts**.

▽ Because the concentration of the remaining body fluid changes relatively little, there is less movement of water between the ICF and ECF than in other forms of dehydration.

▽ Examples include some cases of **blood loss, diarrhea, vomiting, or excessive sweating**, depending on the composition of the fluid lost.

✓ **Isovolemic dehydration = water and electrolytes are lost proportionally.**

✦ 2. Hypovolemic Dehydration

▽ **Hypovolemic dehydration** occurs when **electrolyte loss is proportionally greater than water loss**.

▽ The ECF becomes relatively **dilute**, reducing its osmotic concentration compared with the ICF.

▽ Water therefore tends to move **from the ECF into the ICF**, which can further reduce extracellular and intravascular volume.

▽ This type of fluid disturbance can significantly affect **blood volume and circulation**.

✓ **Hypovolemic dehydration = proportionally greater loss of electrolytes than water.**

✦ 3. Hypervolemic Dehydration

▽ **Hypervolemic dehydration** occurs when **water loss is proportionally greater than electrolyte loss**.

▽ The ECF becomes relatively **concentrated**, increasing its osmotic concentration.

▽ Water then moves **from the ICF into the ECF** in an attempt to reduce the osmotic difference.

▽ As intracellular water is lost, cells become **dehydrated and shrink**.

▽ A common example is significant **water loss without adequate replacement**, such as excessive water loss through sweating or respiration when water intake is insufficient.

✓ **Hypervolemic dehydration = proportionally greater loss of water than electrolytes.**

✦ E. Relationship Between Fluid Volume and Osmolarity

▽ Fluid-volume disturbances and osmotic disturbances often occur together because losing water or electrolytes changes both the **amount of body fluid** and the **concentration of dissolved particles**.

▽ The resulting osmotic changes determine whether water remains in the ECF or shifts between the **ECF and ICF**.

▽ Therefore, when assessing dehydration, it is important to consider both **how much fluid has been lost** and **what type of fluid has been lost**.

✓ **Key idea: Fluid disturbances can involve either fluid excess (hypervolemia) or fluid deficit (hypovolemia). Dehydration can be classified according to whether the loss involves water and electrolytes proportionally, relatively more electrolytes, or relatively more water, which determines how fluid shifts between the ECF and ICF.**

[24/09/2026 7:45 AM] Imaanaa Harmee: ☞ **5. Renal Anatomy, Microanatomy, and Nephron Subtypes**

✦ **A. Four Primary Functions of the Kidneys**

▽ The **kidneys** are paired organs that perform several essential functions in maintaining the body's internal environment. Their activities are closely related to **fluid balance, waste removal, blood pressure, hormone production, and metabolism**.

✦ **1. Regulation**

▽ The kidneys regulate the **osmolarity of body fluids**, helping maintain an appropriate concentration of dissolved substances in the blood and extracellular fluid.

▽ They regulate **extracellular fluid (ECF) volume** by adjusting the amount of water and sodium excreted in urine.

▽ They maintain **electrolyte balance**, including the levels of sodium, potassium, calcium, phosphate, and other ions.

▽ The kidneys contribute to **acid–base balance** by controlling the excretion of hydrogen ions and the conservation or production of bicarbonate.

▽ By regulating sodium and water balance and releasing renin, the kidneys also play an important role in maintaining **arterial blood pressure**.

✦ **2. Excretion**

▽ The kidneys remove **metabolic waste products** from the blood and eliminate them in urine.

▽ Important wastes include **urea, uric acid, creatinine, and other products of metabolism**.

▽ The kidneys also help eliminate **foreign substances**, including certain drugs, drug metabolites, and chemical residues.

▽ Excess amounts of substances such as **water and electrolytes** can also be excreted when they are present in greater amounts than the body requires.

✓ **Excretion prevents the accumulation of potentially harmful substances in the body.**

✦ 3. Secretion and Endocrine Functions

▽ The kidneys act as endocrine organs by producing or releasing substances that have important effects elsewhere in the body.

▽ They produce **erythropoietin (EPO)**, a hormone that stimulates the bone marrow to increase **red blood cell production** when oxygen delivery is inadequate.

▽ The kidneys convert vitamin D into its active hormonal form, **1,25-dihydroxyvitamin D₃ (calcitriol)**, which is important for calcium and phosphate regulation and bone health.

▽ They release **renin**, an enzyme that initiates the renin–angiotensin–aldosterone system and contributes to the regulation of blood pressure and fluid balance.

▽ The kidneys also produce substances such as **prostaglandins**, which participate in local regulation of renal blood flow and other physiological processes.

✦ 4. Metabolism

▽ The kidneys participate in **metabolic processes**, including **gluconeogenesis**.

▽ During prolonged fasting, the kidneys can synthesize **glucose from non-carbohydrate precursors**, helping maintain blood glucose levels.

✓ **Key idea:** The kidneys do much more than produce urine. They regulate the internal environment, remove wastes, perform endocrine functions, and participate in metabolism.

✦ B. Gross Anatomy of the Kidneys

▽ The kidneys are **paired, brownish, bean-shaped organs** located in the posterior part of the abdominal cavity.

▽ They are **retroperitoneal**, meaning they lie behind the parietal peritoneum rather than within the main peritoneal cavity.

▽ The kidneys extend approximately from the level of **T12 to L3 vertebrae**.

▽ The **right kidney is positioned slightly lower** than the left because the large **liver** occupies considerable space on the right side of the upper abdomen.

▽ Each kidney receives blood through the **renal artery** and returns blood to the circulation through the **renal vein**.

✦ C. Three Supporting Tissue Layers of the Kidney

▽ The kidneys are surrounded by several protective and supporting layers that help **protect them from trauma and infection and hold them in their normal anatomical position**.

✦ 1. Renal Capsule

▽ The **renal capsule** is a tough layer of fibrous connective tissue that directly surrounds the kidney.

▽ It helps maintain the kidney's shape and provides a protective barrier against physical damage and the spread of infection into the renal tissue.

✦ 2. Adipose Capsule

▽ Outside the renal capsule is the **adipose capsule**, which consists largely of fatty tissue. [24/09/2026 7:45 AM] Imaanaa Harmee: ▽ This fat provides **cushioning and protection** against mechanical injury.

▽ It also contributes to holding the kidney in its position within the abdominal region.

✦ 3. Renal Fascia

▽ The **renal fascia** is an outer layer of connective tissue that surrounds and anchors the kidney and its surrounding tissues.

▽ It helps **secure the kidneys to surrounding structures and the posterior abdominal wall**.

✓ **Renal capsule → direct protection | Adipose capsule → cushioning | Renal fascia → anchoring.**

✦ D. Internal Microanatomy of the Kidney

▽ Internally, each kidney is organized into a **renal cortex** and **renal medulla**, which have different structural and functional characteristics.

✦ 1. Renal Cortex

▽ The **renal cortex** is the outer portion of the kidney.

▽ It contains the **renal corpuscles**, which include the glomeruli and Bowman's capsules.

▽ It also contains the **proximal and distal convoluted portions of renal tubules** and associated blood vessels.

▽ The cortex is therefore an important site for the early stages of **filtration, reabsorption, and secretion**.

✦ 2. Renal Medulla

▽ The **renal medulla** lies deep to the cortex and is organized into several cone-shaped structures called **renal or medullary pyramids**.

▽ The pyramids contain portions of the **loops of Henle and collecting ducts**, which extend toward the renal papillae.

▽ The medulla is particularly important for establishing and maintaining the **osmotic gradient** that allows the kidneys to produce concentrated urine.

✦ 3. Renal Columns

▽ **Renal columns** are extensions of cortical tissue that project between the medullary pyramids.

▽ They contain important **blood vessels and connective tissue** that extend between regions of the medulla.

✦ 4. Renal Papillae and Calyces

▽ The pointed tip of each renal pyramid is called the **renal papilla**.

▽ Urine produced by the nephron and modified by the collecting ducts passes through the papilla and enters a **minor calyx**.

▽ Several minor calyces join to form **major calyces**, which then drain into the **renal pelvis**.

▽ The renal pelvis collects urine and funnels it into the **ureter**, which carries urine toward the urinary bladder.

✓ **Urine pathway:** Renal papilla → minor calyx → major calyx → renal pelvis → ureter.

✦ E. The Nephron — Functional Unit of the Kidney

▽ The **nephron** is the fundamental structural and functional unit responsible for the formation of urine.

▽ Each kidney contains approximately **1 million nephrons**.

▽ Nephrons collectively perform the major processes involved in urine formation: **glomerular filtration, tubular reabsorption, and tubular secretion**.

▽ Each nephron has two major components:

▽ **1. Glomerulus:** A specialized network of capillaries that receives blood from the afferent arteriole and performs the initial **filtration of plasma**.

▽ **2. Renal tubule:** A long tubular structure through which the filtrate passes. As the filtrate moves through the tubule, substances are selectively **reabsorbed into the blood or secreted into the tubular fluid**.

[24/09/2026 7:45 AM] Imaanaa Harmee: ✦ **F. Tubular Flow Pathway**

▽ After filtration occurs at the glomerulus, the filtrate follows a specific pathway through the nephron and collecting system.

▽ **Glomerulus → Proximal Convoluted Tubule → Loop of Henle → Distal Convoluted Tubule → Collecting Tubule → Collecting Duct**

▽ **Proximal Convoluted Tubule (PCT):** Located mainly in the cortex and responsible for extensive reabsorption of water, electrolytes, glucose, amino acids, and other useful substances.

▽ **Loop of Henle:** Extends into the medulla and is important for establishing the osmotic conditions needed for regulating water reabsorption and concentrating urine.

▽ **Distal Convoluted Tubule (DCT):** Located mainly in the cortex and participates in the fine regulation of electrolytes and acid–base balance.

▽ **Collecting Tubules and Collecting Ducts:** Receive fluid from multiple nephrons and carry it toward the renal papilla. They play an important role in the final regulation of **water and electrolyte excretion**.

✓ **The nephron modifies the filtered fluid as it moves through the renal tubule, ultimately producing urine.**

✦ **G. Two Major Types of Nephrons**

▽ Nephrons are commonly classified into **cortical nephrons** and **juxtamedullary nephrons** according to the location of their renal corpuscles and the length of their loops of Henle.

✦ **1. Cortical Nephrons**

▽ The glomeruli of **cortical nephrons** are located mainly in the **outer region of the renal cortex**.

▽ They have relatively **short loops of Henle** that extend only a limited distance into the outer medulla.

▽ They represent the **vast majority of nephrons**.

▽ Their tubular structures are closely associated with a dense network of **peritubular capillaries**, which receives substances reabsorbed from the renal tubules and supplies substances for tubular secretion.

▽ Blood flow through the cortical circulation is relatively rapid compared with the specialized medullary circulation.

▽ The cortical interstitial fluid has an osmolarity close to **300 mOsm/L** under normal conditions.

✓ **Cortical nephrons → outer cortex + short loops of Henle + abundant peritubular capillaries.**

✦ 2. Juxtamedullary Nephrons

▽ **Juxtamedullary nephrons** have renal corpuscles located close to the **corticomedullary junction**, in the deeper portion of the cortex.

▽ They possess **long loops of Henle** that extend deeply into the renal medulla.

▽ Their specialized blood vessels, particularly the **vasa recta**, extend alongside the long loops of Henle.

▽ Blood flow through the medullary circulation is relatively slow, which helps preserve the high osmolarity of the **medullary interstitial fluid**.

▽ This high medullary osmotic gradient is essential for the kidney's ability to **concentrate urine and conserve water**.

▽ Juxtamedullary nephrons therefore play a particularly important role in maintaining the **medullary osmotic gradient** and producing concentrated urine.

✓ **Juxtamedullary nephrons → deep cortex + long loops of Henle + vasa recta + urine concentration.**

[24/09/2026 7:45 AM] Imaanaa Harmee: ✦ **H. Glomerular Arterioles and Blood Flow**

▽ Each nephron receives blood through an **afferent arteriole**.

▽ The afferent arteriole carries blood into the **glomerulus**, where it forms a specialized tuft

of capillaries.

▽ The relatively high pressure within these capillaries promotes **glomerular filtration**, allowing water and small dissolved substances to pass into Bowman's capsule while retaining most blood cells and large plasma proteins.

▽ Blood that remains in the glomerular capillaries leaves through the **efferent arteriole**.

▽ The efferent arteriole then gives rise to different capillary networks depending on the nephron type.

▽ In **cortical nephrons**, it mainly forms the **peritubular capillary network** surrounding the renal tubules.

▽ In **juxtamedullary nephrons**, it contributes substantially to the **vasa recta**, which descend into the medulla and are important for maintaining the medullary osmotic gradient.

✓ **Blood pathway:** Renal artery → afferent arteriole → glomerulus → efferent arteriole → peritubular capillaries/vasa recta → renal vein.

✓ **Key idea:** The kidney is organized into the cortex, medulla, renal tubules, and collecting system. The nephron performs filtration and tubular processing, while cortical nephrons mainly support routine tubular processing and juxtamedullary nephrons are especially important for concentrating urine and conserving water.

[24/09/2026 7:45 AM] Imaanaa Harmee: ☺ 6. Renal Blood Flow & Serial Circulation

✦ A. Renal Blood Flow (RBF) Dynamics

▽ The kidneys receive a remarkably large amount of blood compared with their relatively small size. **Renal blood flow (RBF)** is normally about **21–22% of cardiac output**, corresponding to approximately **1,200 mL/min** in a typical adult.

▽ This high blood flow is greater than what would be required simply to provide oxygen and nutrients for renal metabolism.

▽ The major reason for the high renal blood flow is to deliver a large amount of **plasma to the glomeruli**, allowing the kidneys to perform continuous and efficient **glomerular filtration**.

▽ Renal blood flow is determined by the relationship between the **pressure gradient** driving blood through the kidneys and the **vascular resistance** opposing that flow.

▽ This relationship can be expressed using the basic flow equation:

▽ More specifically, $Q = \Delta P/R$, where **Q** represents blood flow, **ΔP** represents the pressure difference between the renal artery and renal vein, and **R** represents total renal vascular

resistance.

▽ Therefore, increasing the pressure gradient tends to increase renal blood flow, whereas increasing renal vascular resistance tends to decrease it.

✓ **The kidneys receive high blood flow primarily to support their high filtration requirements.**

✦ B. Anatomical Path of Renal Blood Flow

▽ Blood enters each kidney through the **renal artery**, which enters at the kidney's **hilum**.

▽ The renal artery progressively divides into smaller arteries as it travels through the kidney.

▽ The major pathway is:

▽ **Renal artery → interlobar arteries → arcuate arteries → interlobular (radial) arteries → afferent arterioles → glomerular capillaries → efferent arterioles → peritubular capillaries/vasa recta → interlobular veins → arcuate veins → interlobar veins → renal vein**

▽ **Interlobar arteries** travel between the renal pyramids.

▽ **Arcuate arteries** arch along the boundary between the cortex and medulla.

▽ **Interlobular arteries** extend into the cortex and give rise to the **afferent arterioles**.

▽ The **afferent arteriole** carries blood into the glomerulus.

▽ After filtration, blood leaves the glomerulus through the **efferent arteriole** and enters the second capillary network.

▽ In cortical nephrons, this network is mainly the **peritubular capillaries**.

▽ In juxtamedullary nephrons, specialized vessels called the **vasa recta** extend into the medulla.

▽ Venous blood then returns through progressively larger veins and ultimately leaves the kidney through the **renal vein**.

✓ **The renal circulation carries blood through two capillary beds before returning it to the systemic venous circulation.**

✦ C. The Serial Capillary Bed Arrangement

▽ One of the most distinctive features of renal circulation is the presence of **two capillary beds arranged in series**.

▽ These two capillary networks are separated by an **efferent arteriole**, rather than being connected directly by a venule.

▽ The arrangement allows the kidney to maintain different pressure conditions in the two capillary beds and perform two different functions: **filtration and reabsorption**.

✦ 1. Glomerular Capillary Bed

▽ The **glomerulus** is the first capillary bed.

▽ It is a **high-pressure capillary system**, with glomerular hydrostatic pressure of approximately **60 mm Hg** under typical conditions.

▽ This relatively high pressure promotes the movement of water and small dissolved substances from the blood into **Bowman's capsule**, producing the initial filtrate.

▽ The glomerular capillary bed is unusual because it lies **between two arterioles**:

▽ **Afferent arteriole → glomerular capillaries → efferent arteriole**

▽ The diameter and resistance of these arterioles can be adjusted to influence **glomerular pressure and filtration rate**.

✓ **High pressure in the glomerulus favors filtration.**

[24/09/2026 7:45 AM] Imaanaa Harmee: ✦ 2. Peritubular Capillary Bed

▽ Blood leaves the glomerulus through the **efferent arteriole** and then enters the second capillary network.

▽ The **peritubular capillaries** surround the renal tubules, particularly in the cortex.

▽ They are a relatively **low-pressure capillary system**, with hydrostatic pressure of approximately **13 mm Hg**.

▽ Their lower hydrostatic pressure, together with plasma oncotic pressure, favors the movement of water and solutes **from the renal tubules/interstitial space back into the blood**.

▽ This supports **tubular reabsorption**, allowing useful substances such as water, electrolytes, glucose, and other molecules to return to the circulation.

▽ In juxtamedullary nephrons, the specialized **vasa recta** perform a similar exchange function around the long loops of Henle and are particularly important for maintaining the

medullary osmotic gradient.

✓ **Low pressure in the peritubular circulation favors reabsorption.**

✦ D. Role of the Efferent Arteriole

▽ The **efferent arteriole** connects the two renal capillary beds and acts as an important resistance vessel.

▽ By changing its degree of constriction, it can influence pressure in the **glomerular capillaries** and blood flow into the **peritubular circulation**.

▽ Changes in efferent arteriolar resistance therefore affect both **glomerular filtration** and the conditions surrounding the renal tubules.

▽ This arrangement allows the kidney to independently regulate the pressures needed for **filtration at the glomerulus** and **reabsorption around the tubules**.

✓ **Afferent arteriole controls blood entering the glomerulus, while the efferent arteriole controls blood leaving it and entering the second capillary bed.**

✦ E. Core Roles of Renal Blood Flow

▽ **1. Supports glomerular filtration:** Adequate blood flow delivers plasma to the glomeruli so that filtration can occur continuously.

▽ **2. Supports waste excretion:** Filtration and subsequent tubular processing allow metabolic waste products and excess substances to be removed from the blood.

▽ **3. Regulates body-fluid composition:** By adjusting filtration, reabsorption, and excretion, the kidneys can rapidly influence **ECF volume and electrolyte composition**.

▽ **4. Provides oxygen and nutrients:** Renal blood flow supplies kidney tissues with the oxygen and nutrients needed for their metabolic activities.

▽ **5. Contributes to circulatory adaptation:** During severe systemic stress or shock, sympathetic activation can greatly reduce renal blood flow, helping redirect limited circulating blood toward organs such as the **heart and brain**.

✓ **Renal blood flow supports filtration, waste removal, fluid regulation, tissue metabolism, and circulatory adaptation.**

✦ F. Distribution of Renal Blood Flow

▽ Renal blood flow is not distributed equally throughout the kidney.

- ▽ Approximately **94% of renal blood flow reaches the cortex.**
- ▽ Only about **6% reaches the renal medulla.**
- ▽ The high cortical blood flow reflects the large number of **glomeruli and cortical tubules** located in this region.
- ▽ The relatively low medullary blood flow is especially important because excessive blood flow through the medulla could disrupt the **osmotic gradient** required for producing concentrated urine.
- ✓ **Cortex $\approx$ 94% of RBF | Medulla $\approx$ 6% of RBF.**

✦ G. Physiological Control of Renal Blood Flow and GFR

▽ Both **renal blood flow (RBF)** and **glomerular filtration rate (GFR)** are carefully regulated because excessive or inadequate filtration can disturb body-fluid homeostasis.
[24/09/2026 7:45 AM] Imaanaa Harmee: ✦ **1. Sympathetic Nervous System Control**

- ▽ The renal blood vessels receive extensive **sympathetic nerve innervation.**
- ▽ Under normal resting conditions, sympathetic activity to the kidneys is relatively low.
- ▽ During severe activation of the sympathetic nervous system, such as **major trauma, severe stress, or significant blood loss**, renal blood vessels undergo marked **vasoconstriction.**
- ▽ Vasoconstriction of the **renal arteries and especially the afferent arterioles** reduces renal blood flow.
- ▽ Strong sympathetic activation also decreases **GFR**, thereby reducing urine formation and helping conserve water and sodium.
- ▽ This response is useful during severe acute stress because preserving circulating blood volume and redirecting blood toward vital organs becomes more important than maintaining normal renal perfusion.
- ✓ **Strong sympathetic activation $\rightarrow$ renal vasoconstriction $\rightarrow$ $\downarrow$ RBF $\rightarrow$ $\downarrow$ GFR $\rightarrow$ conservation of body fluid.**

✦ 2. Angiotensin II Control

- ▽ **Angiotensin II** is a powerful vasoconstrictor produced as part of the **renin–angiotensin–**

aldosterone system (RAAS).

▽ Its concentration increases when conditions such as **reduced renal perfusion or reduced arterial pressure** activate renin release.

▽ Angiotensin II constricts both afferent and efferent vessels, but at appropriate concentrations it has a relatively greater effect on the **efferent arteriole**.

▽ Moderate efferent arteriolar constriction helps maintain **glomerular capillary pressure**, allowing GFR to remain relatively stable even when renal perfusion pressure falls.

▽ At the same time, efferent arteriolar constriction reduces blood flow through the downstream peritubular circulation.

▽ The resulting changes in peritubular capillary forces favor increased **reabsorption of sodium and water**, helping restore extracellular fluid volume and arterial pressure.

▽ Very high levels of angiotensin II, however, can cause substantial renal vasoconstriction and eventually **reduce renal blood flow and GFR**.

✓ **Angiotensin II helps preserve filtration and promote sodium/water retention when blood pressure or circulating volume falls.**

✦ H. Relationship Between RBF and GFR

▽ **Renal blood flow (RBF)** refers to the amount of blood passing through the kidneys per unit of time.

▽ **Glomerular filtration rate (GFR)** refers specifically to the amount of filtrate formed by all the glomeruli per unit of time.

▽ RBF and GFR are closely related, but they are **not the same measurement**.

▽ Changes in the resistance of the **afferent and efferent arterioles** can alter glomerular capillary pressure and therefore influence GFR.

▽ The kidney continuously adjusts vascular resistance to maintain appropriate filtration while also protecting overall body-fluid homeostasis.

✓ **Key idea: Renal circulation is unique because blood passes through a high-pressure glomerular capillary bed followed by a low-pressure peritubular/vasa recta network, with the efferent arteriole between them. This arrangement allows the kidney to efficiently perform filtration followed by tubular reabsorption, while neural and hormonal mechanisms regulate RBF and GFR according to the body's needs.**

[24/09/2026 7:45 AM] Imaanaa Harmee: ☞ **7. The Juxtaglomerular Apparatus (JGA) & Autoregulation**

✦ A. Juxtaglomerular Apparatus (JGA)

▽ The **juxtaglomerular apparatus (JGA)** is a specialized region of the nephron where the **distal tubule comes into close contact with the vascular pole of its own glomerulus.**

▽ It is located where the distal tubule passes near the **afferent arteriole** and the **efferent arteriole.**

▽ The JGA functions as an important regulatory system that continuously monitors conditions within the nephron and helps regulate **renal blood flow (RBF), glomerular filtration rate (GFR), blood pressure, and sodium balance.**

▽ It connects information about the composition and flow of tubular fluid with changes in the renal circulation.

✓ **The JGA acts as a communication and control center between the renal tubule and the glomerular blood vessels.**

✦ B. Two Critical JGA Cell Types

✦ 1. Macula Densa Cells

▽ The **macula densa** consists of specialized epithelial cells located in the wall of the distal tubule.

▽ These cells function primarily as **chemoreceptors**, monitoring the concentration of **NaCl in the tubular fluid.**

▽ They also provide information about the rate at which filtrate is reaching the distal nephron.

▽ When the GFR decreases, less filtrate and usually less NaCl reach the macula densa.

▽ When the GFR increases, more NaCl is delivered to the macula densa.

▽ Based on these changes, the macula densa releases local chemical signals that influence the **afferent arteriole** and the activity of **juxtaglomerular cells.**

✓ **Macula densa = senses tubular NaCl delivery and helps regulate GFR.**

✦ 2. Juxtaglomerular Cells (JGCs)

▽ **Juxtaglomerular cells** are modified smooth-muscle cells located mainly in the wall of the **afferent arteriole** near the glomerulus.

▽ They function as **mechanoreceptors**, responding to changes in the pressure or stretch of the afferent arteriole.

▽ When renal perfusion pressure falls, these cells increase secretion of the enzyme **renin** into the circulation.

▽ Renin initiates the **renin–angiotensin–aldosterone system (RAAS)**, which helps restore blood pressure and extracellular fluid volume.

▽ These cells also contribute to the regulation of the diameter and resistance of the renal arterioles.

✓ **Juxtaglomerular cells = sense reduced renal perfusion pressure and release renin.**

✦ C. JGA Response to Low GFR and Low Glomerular Pressure

▽ When renal perfusion pressure or GFR decreases, the macula densa detects **reduced NaCl delivery** to the distal tubule.

▽ The JGA responds through several coordinated mechanisms that attempt to restore **glomerular pressure, GFR, and systemic blood pressure.**

✦ 1. Activation of the Renin–Angiotensin–Aldosterone System (RAAS)

▽ Reduced renal perfusion stimulates **juxtaglomerular cells to release renin.**

▽ **Renin** is an enzyme that acts on **angiotensinogen**, a plasma protein produced primarily by the liver.

▽ Renin cleaves angiotensinogen to form **angiotensin I.**

▽ Angiotensin I is then converted to **angiotensin II**, mainly through the action of **angiotensin-converting enzyme (ACE).**

▽ **Angiotensin II** is a powerful vasoconstrictor and has important effects on both the kidneys and the systemic circulation.

✦ 2. Angiotensin II and the Efferent Arteriole

▽ Angiotensin II preferentially constricts the **efferent arteriole** at appropriate concentrations.

▽ Because the efferent arteriole is the vessel through which blood leaves the glomerulus, its

constriction increases resistance to blood leaving the glomerular capillaries.

▽ This can help maintain or increase **glomerular capillary hydrostatic pressure**, helping preserve GFR when renal perfusion pressure falls.

▽ However, severe or prolonged vasoconstriction can eventually reduce renal blood flow and GFR, so the effect depends on the degree of angiotensin II activation.

✓ **Moderate efferent constriction helps maintain glomerular pressure when renal perfusion falls.**

[24/09/2026 7:45 AM] Imaanaa Harmee: ✦ 3. Aldosterone and Sodium Reabsorption

▽ Angiotensin II stimulates the **adrenal cortex** to release the hormone **aldosterone**.

▽ Aldosterone increases **sodium reabsorption** in the distal nephron, particularly in the distal tubule and collecting duct.

▽ Because water tends to follow retained sodium, sodium retention promotes **water retention**.

▽ This increases extracellular fluid volume and helps restore **blood volume and arterial pressure**.

✓ **RAAS activation → sodium retention + water retention → increased ECF volume and blood pressure.**

✦ 4. Systemic Vasoconstriction

▽ Angiotensin II also produces **vasoconstriction throughout the systemic circulation**.

▽ Increased systemic vascular resistance helps raise **arterial blood pressure**.

▽ Therefore, the RAAS response does not simply act on the kidney; it helps restore circulation throughout the body.

✦ 5. Afferent Arteriolar Vasodilation

▽ Reduced NaCl delivery at the macula densa also promotes local signals that cause the **afferent arteriole to dilate**.

▽ Afferent dilation allows more blood to enter the glomerulus.

▽ When combined with appropriate efferent arteriolar constriction, this helps restore **glomerular capillary pressure and GFR**.

✓ **Low GFR** → ↓ **NaCl at macula densa** → **afferent dilation + renin release** → **support of GFR and blood pressure.**

✦ **D. Clinical Connection: ACE Inhibitors**

▽ **ACE inhibitors** reduce the conversion of angiotensin I to angiotensin II.

▽ This decreases the vasoconstrictor effects of angiotensin II and reduces aldosterone stimulation.

▽ As a result, ACE inhibitors can lower **systemic blood pressure** and reduce sodium and water retention.

▽ Because angiotensin II also helps maintain glomerular pressure through efferent arteriolar constriction, blocking this pathway can alter **GFR**, particularly in certain patients with reduced renal perfusion.

✓ **ACE inhibition lowers the activity of the RAAS and is therefore important in the treatment of several cardiovascular and renal conditions.**

✦ **E. JGA Response to High Glomerular Pressure — Myogenic Response**

▽ When systemic blood pressure rises suddenly, pressure within the renal circulation also increases.

▽ The increased pressure stretches the smooth-muscle wall of the **afferent arteriole**.

▽ The afferent arteriole responds automatically by contracting. This is called the **myogenic response**.

▽ **Myogenic vasoconstriction** reduces blood flow entering the glomerulus.

▽ This prevents excessive increases in glomerular capillary pressure and helps keep **GFR relatively stable** despite short-term changes in arterial pressure.

▽ This is an example of **renal autoregulation**, meaning the kidney can adjust its own vascular resistance without requiring direct changes from the systemic nervous system.

✓ **High renal pressure** → **afferent arteriole constriction** → **reduced glomerular pressure** → **protection of GFR.**

✦ **F. Limits of Myogenic Autoregulation**

▽ Myogenic autoregulation is particularly effective against **short-term or transient**

changes in blood pressure.

- ▽ It cannot completely prevent the effects of **long-standing severe hypertension**.
- ▽ Persistent hypertension can progressively damage renal blood vessels and glomerular structures, eventually impairing normal renal function.
- ▽ Therefore, autoregulation provides important short-term protection but does not make the kidneys immune to chronic blood-pressure elevation.

✓ **Autoregulation buffers changes in renal pressure; it does not completely protect the kidneys from chronic hypertension.**

✦ G. Hormonal and Local Regulators of RBF and GFR

▽ Renal blood flow and GFR are influenced by several hormones and locally produced substances that either **constrict or dilate renal blood vessels**.

[24/09/2026 7:45 AM] Imaanaa Harmee: ✦ **1. Major Vasoconstrictors**

- ▽ **Angiotensin II:** Produces vasoconstriction and, at appropriate concentrations, preferentially constricts the efferent arteriole to help maintain glomerular pressure.
- ▽ **Endothelin:** A potent vasoconstrictor produced by vascular endothelial cells and other tissues. Increased endothelin activity can reduce renal blood flow.

✦ 2. Major Vasodilators

- ▽ **Nitric oxide (NO):** Promotes relaxation of vascular smooth muscle and helps maintain renal blood flow.
 - ▽ **Prostaglandins:** Locally produced prostaglandins, including **PGE₂**, help counteract excessive renal vasoconstriction and support renal perfusion.
 - ▽ **Atrial natriuretic peptide (ANP):** Promotes renal effects that favor sodium and water excretion and can influence glomerular filtration and renal vascular tone.
- ✓ **Vasoconstrictors generally reduce renal blood flow, while vasodilators help preserve or increase renal perfusion.**

✦ H. Overall JGA Feedback Mechanism

▽ The JGA continuously links **tubular NaCl delivery, renal perfusion pressure, renal blood flow, and GFR**.

▽ When GFR becomes too low, the macula densa detects reduced NaCl delivery and initiates responses that help restore filtration.

▽ These responses include **renin release, activation of angiotensin II, efferent arteriolar constriction, aldosterone release, systemic vasoconstriction, and afferent arteriolar dilation.**

▽ When renal pressure rises suddenly, the **myogenic response** constricts the afferent arteriole and limits excessive increases in renal blood flow and GFR.

✓ **Key idea:** The JGA is a major autoregulatory system of the kidney. The macula densa monitors tubular NaCl delivery, while juxtaglomerular cells respond to renal vascular pressure and release renin. Together, these mechanisms help stabilize RBF and GFR and also contribute to the regulation of blood pressure, sodium balance, and body-fluid volume.

[24/09/2026 7:45 AM] Imaanaa Harmee: ☞ 8. Cardio-Renal Integration & Tubule Histology

✦ A. Atrial Stretch Reflex Loop — Cardio-Renal Integration

▽ The kidneys and cardiovascular system work together to maintain **blood volume, blood pressure, and extracellular fluid balance.**

◆ When **blood volume increases**, more blood returns to the heart, causing increased stretching of the walls of the **atria**, particularly the right atrium.

▽ Atrial stretch activates cardiovascular receptors and promotes hormonal responses that encourage the body to **remove excess sodium and water.**

✦ 1. Effect on ADH

▽ Increased blood volume and atrial stretch contribute to suppression of **antidiuretic hormone (ADH)** release.

▽ ADH normally promotes water reabsorption by the kidneys. Therefore, reduced ADH activity allows **more water to remain in the tubular fluid and be excreted in urine.**

▽ This helps prevent further expansion of blood volume.

✦ 2. Release of ANP

▽ Stretch of the atrial walls stimulates the release of **atrial natriuretic peptide (ANP)** from specialized cardiac muscle cells.

▽ ANP acts on the kidneys and blood vessels to promote the elimination of **sodium and water**.

▽ It promotes renal effects that increase sodium excretion, a process known as **natriuresis**, and increases water excretion, known as **diuresis**.

✦ 3. Effects of ANP on the Kidney

▽ ANP can promote **afferent arteriolar dilation** and influence glomerular filtration, helping increase the amount of filtrate formed.

▽ ANP also reduces sodium reabsorption in parts of the renal tubule.

▽ Increased filtration together with reduced sodium reabsorption causes more **sodium and water to remain in the tubular fluid**.

▽ The increased urinary loss of sodium and water helps reduce **extracellular fluid volume, blood volume, and blood pressure** toward normal.

✓ **Increased blood volume → atrial stretch → ↑ ANP and ↓ ADH → ↑ sodium and water excretion → ↓ blood volume and blood pressure.**

✦ B. Low Blood Volume — Systemic Response

▽ When blood volume falls significantly, venous return to the heart decreases and there is **less stretching of the atrial walls**.

▽ The body interprets this reduced stretch as a signal that circulating volume needs to be preserved.

▽ One important response is increased release of **ADH (vasopressin)**.

✦ 1. ADH and Water Conservation

▽ ADH acts primarily on the **collecting ducts of the kidneys**, increasing their permeability to water.

▽ This allows more water to be reabsorbed from the tubular fluid back into the circulation.

▽ As a result, less water is lost in urine and more water is retained by the body.

✦ 2. ADH and Blood Pressure

▽ At sufficiently high concentrations, ADH also acts as a **vasoconstrictor** on systemic arterioles.

▽ Vasoconstriction increases systemic vascular resistance and can help raise **arterial blood pressure**.

▽ Therefore, ADH contributes to both **water conservation** and, when strongly activated, **blood-pressure support**.

✓ **Low blood volume** → ↑ ADH → ↑ **renal water reabsorption** + **systemic vasoconstriction** → **preservation of volume and blood pressure**.

✦ C. Tubule Cell Histological Specializations

▽ Different parts of the nephron have different functions, so their epithelial cells have specialized structures that match their physiological roles.

▽ The amount of **microvilli, mitochondria, cell thickness, and membrane specialization** varies according to how actively each segment transports substances.

✦ 1. Proximal Convoluted Tubule (PCT)

▽ PCT cells are mainly **simple cuboidal epithelial cells** with highly specialized surfaces.

▽ Their apical surface contains a dense **brush border of microvilli**.

▽ The microvilli greatly increase the surface area available for contact with the tubular fluid, allowing extensive **reabsorption**.

▽ PCT cells contain numerous **mitochondria**, providing ATP needed for active transport processes.

[24/09/2026 7:45 AM] Imaanaa Harmee: ▽ The PCT performs a very large proportion of renal tubular reabsorption, including reabsorption of **sodium, water, glucose, amino acids, bicarbonate, and other filtered substances**.

▽ It also participates in **tubular secretion**, transferring selected substances from the blood into the tubular fluid.

✓ **PCT = abundant microvilli + many mitochondria** → **very active reabsorption and secretion**.

✦ 2. Thin Segment of the Loop of Henle

▽ The thin portions of the loop of Henle are lined mainly by **simple squamous epithelial**

cells.

- ▽ These cells are thin and contain relatively few organelles, including fewer mitochondria than the metabolically active PCT cells.
 - ▽ Their structure provides a short diffusion distance across the epithelium.
 - ▽ The **thin descending limb** is highly permeable to water, allowing water to move out of the tubule by **osmosis** when the surrounding medullary interstitial fluid is more concentrated.
 - ▽ This passive water movement contributes to the concentration of tubular fluid as it descends into the renal medulla.
- ✓ **Thin descending limb = thin epithelium + high water permeability → mainly passive water movement.**

✦ 3. Thick Segment of the Loop of Henle and Distal Convoluted Tubule

- ▽ The **thick ascending limb** and **distal convoluted tubule (DCT)** contain mainly **cuboidal epithelial cells**.
 - ▽ Compared with PCT cells, they have **fewer microvilli**, giving them a less prominent brush border.
 - ▽ They also have substantial metabolic activity and mitochondria because active transport is important in these segments.
 - ▽ The **thick ascending limb** actively transports electrolytes out of the tubular fluid but is relatively impermeable to water. This helps establish the medullary osmotic gradient.
 - ▽ The **DCT** performs additional regulation of electrolyte composition and acid–base balance and participates in the fine control of sodium and other ions.
- ✓ **Thick ascending limb = active electrolyte transport with little water movement.**

✦ 4. Collecting Duct Cells

- ▽ The collecting ducts are lined mainly by **simple cuboidal to columnar epithelial cells**, depending on their location.
- ▽ Their cells have fewer microvilli than PCT cells because they are not specialized for the massive bulk reabsorption performed by the PCT.
- ▽ The collecting ducts are important for the **final regulation of water and electrolyte excretion**.

▽ Their permeability to water is strongly regulated by **ADH**.

▽ When ADH is present, collecting-duct cells insert **aquaporin-2 water channels** into their apical membranes, increasing water reabsorption.

▽ When ADH levels are low, the collecting ducts become much less permeable to water, allowing more water to remain in the urine.

✓ **Collecting duct = final adjustment of urine composition, especially under the control of ADH.**

✦ D. Relationship Between Histology and Function

▽ The microscopic structure of each nephron segment directly reflects its physiological function.

▽ **PCT:** Large surface area and abundant mitochondria → extensive active reabsorption.

▽ **Thin descending limb:** Thin squamous cells → efficient passive water movement.

▽ **Thick ascending limb:** Specialized transport cells → active electrolyte reabsorption with minimal water movement.

▽ **DCT:** Specialized cuboidal cells → fine regulation of electrolytes and acid–base balance.

▽ **Collecting duct:** Hormone-responsive cells → final regulation of water and urine concentration.

✓ **Key idea: Nephron histology is closely matched to function. Segments that perform extensive active transport contain more metabolic machinery, while thin segments are structurally adapted for passive diffusion. The collecting duct provides the final, hormonally regulated control of water and electrolyte excretion.**

[24/09/2026 7:45 AM] Imaanaa Harmee: ☞ **9. Mechanisms of Urine Formation & GFR Dynamics**

✦ A. Three Core Processes of Urine Formation

▽ Urine formation occurs through three major renal processes: **glomerular filtration, tubular reabsorption, and tubular secretion.**

✦ 1. Glomerular Filtration

▽ Glomerular filtration is the first step in urine formation.

▽ Blood enters the glomerular capillaries under relatively high hydrostatic pressure. This pressure pushes **water and small dissolved substances** out of the plasma and into Bowman's capsular space.

▽ The resulting fluid is called **glomerular filtrate** and initially resembles plasma in its small dissolved substances, but it normally contains very little protein and essentially no blood cells.

✓ **Filtration moves substances from blood → Bowman's capsule.**

✦ 2. Tubular Reabsorption

▽ As filtrate travels through the renal tubule, the body selectively takes back substances that it needs.

▽ Water and useful solutes such as **sodium, glucose, amino acids, and bicarbonate** can be reabsorbed from the tubular fluid into the surrounding interstitial fluid and then into the peritubular capillaries.

▽ Reabsorption prevents valuable substances from being unnecessarily lost in urine.

✓ **Reabsorption moves useful substances from renal tubule → blood.**

✦ 3. Tubular Secretion

▽ Tubular secretion moves selected substances from the blood into the renal tubular fluid.

▽ This process is especially important for eliminating substances that need to be removed or regulated, including **hydrogen ions, potassium, creatinine, and certain drugs and organic acids/bases**.

▽ Secretion therefore provides an additional route for substances to enter the urine after filtration.

✓ **Secretion moves selected substances from blood → renal tubule.**

✦ B. Why the Kidneys Use a Very High Filtration Rate

▽ The kidneys filter an enormous volume of fluid each day. A typical GFR of about **125 mL/min** corresponds to approximately **180 L/day** of filtrate.

▽ Yet only a small fraction of this filtered fluid becomes final urine. Normally, **more than 99% of the filtered water is reabsorbed**, leaving roughly **1–2 L/day** of urine in many healthy adults, depending on hydration and other conditions.

▽ This may seem inefficient, but the high filtration rate provides an important advantage: the kidneys can continuously expose large amounts of plasma water and solutes to the nephron's regulatory mechanisms.

✦ 1. Rapid Waste Removal

▽ Waste products are continuously delivered to the kidneys in the blood.

▽ By filtering a large volume of plasma, the kidneys can rapidly process substances such as **urea and other metabolic wastes**.

▽ Useful substances can then be selectively reclaimed through tubular reabsorption.

✦ 2. Repeated Processing of ECF

▽ Because the kidneys filter a volume comparable to the body's extracellular fluid many times over during a day, they have repeated opportunities to regulate **electrolytes, water, acid–base balance, and waste products**.

✓ **High filtration + selective reabsorption allows the kidneys to remove wastes efficiently without losing most of the body's useful water and solutes.**

✦ C. Glomerular Filtration Rate (GFR)

▽ **GFR** is the volume of filtrate formed by both kidneys per unit of time.

▽ A commonly used average value is approximately **125 mL/min**, or about **180 L/day**.

▽ GFR is one of the most important indicators of overall kidney filtration function.

▽ GFR depends on the **net filtration pressure** across the glomerular filtration membrane and on the filtration properties of that membrane.

✦ 1. GFR and Filtration Coefficient

▽ The relationship can be expressed as:

$$\text{GFR} = K_f \times \text{NFP}$$

▽ **K_f**, the filtration coefficient, reflects the ability of the glomerular capillaries to filter fluid.

▽ It depends mainly on two properties:

▽ **Filtration surface area** — more available surface area provides more opportunity for filtration.

[24/09/2026 7:45 AM] Imaanaa Harmee: ▽ **Membrane permeability** — a more permeable filtration barrier allows fluid to cross more easily.

✓ **GFR depends on both the pressure driving filtration and the physical characteristics of the filtration membrane.**

✦ **D. Starling Forces Governing Glomerular Filtration**

▽ Filtration across the glomerulus is determined by opposing **hydrostatic and oncotic pressures**.

▽ Some pressures push fluid **out of the glomerular capillaries**, favoring filtration.

▽ Other pressures oppose filtration by pushing fluid toward or retaining fluid within the capillary.

✦ **1. Glomerular Capillary Hydrostatic Pressure (GCHP)**

▽ Approximately **60 mm Hg**.

▽ This is the pressure of blood within the glomerular capillaries.

▽ It pushes water and small solutes **out of the capillaries and into Bowman's capsule**.

▽ It is therefore the major pressure favoring filtration.

✓ **GCHP → favors filtration.**

✦ **2. Bowman's Capsular Hydrostatic Pressure**

▽ Approximately **18 mm Hg**.

▽ Fluid already present within Bowman's capsule produces pressure that opposes additional filtration.

▽ It pushes back against fluid leaving the glomerular capillaries.

✓ **Capsular hydrostatic pressure → opposes filtration.**

✦ **3. Glomerular Capillary Oncotic Pressure (GCOP)**

▽ Approximately **32 mm Hg**.

▽ Plasma proteins remain within the glomerular capillaries and create **colloid osmotic/oncotic pressure**.

▽ These proteins attract water toward the blood, opposing movement of fluid into Bowman's capsule.

✓ **GCOP → opposes filtration.**

✦ 4. Bowman's Capsular Oncotic Pressure

▽ Normally approximately **0 mm Hg** because significant amounts of plasma protein are normally prevented from entering the filtrate.

▽ Therefore, there is essentially no protein-generated oncotic force within Bowman's capsular space under normal conditions.

✓ **Bowman's capsular oncotic pressure → normally negligible.**

✦ E. Net Filtration Pressure (NFP)

▽ The pressures favoring filtration are compared with the pressures opposing filtration.

▽ Using the values in your lecture:

$$\text{NFP} = (\text{GCHP} + \text{Bowman's capsular oncotic pressure}) - (\text{Bowman's capsular hydrostatic pressure} + \text{GCOP})$$

▽ Therefore:

$$\text{NFP} = (60 + 0) - (18 + 32) = 10 \text{ mm Hg}$$

▽ A positive NFP means that the forces favoring filtration are greater than the forces opposing it.

✓ **$\text{NFP} \approx 10 \text{ mm Hg}$ → fluid is driven from glomerular capillaries into Bowman's capsule.**

✦ F. Regulation of GFR Through Arteriolar Resistance

▽ Because glomerular capillary hydrostatic pressure is a major determinant of GFR, changes in the resistance of the **afferent and efferent arterioles** can alter filtration.

✦ 1. Afferent Arteriole

- ▽ The afferent arteriole carries blood **into the glomerulus**.
- ▽ **Afferent constriction** increases resistance before blood reaches the glomerulus.
- ▽ This decreases glomerular capillary pressure and usually decreases **both renal blood flow and GFR**.
- ▽ Afferent dilation has the opposite general effect, increasing blood flow into the glomerulus and tending to increase GFR.
- ✓ **Afferent constriction** → ↓ RBF → ↓ glomerular pressure → ↓ GFR.

✦ 2. Efferent Arteriole

- ▽ The efferent arteriole carries blood **away from the glomerulus**.
- ▽ Moderate efferent constriction can increase pressure within the glomerular capillaries and help maintain GFR when renal perfusion pressure falls.
- ▽ However, very strong efferent constriction can eventually reduce renal blood flow substantially and increase the concentration of proteins in glomerular blood, so GFR may eventually decrease.
- ✓ **The effect of efferent resistance on GFR depends on the degree of constriction and the surrounding physiological conditions.**

✦ G. Physical Filtration Membrane

- ▽ The glomerular filtration barrier is a highly specialized structure that allows water and small solutes to pass while retaining **blood cells and most plasma proteins**.

- ▽ It is commonly described as having three functional layers:

[24/09/2026 7:45 AM] Imaanaa Harmee: ✦ 1. Fenestrated Glomerular Endothelium

- ▽ The inner endothelial cells contain numerous small pores called **fenestrations**.
- ▽ These allow water and small dissolved substances to pass while preventing blood cells from crossing the capillary wall.

✦ 2. Glomerular Basement Membrane

- ▽ The basement membrane forms a major physical and electrical barrier to filtration.
- ▽ It contributes to the selective restriction of larger molecules and negatively charged

plasma proteins.

✦ 3. Podocytes and Filtration Slits

- ▽ Podocytes are specialized cells forming the visceral layer of Bowman's capsule.
- ▽ Their branching extensions end in small structures called **pedicels**.
- ▽ Spaces between adjacent pedicels form **filtration slits**, through which filtrate passes toward Bowman's capsular space.

✦ 4. Direction of Filtrate Movement

- ▽ The general pathway is:

Glomerular blood → fenestrated endothelium → basement membrane → podocyte filtration slits → Bowman's capsular space

- ✓ The filtration barrier is selective rather than simply acting like an open sieve.

✦ H. Parietal Layer of Bowman's Capsule

- ▽ Bowman's capsule has two layers.
- ▽ The **visceral layer** is formed by podocytes and participates directly in the filtration barrier.
- ▽ The **parietal layer** forms the outer wall of the capsule and is composed mainly of **simple squamous epithelial cells**.
- ▽ The capsular space lies between these layers and receives the newly formed filtrate.
- ✓ **Visceral layer = podocytes and filtration. Parietal layer = outer wall of Bowman's capsule.**

✦ I. Filterability of Solutes

- ▽ Not every substance in the blood is filtered equally.
- ▽ The ability of a substance to cross the glomerular filtration barrier depends mainly on its **size and electrical charge**, as well as the properties of the filtration barrier.

✦ 1. Freely Filtered Substances

▽ Small substances that are normally freely filtered include:

- ▽ Water
- ▽ Sodium and other electrolytes
- ▽ Glucose
- ▽ Amino acids
- ▽ Urea

▽ Many other small dissolved molecules

▽ These substances are small enough to pass through the filtration barrier relatively easily.

✦ 2. Blood Cells

▽ Red blood cells, white blood cells, and platelets are normally retained within the circulation.

▽ Their large size prevents them from crossing the normal glomerular filtration barrier.

✦ 3. Plasma Proteins

▽ Most plasma proteins are also retained in the blood.

▽ **Albumin**, for example, is relatively large and negatively charged, making it poorly filtered under normal conditions.

✓ **Normal filtrate contains water and small solutes but essentially no blood cells and very little protein.**

✦ J. Effect of Molecular Size

▽ Smaller molecules generally pass through the filtration barrier more readily than larger molecules.

▽ As molecular size increases, passage through the filtration barrier becomes progressively more restricted.

▽ Therefore, molecular weight is an important determinant of filterability.

✓ **Smaller molecules → generally easier filtration.**

✦ K. Effect of Molecular Charge

▽ The glomerular filtration barrier contains negatively charged components.

▽ Negatively charged molecules experience greater electrical repulsion from these parts of the barrier.

▽ Because albumin is both relatively large and negatively charged, it is normally retained in the bloodstream.

▽ Changes that damage the filtration barrier can increase protein leakage into urine, producing **proteinuria**.

✓ **Normal glomerular filtration depends on both size selectivity and charge selectivity.**

[24/09/2026 7:45 AM] Imaanaa Harmee: ✦ **L. Why the Glomerulus Is Highly Efficient**

▽ The glomerulus is particularly well designed for filtration compared with ordinary systemic capillary beds.

✦ **1. High Capillary Hydrostatic Pressure**

▽ Glomerular capillary pressure is relatively high, providing a strong force driving fluid out of the blood.

✦ **2. Highly Permeable Filtration Barrier**

▽ The specialized fenestrated endothelium, basement membrane, and podocytes provide a large and effective filtration surface.

✦ **3. Favorable Net Filtration Pressure**

▽ The combination of high glomerular hydrostatic pressure and the opposing pressures produces a positive **net filtration pressure**.

✦ **4. Specialized Vascular Arrangement**

▽ The glomerulus is positioned between an **afferent arteriole and an efferent arteriole**.

▽ This arrangement allows the kidney to regulate glomerular pressure by changing the resistance of these arterioles.

✓ **The glomerulus is an efficient filtration system because of its high pressure, specialized filtration membrane, large filtration capacity, and adjustable arteriolar blood supply.**

[24/09/2026 7:45 AM] Imaanaa Harmee: ☞ **10. Clearance, Tubular Reabsorption, and Transport Maximums**

✦ A. Renal Clearance

▽ **Renal clearance** describes the volume of blood plasma from which the kidneys effectively remove a particular substance per unit of time.

▽ Clearance is expressed in **mL/min** and is useful for evaluating how efficiently the kidneys handle different substances.

▽ Depending on the substance being studied, clearance can provide information about **glomerular filtration, tubular reabsorption, tubular secretion, and renal blood flow**.

✦ 1. Clearance Formula

▽ The basic clearance equation is:

$$C_s = (U_s \times V) / P_s$$

C_s = clearance of the substance in mL/min.

U_s = concentration of the substance in urine.

V = urine flow rate in mL/min.

P_s = concentration of the substance in plasma.

▽ A high clearance means that the kidneys are removing a relatively large amount of that substance from plasma.

✓ **Clearance tells us how effectively the kidneys remove a particular substance from plasma.**

✦ 2. Inulin and Creatinine

▽ **Inulin** is a classic substance for measuring GFR because it is freely filtered and is normally **neither reabsorbed nor secreted** by the renal tubules.

▽ Therefore, the amount filtered is essentially equal to the amount excreted.

▽ **Creatinine** is also widely used clinically as an indicator of GFR because it is produced by normal metabolism and is filtered at the glomerulus.

▽ Creatinine is not perfectly identical to inulin because a small amount can be secreted by the tubules, so creatinine clearance tends to slightly overestimate true GFR.

✓ **Inulin = ideal theoretical filtration marker; creatinine = practical clinical marker.**

✦ B. Tubular Reabsorption

▽ Tubular reabsorption is the movement of substances from the **renal tubular fluid back into the blood**.

▽ It allows the kidneys to remove useful substances from the filtrate before urine is excreted.

▽ Overall, the kidneys normally reabsorb **more than 99% of filtered water**, along with large amounts of filtered electrolytes and other useful substances.

▽ Reabsorption can occur through both **passive processes** and **active transport mechanisms**.

▽ The amount reabsorbed varies greatly between nephron segments and depends on the body's physiological needs.

✓ **Filtration creates a large volume of filtrate; reabsorption retrieves most of what the body needs.**

✦ C. Major Sites of Tubular Reabsorption

✦ 1. Proximal Convoluted Tubule (PCT)

▽ The PCT is the major site of bulk reabsorption.

▽ Its epithelial cells contain abundant mitochondria and an extensive apical **brush border of microvilli**, providing both energy and a large surface area for transport.

▽ Approximately **65% of filtered water and NaCl** is reabsorbed in the PCT under typical conditions.

▽ The PCT also reabsorbs essentially all filtered **glucose and amino acids** under normal plasma concentrations.

▽ Large amounts of **bicarbonate, phosphate, potassium, and other solutes** are also handled here.

▽ Water follows many reabsorbed solutes by osmosis, so PCT reabsorption is generally close to **isosmotic**.

✓ **PCT = major bulk-reabsorption segment.**

✦ 2. Descending Limb of the Loop of Henle

▽ The descending limb is highly permeable to **water**.

▽ As tubular fluid travels deeper into the renal medulla, it encounters an increasingly hyperosmotic interstitium.

▽ Water therefore moves out of the tubular fluid and into the surrounding interstitium by **osmosis**.

▽ Approximately **20% of filtered water** may be reabsorbed through the descending limb, depending on the segment and conditions.

✓ **Descending limb = water leaves; tubular fluid becomes progressively more concentrated.**

✦ 3. Ascending Limb and Distal Convuluted Tubule

▽ The ascending limb reabsorbs significant amounts of **Na⁺ and Cl⁻** while being essentially impermeable to water.

▽ This means salt leaves the tubular fluid without water following it.

[24/09/2026 7:50 AM] Imaanaa Harmee: ▽ The DCT continues the fine regulation of electrolytes and contributes to the adjustment of the final composition of urine.

✓ **Ascending limb = salt reabsorption without water.**

✦ D. Molecular Transport in the PCT

▽ PCT reabsorption depends on coordinated transport proteins located on the **apical and basolateral membranes** of tubular cells.

✦ 1. Sodium-Dependent Co-transport

▽ The sodium concentration gradient across the PCT cell membrane provides an important driving force for reabsorption.

▽ Sodium enters PCT cells from the tubular lumen through several transport systems.

▽ Sodium can be coupled with nutrients such as **glucose and amino acids**, allowing these valuable substances to be reabsorbed from the filtrate.

▽ Under normal conditions, essentially **100% of filtered glucose and amino acids** are reabsorbed, primarily in the PCT.

✦ 2. Sodium-Hydrogen Exchange

- ▽ Another important PCT mechanism is the **Na⁺/H⁺ exchanger**.
- ▽ Sodium moves into the PCT cell while hydrogen ions are secreted into the tubular lumen.
- ▽ Hydrogen secretion contributes to **acid–base regulation** and is closely connected with bicarbonate reabsorption.
- ▽ The hydrogen ions can combine with filtered bicarbonate through reactions that ultimately allow bicarbonate to be reclaimed by the body.
- ✓ **PCT transport simultaneously conserves valuable nutrients and contributes to acid–base regulation.**

✦ E. Tubular Transport Maximum (T_m)

- ▽ Many substances are transported by specialized **carrier proteins**.
- ▽ Carrier proteins have a finite capacity. When all available transporters are occupied, the system has reached its **tubular transport maximum (T_m)**.
- ▽ The T_m represents the maximum rate at which a substance can be transported by the tubular cells under specified conditions.

✦ 1. What Happens When T_m Is Exceeded?

- ▽ When the filtered load is below the transport capacity, the tubules can reabsorb essentially all of the substance.
- ▽ When the filtered load becomes greater than the available transport capacity, the carriers become **saturated**.
- ▽ Any substance remaining in the tubular fluid cannot be reabsorbed fast enough and therefore remains in the urine.
- ✓ **Filtered load > T_m → transporter saturation → excess substance appears in urine.**

✦ F. Glucose and Glucosuria

- ▽ Glucose provides a classic example of a substance with a tubular transport maximum.
- ▽ Normally, filtered glucose is almost completely reabsorbed in the PCT.
- ▽ As plasma glucose concentration rises, the amount of glucose filtered by the kidneys also

increases.

▽ Eventually, the glucose transporters approach saturation.

▽ The renal threshold for glucose is commonly described as approximately **200 mg/dL** in many teaching contexts, although the exact threshold varies among individuals.

▽ Once filtered glucose exceeds the reabsorptive capacity, glucose begins to remain in the tubular fluid and is excreted in urine

▽ The presence of glucose in urine is called **glucosuria**.

▽ This can occur when blood glucose becomes sufficiently elevated, such as in **poorly controlled diabetes mellitus**.

✓ **High plasma glucose → increased filtered glucose load → transporter saturation → glucosuria.**

✦ G. Loop of Henle — Segmental Specialization

▽ The loop of Henle has different epithelial properties in its descending and ascending portions.

▽ These differences are essential for establishing the **medullary osmotic gradient** used to concentrate urine.

✦ 1. Thin Descending Segment

▽ The thin descending segment is lined mainly by **simple squamous epithelial cells**.

▽ It has few mitochondria and little metabolic activity compared with the PCT.

▽ It has no prominent brush border.

▽ Its major characteristic is **high permeability to water**.

▽ Water leaves the tubular fluid passively when the surrounding medullary interstitium is more concentrated.

✓ **Thin descending limb = passive water reabsorption.**

[24/09/2026 7:50 AM] Imaanaa Harmee: ✦ 2. Thin Ascending Segment

▽ The thin ascending segment has a thin epithelial lining.

▽ Unlike the descending limb, it is essentially **impermeable to water**.

▽ Solutes can move across portions of the thin ascending limb, contributing to the overall medullary concentration gradient.

✓ **Thin ascending limb = little or no water movement outward.**

✦ 3. Thick Ascending Limb (TAL)

▽ The thick ascending limb contains metabolically active cuboidal epithelial cells with many mitochondria.

▽ It is **essentially impermeable to water**, so water cannot follow the salt being transported out of the tubule.

▽ The TAL actively reabsorbs substantial amounts of **Na⁺, K⁺, and Cl⁻**.

✦ 4. NKCC2 Transporter

▽ A major transporter in the TAL is the **Na⁺-K⁺-2Cl⁻ cotransporter (NKCC2)** on the apical membrane.

▽ It moves sodium, potassium, and two chloride ions from the tubular fluid into the epithelial cell.

▽ These ions are subsequently transported toward the interstitial fluid.

▽ The TAL reabsorbs approximately **25% of filtered NaCl**.

▽ It also contributes to the handling of **calcium and magnesium**, partly through electrical forces created by ion transport.

✓ **TAL = active sal reabsorption + no significant water reabsorption.**

✦ 5. Loop Diuretics

▽ Loop diuretic drugs act on the thick ascending limb by inhibiting **NKCC2**.

▽ Blocking this transporter reduces NaCl reabsorption.

▽ More sodium and chloride remain in the tubular fluid, and water excretion increases.

▽ These drugs therefore produce a strong **diuretic effect**.

✓ **NKCC2 inhibition → ↓ NaCl reabsorption → ↑ urinary salt and water loss.**

✦ H. Obligatory Water Reabsorption

▽ Salt transport by the ascending limb contributes to the high osmolarity of the renal medulla.

▽ The medullary interstitium becomes progressively more concentrated as solute accumulates deeper in the medulla.

▽ When tubular fluid passes through the descending limb, water moves out toward this hyperosmotic interstitium.

▽ This water movement occurs because of the **osmotic gradient** between the tubular fluid and surrounding medulla.

▽ This is sometimes described as **obligatory water reabsorption**, because a significant amount of water leaves the nephron as a consequence of the established osmotic environment.

▽ However, the amount of water ultimately excreted can still be regulated later, especially in the collecting ducts under the influence of **ADH**.

✓ **Ascending-limb salt transport creates the osmotic environment that drives water out of the descending limb.**

✦ I. Counter-Current System

▽ The kidney's ability to produce concentrated urine depends heavily on a counter-current arrangement involving the **loop of Henle and vasa recta**.

▽ The two structures have different but complementary functions.

▽ The **loop of Henle acts primarily as a counter-current multiplier**, while the **vasa recta acts primarily as a counter-current exchanger**.

✦ 1. Counter-Current Multiplier

▽ Tubular fluid flows downward through the descending limb and upward through the ascending limb.

▽ Because the two limbs have different permeability properties, especially regarding water and salt, their opposing flow helps establish a progressively increasing osmotic gradient in the renal medulla.

▽ The thick ascending limb actively removes NaCl from tubular fluid without allowing water to follow.

▽ This makes the surrounding medullary interstitium increasingly concentrated.

▽ Water then leaves the descending limb, concentrating the tubular fluid.

▽ As the fluid continues through the loop, repeated cycles of salt transport and water movement progressively establish the **medullary osmotic gradient**.

▽ Osmolarity is relatively low near the cortex, around **300 mOsm/L**, and can reach approximately **1200–1400 mOsm/L** deep in the inner medulla under conditions that favor maximal urine concentration.

✓ **Counter-current multiplication creates and maintains a strong corticomedullary osmotic gradient.**

[24/09/2026 7:50 AM] Imaanaa Harmee: ✦ 2. Counter-Current Exchange — Vasa Recta

▽ The **vasa recta** are specialized capillaries associated mainly with juxtamedullary nephrons.

▽ Their hairpin-shaped arrangement and slow blood flow help preserve the medullary osmotic gradient.

▽ Blood moving downward through the descending vasa recta encounters increasingly concentrated interstitial fluid.

▽ Water moves out of the blood, while NaCl and other solutes move into it.

▽ As blood travels upward through the ascending vasa recta, the surrounding interstitium becomes progressively less concentrated.

▽ Water moves back into the blood, while solutes move back toward the medullary interstitium.

▽ This exchange minimizes the net removal, or **washout**, of medullary solutes.

✓ **Vasa recta = preserve the medullary gradient while still supplying blood to the medulla.**

✦ J. Four Major Factors Supporting Medullar Hyperosmolality

▽ The high osmolarity of the renal medulla depends on several interacting mechanisms.

✦ 1. Counter-Current Arrangement

▽ The parallel, opposite-flowing limbs of the loop of Henle allow the osmotic gradient to be progressively amplified.

✦ 2. Ascending-Limb Salt Transport

▽ Na^+ , Cl^- , and K^+ movement out of the ascending limb contributes strongly to the accumulation of solute in the medullary interstitium.

✦ 3. Urea Recycling

▽ Urea contributes substantially to the high osmolarity of the inner medulla.

▽ Under the influence of ADH, urea permeability increases in portions of the inner medullary collecting duct.

▽ Urea moves into the medullary interstitium and can enter parts of the loop of Henle, creating a recycling process that helps maintain inner-medullary hyperosmolality.

✦ 4. Slow Vasa Recta Blood Flow

▽ Blood flow through the medulla is relatively low.

▽ Slow flow, together with counter-current exchange, reduces the tendency of circulating blood to carry medullary solutes away.

✓ **The medullary gradient depends on salt transport, urea recycling, counter-current architecture, and slow medullary blood flow.**

✦ K. Importance of Medullary Hyperosmolality

▽ The hyperosmotic renal medulla is essential for the kidney's ability to conserve water.

▽ When the body is dehydrated, **ADH levels increase**.

▽ ADH increases water permeability of the collecting ducts.

▽ Water can then move from the collecting duct into the hyperosmotic medullary interstitium and subsequently into the circulation.

▽ The stronger the medullary osmotic gradient, the greater the kidney's ability to reclaim water from the collecting duct.

▽ This allows the kidneys to produce a **small volume of concentrated urine** when water needs to be conserved.

✓ **Medullary hyperosmolality + ADH → increased collecting-duct water reabsorption → concentrated urine and conservation of body water.**

[24/09/2026 7:50 AM] Imaanaa Harmee:

GIT-Physiology

Dr. Doc

[24/09/2026 7:50 AM] Imaanaa Harmee:

GASTROINTESTINAL SYSTEM(GIT)

🌀 *Objectives*

- ▽ Structure
- ▽ Function
- ▽ Regulation
- ▽ Blood Supply

1. ACTIVITIES OF THE GI SYSTEM

★ The gastrointestinal (GI) system performs a **coordinated series of activities** that allow the body to take in food, digest it, absorb nutrients, and eliminate waste.

◇ **Major Activities**

◆ ***Ingestion***

- Intake of food into the digestive tract.
- **Feeding and satiety centers in the hypothalamus** 🧠 help regulate food intake.

◆ **Mastication (Chewing)**

- Mechanical breakdown of food by the teeth.
- **Increases** 📏 **the surface area** available for enzymatic digestion.

◆ ***Swallowing (Deglutition)***

- Propels the food **bolus** from the mouth → pharynx → esophagus → stomach.

◆ ***Chemical Digestion***

- Enzymatic breakdown of complex nutrients into smaller absorbable molecules.

◆ ***Secretion***

→ Release of:

 Digestive enzymes

 HCl

 HCO_3^-

 Mucus

Regulatory hormones

◆ *Absorption*

→ Movement of nutrients, water, and electrolytes across the intestinal mucosa **into: Blood**
+ **Lymphatic vessels**

◆ **Elimination (Defecation)**

→ Removal of indigestible material and feces through the anal canal.

🕒 **Regional Functions**

 Mouth

↓

Ingestion + mastication + **initial chemical digestion**

Esophagus

↓

Food propulsion

Stomach

↓

Temporary Storage + mixing + chemical digestion + Secretion + controlled emptying

Small intestine ★

↓

Major site of digestion + secretion + nutrient absorption

Large intestine

↓

Water/electrolyte absorption + fecal  storage + elimination

GI MOTILITY

① **Peristalsis** → **PROPULSION** 

★ A coordinated, wave-like contraction that moves intestinal contents forward.

 Food bolus → → → →

② Segmentation → MIXING

▽ Localized contractions that:

★ Mix chyme + digestive secretions → improve digestion & absorption

★ **Exam Point:**

Peristalsis = propulsion

Segmentation = mixing

[24/09/2026 7:50 AM] Imaanaa Harmee:

2. Structural Organization & Histology of the Digestive System

▽ Classification of Digestive Organs

Principal Structures — Alimentary Canal

▽ A continuous muscular tube extending from the mouth to the anus:

✓ Mouth → Pharynx → Esophagus → Stomach → Small intestine → Large intestine → Anus

Accessory Structures

 Teeth

 Tongue

Salivary glands

▽ Liver

▽ Gallbladder

▽ Pancreas

🔧 These structures assist in **mechanical breakdown or produce digestive secretions.**

Histology of the Alimentary Canal — 4 Layers

From the lumen outward:

🕒 **LUMEN → MUCOSA → SUBMUCOSA → MUSCULARIS EXTERNA → SEROSA**

1️⃣ **Mucosa**

▽ The innermost moist epithelial membrane lining the lumen.

Functions

-  Secretion of protective mucus
-  Absorption of digestive end products
-  Protection against pathogens

Three Sub-layers

A. Epithelium

★ **Mouth, esophagus and anal canal** → mainly **non-keratinized stratified squamous epithelium**

★ **Stomach and intestines** → **simple columnar epithelium**

⚙️ Specialized cells perform secretion, absorption and hormonal functions.

B. Lamina Propria

▽ Loose connective tissue containing:

Blood vessels
Lymphatic vessels
Immune/lymphoid tissue

 It nourishes the epithelium and participates in **mucosal defense**. 

C. Muscularis Mucosae

 **A thin layer of smooth muscle responsible for local mucosal movement.**

Submucosa

 **Connective tissue containing:**

Blood vessels
Lymphatic vessels
Glands
~ Nerve fibers

★ **Submucosal Plexus — Meissner's Plexus**

 **Primarily regulates:**

△ *GI secretion + local blood flow*

3 Muscularis Externa

△ Responsible mainly for **mixing and propulsion**.

Muscle arrangement

🌀 Inner → Circular muscle

↗ ↘ Outer → Longitudinal muscle

Skeletal muscle is present in:

Mouth

Pharynx

Upper esophagus

External anal sphincter

★ **Myenteric Plexus — Auerbach's Plexus**

↗ ↘ Located between the circular and longitudinal muscle layers.

🌀 **Main function:**

★ GI MOTILITY

4 Serosa

▽ **The outermost layer consisting of:**

👉 **Connective tissue**

👉 **Simple squamous epithelium**

👉 **Below the diaphragm, it forms the visceral peritoneum.**

[24/09/2026 7:50 AM] Imaanaa Harmee:

3. Detailed Anatomy of the Oral Cavity & Accessory Structures

☞ **Oral Cavity, Lips & Cheeks**

▽ The oral cavity is bounded by:

👉 Lips + cheeks + palate + tongue

✎ It is primarily lined by **stratified squamous epithelium**.

▽ The gums, hard palate and dorsum of the tongue are more resistant to mechanical abrasion.

▽ Vestibule

☞ Space between:
Lips/cheeks ↔ teeth/gums

▽ Oral Cavity Proper

☞ Space inside the arches of the teeth and gums.

▽ Lips & Cheeks

✓ Lips → orbicularis oris

✓ Cheeks → buccinator

✓ **Labial Frenulum:** A median fold attaching the lip to the adjacent gum.

☪ Palate & Tongue

▽ Hard Palate

★ Anterior

★ Bone-supported

★ Corrugated surface

★ Helps the tongue manipulate food during chewing

▽ Soft Palate

★ Posterior

★ Mobile muscular fold

★ Terminates in the **uvula**

Elevates during swallowing to **close the nasopharynx**

☪ Tongue

▽ The tongue:

☞ Manipulates food
Mixes food with saliva

☞ Forms the **bolus**

☞ Initiates swallowing

☞ Assists speech

☞ Contains taste receptors

☞ **Tongue Papillae**

☞ **Filiform Papillae**

Provide roughness and friction.

☞ **Fungiform Papillae**

Scattered over the tongue and appear reddish.

☞ **Circumvallate Papillae**

Large papillae arranged in a V-shaped row near the posterior tongue.

★ **Sulcus Terminalis**

▽ A groove dividing the tongue into:

☞ Anterior 2/3 → Oral part

☞ Posterior 1/3 → Pharyngeal part

💧 **Salivary Glands & Saliva** **Daily Saliva Production**

☞ Approximately:

1,000–1,500 mL/day

★ **Three Pairs of Major Salivary Glands**

1️⃣ **Parotid Glands**

📍 **Location:**

☞ **Anterior and inferior to the ears**

☞ The parotid/Stensen duct crosses the masseter, pierces the buccinator and opens opposite the second maxillary molar.

2️⃣ **Submandibular Glands**

▽ Located beneath the posterior floor of the mouth.
Wharton's duct opens near the base of the lingual frenulum.

3️⃣ **Sublingual Glands**

▽ Located in the floor of the mouth.

★ Drain through numerous small ducts, including Rivinus ducts.

Minor Salivary Glands

☞ Scattered throughout the oral mucosa.

✓ **Composition of Saliva**
Approximately 97–99.5% water.

▽ Contains:

Na⁺

K⁺

Cl⁻

PO₄²⁻

HCO₃⁻

Salivary amylase

Mucin

Lysozyme

Defensins

IgA

Urea

Uric acid

▽ **Saliva is generally hypo-osmotic and has a pH around 6–7.**

★ Important enzyme

👉 Salivary amylase → begins starch digestion

🧠 **Regulation of Salivary Secretion**

▽ Mainly controlled by the autonomic nervous system, especially **parasympathetic activity**.

★ Stimuli include:

Taste

Mechanical stimulation

Sight of food

Smell of food

Thought of food

Sound associated with food

[24/09/2026 7:50 AM] Imaanaa Harmee:

🦷 **Teeth: Classification, Structure & Pathology**

▽ Humans have **two dentitions**.

★ **Primary/Deciduous Dentition**

▽ 20 teeth 🦷

▽ Erupt approximately between 6–24 months.

▽ Usually shed between 6–12 years.

Formula

$$2 I + 1 C + 2 M / 2 I + 1 C + 2 M \times 2 = 20$$

★ Permanent Dentition

▽ 32 teeth

Formula

$$2 I + 1 C + 2 PM + 3 M / 2 I + 1 C + 2 PM + 3 M \times 2 = 32$$

🦷 *Functional Tooth Types*

✂ Incisors

✍ Cutting and nipping.

🦷 Canines

✂ Tearing and piercing.

☢ Premolars & Molars

🔨 Crushing and grinding.

🦷 Anatomical Structure of a Tooth

👑 Crown

✍ Exposed portion above the gingiva.

✍ Covered by **enamel, the hardest substance in the human body.**

Root

△ **Embedded within the alveolar bone.**

🌀 Anchored by:

🔗 Cementum + periodontal ligament

▽ This forms a **gomphosis joint.**

🌀 Neck

▽ Constricted region between the crown and root.

○ Dentin

✓ Bonelike tissue forming most of the tooth.

○ **Produced by odontoblasts.**

○ **Pulp Cavity**

Contains:

✓ Nerves

✓ Blood vessels

✓ Connective tissue

The pulp continues through the root canal and communicates through the apical foramen.

🦷 **Dental Diseases** 🚫

★ **Gingivitis** Not Gumitis 😊

👉 Inflammation of the gingiva, commonly associated with **plaque/calculus accumulation.**

★ **Periodontitis**

👉 Inflammatory destruction involving the periodontal ligament and alveolar bone.

★ **Dental Caries**

▽ Demineralization of enamel and dentin by acids produced when plaque bacteria metabolize dietary carbohydrates.

[24/09/2026 7:50 AM] Imaanaa Harmee:

4. Regulation & Neural-Hormonal Control of the GIT

🧠 **Gastrointestinal Receptors**

▽ Mechanoreceptors

Detect:

🌀 Stretch

🌀 Distension

🌀 Mechanical contraction

Chemoreceptors

Detect:

🌀 Osmolarity

🌀 pH

- ⚙ Chemical irritation
- ⚙ Digested proteins
- ⚙ Lipids
- ⚙ Carbohydrates

Thermoreceptors

- ⚙ Respond to temperature changes.

Nociceptors

- ⚙ **Detect pain** associated with tissue injury or severe ischemia.

🧠 Neural Control of the GIT

▽ Enteric Nervous System — ENS

- 👉 The ENS is an intrinsic nervous system located within the gut wall.

★ Myenteric Plexus — Auerbach

▽ Main function → MOTILITY

Controls:

- ☒ Smooth muscle tone
- ☒ Peristalsis
- ☒ GI movements

★ Submucosal Plexus — Meissner

Main function → SECRETION

Controls:

- ☒ Mucosal secretion
- ☒ Local blood flow

⚡ Enteric Neurotransmitters

+ Excitatory

Acetylcholine (ACh)
Substance P

⊖ Inhibitory

VIP

● Nitric oxide (NO)

● ATP

Other important signaling molecules include:

Serotonin

Enkephalins
Somatostatin
Dopamine

Extrinsic Autonomic Control

Parasympathetic Nervous System

Generally:

- ↑ GI motility
- ↑ GI secretion
- ↑ GI blood flow

Cranial Division

▽ Mainly **through the vagus nerve.**

Supplies much of:

- ✓ Esophagus
- ✓ Stomach
- ✓ Pancreas
- ✓ Intestine
- ✓ Proximal large intestine

Sacral Division

▽ **Originates from: S2–S4**

Through pelvic nerves.

Supplies:

- ✓ Distal colon
- ✓ Sigmoid colon
- ✓ Rectum
- ✓ Anal region

Important in the defecation reflex.

Sympathetic Nervous System

Generally:

- ↓ GI motility
- ↓ GI secretion
- ↓ GI blood flow

Sympathetic fibers originate mainly from: T5–L2

^ They pass through sympathetic pathways and synapse in prevertebral ganglia such as: Celiac ganglion, Mesenteric ganglia

⚡ Postganglionic neurons mainly release:
Norepinephrine (NE)

⊖ Strong sympathetic stimulation can markedly inhibit or even temporarily arrest GI motility.

[24/09/2026 7:50 AM] Imaanaa Harmee:

📁 Gastrointestinal Reflexes

▽ **GI reflexes** can be divided into **three** major groups.

1️⃣ Reflexes entirely within the ENS

These control:

- ⌚ Local secretion
- ⌚ Peristalsis
- ⌚ Mixing contractions
- ⌚ Local inhibition

2️⃣ Gut → Prevertebral Ganglia → Gut

Examples:

📄 Gastrocolic Reflex

△ Stomach activity stimulates colonic activity and can promote defecation.

📄 Enterogastric Reflex

△ Signals from the small intestine inhibit gastric motility and gastric emptying.

📄 Colonoileal Reflex

Signals from the colon inhibit ileal emptying into the cecum.

3️⃣ Gut → Spinal Cord/Brainstem → Gut

These include **long reflexes** involved in:

Defecation

Pain

More complex GI responses

[24/09/2026 7:50 AM] Imaanaa Harmee:

🔑 Hormonal Control of GI Functions

🔑 GI hormones are secreted mainly by **enteroendocrine cells**.

★ **CCK**

📌 Secreted mainly **from the duodenum and jejunum.**

✚ Stimulated especially by:

△ **Fat + protein products**

Actions:

✓ **Contracts gallbladder**

✓ **Promotes bile delivery**

✓ **Stimulates pancreatic digestive enzyme secretion**

✓ **Slows gastric emptying**

★ **Secretin**

△ Released from the duodenum in response to:
Acidic chyme

Main action:

△ ↑ Pancreatic HCO_3^- secretion
This helps **neutralize gastric acid entering the duodenum.**

★ **GIP**

△ Released from the **upper small intestine.**

✚ Stimulated by:

Carbohydrates

Fat

Effects:

📌 Decreases gastric motor activity.

☹ **Slows gastric emptying**

☑ **Enhances insulin secretion**

★ **Gastrin**

△ Produced mainly by **G cells of the stomach.**

Stimulated by:

▽ **Peptides**

☞ Gastric distension

Major actions:

↑ HCl secretion

↑ Gastric mucosal growth

[24/09/2026 7:50 AM] Imaanaa Harmee:

5. Splanchnic Circulation & GI Blood Supply

Splanchnic Circulation

*The splanchnic circulation includes blood flow through:
GI tract + spleen + pancreas + liver*

● Hepatic Portal System

● Venous blood from the:
Gut + spleen + pancreas
enters the:

Hepatic Portal Vein

↓

Liver Sinusoids

↓

Kupffer cells remove bacteria, debris and foreign particles

↓

Hepatic Veins

↓

Inferior Vena Cava

◇ This arrangement allows substances absorbed from the digestive tract to reach the liver before entering systemic circulation.

Arterial Supply of the GI Tract

● Celiac Artery

△ Supplies mainly the:

☑ Foregut organs, including the stomach.

● Superior Mesenteric Artery

△ Supplies much of the:

☑ Small intestine + proximal large intestine

🔴 Inferior Mesenteric Artery

△ Supplies much of the:

☑ Distal large intestine

Arterial branches penetrate the gut wall and supply the:

Submucosa

Muscular layers

Mucosa

Intestinal villi

💧 Regulation of GI Blood Flow

🧠 Mealtime Hyperemia

✂ Blood flow to the GI tract can increase substantially during active digestion.

▽ Major vasodilator signals include:

↓ O₂

↑ Metabolic waste

↑ Tissue metabolic activity

GI hormones and mediators also contribute:

CCK

VIP

Gastrin

Secretin

Bradykinin

Nitric oxide

⚡ Autonomic Regulation

Parasympathetic

Generally:

↑ GI blood flow

🔴 Sympathetic

Generally:

Vasoconstriction → ↓ GI blood flow

[24/09/2026 7:50 AM] Imaanaa Harmee:

🧠 ENDOCRINE SYSTEM

▽ Homeostasis

▽ Hormones

▽ Receptors

- ▽ Feedback •
- ▽ Regulation

1 HOMEOSTASIS & GENERAL PRINCIPLES OF ENDOCRINE CONTROL

Homeostasis

 Homeostasis is the maintenance of a relatively stable biochemical and physical internal environment despite continuous internal and external changes.

The basic concept

Environmental change



Body detects the change



Regulatory mechanisms respond



Internal conditions are restored toward normal



★ **HOMEOSTASIS**

⊖ When compensation fails:

Loss of homeostasis → Pathophysiology → Illness → Potential death

Hormones — The Chemical Messengers

△ **The endocrine system maintains homeostasis primarily through hormones.**

★ Hormones

☒ Hormones are chemical messengers released by endocrine cells into the:

Blood / extracellular fluid

They can therefore reach many tissues throughout the body.

But: ⚠ **Only target cells respond**

△ A hormone can circulate throughout the body, but it affects a cell only when that cell possesses the appropriate receptor.

 Remember

Hormone + Specific receptor = Cellular response

[24/09/2026 7:50 AM] Imaanaa Harmee: 🌀 **TARGET CELL CONCEPT**

★ **Important**

△ The same hormone may produce different physiological effects in different tissues because target cells can differ in:

**Receptor type +
intracellular signaling machinery**

🔗 INTERCELLULAR COMMUNICATION

◇ Cells communicate with one another to:

- ◆ Coordinate body functions
- ◆ Respond to environmental changes
- ◆ Maintain homeostasis
- ◆ Regulate growth and metabolism

■ Major signaling patterns include:

- △ Endocrine → **Distant target through blood**
- △ Paracrine → Nearby cells
- △ Autocrine → Same cell that released the signal

2️⃣ ENDOCRINE SYSTEM vs NERVOUS SYSTEM

✅ Both systems coordinate body functions and contribute to homeostasis.

⚡ Nervous System

△ Uses:

Action potentials + neurotransmitters

△ Therefore, responses are generally:

FAST + PRECISE + SHORT-LIVED

🩺 Endocrine System

△ Uses:

Hormones transported through blood

△ Therefore, responses are generally:

SLOWER + BROADER + LONGER-LASTING

[24/09/2026 7:50 AM] Imaanaa Harmee:

3️⃣ MAJOR FUNCTIONS OF HORMONES

△ Hormones regulate four major physiological areas.

♥ 1. REPRODUCTION

Hormones regulate:

- ◆ Gametogenesis
- ◆ Sexual development
- ◆ Sexual desire
- ◆ Coitus-related functions
- ◆ Fertilization
- ◆ Reproductive cycles

🌱 2. GROWTH & DEVELOPMENT

△ Hormones coordinate:

Growth rate + tissue maturation + development

▽ Important endocrine regulators include:

GH → IGF-1 → Growth

▪Thyroid hormones → Normal development

▪Sex hormones → Pubertal development

⚡ 3. ENERGY METABOLISM

△ Hormones control:

Production → Utilization → Storage of energy

Examples include regulation of:

- ◆ Glucose
- ◆ Lipids
- ◆ Proteins
- ◆ Basal metabolic activity

🏠 4. INTERNAL HOMEOSTASIS

Hormones help maintain:

💧 Water & Electrolytes

✓ Maintenance of body-fluid composition.

♥ Arterial Blood Pressure

Regulation of vascular tone and body-fluid volume.

🔑 Body Temperature
Control of metabolic heat production and dissipation.

🔑 Structural Mass
Regulation of:
Bone + Muscle + Adipose tissue

🧠 Emotional & Behavioral States
Hormones can influence mood, behavior and physiological responses to stress.

4 CLASSIFICATION OF HORMONES

Hormones can be classified according to:

- A. Chemical nature
- and
- B. Solubility & receptor location

🧬 A. CLASSIFICATION BY CHEMICAL NATURE

1 AMINES

Derived primarily from:

Tyrosine

Two important groups:

💧 Catecholamines

Tyrosine → Epinephrine + Norepinephrine + Dopamine

🌿 Thyroid Hormones

Tyrosine + Iodine → T₃ + T₄

2 PEPTIDES / POLYPEPTIDES / PROTEINS

Made from:

Amino acids

Examples:

Insulin • GH • ADH • Oxytocin • ACTH

3 STEROID HORMONES

Derived from:

★ Cholesterol

They possess a characteristic steroid nucleus.

Examples include:

◆ Adrenocorticoids

◆ Sex hormones

◆ Active vitamin D metabolites

4 EICOSANOIDS

Derived from:

Polyunsaturated fatty acids

Important groups include:

- ◆ Prostaglandins
- ◆ Prostacyclins
- ◆ Leukotrienes
- ◆ Thromboxanes

💧 📖 B. CLASSIFICATION BY SOLUBILITY

The most important distinction:

💧 WATER-SOLUBLE HORMONES

Hydrophilic

Examples:

- ◆ Peptide/protein hormones
- ◆ Catecholamines

They generally cannot cross the lipid bilayer.

Therefore:

Hormone → Cell-surface receptor → Second messenger → Cellular response

📖 LIPID-SOLUBLE HORMONES

Lipophilic

Examples:

- ◆ Steroid hormones
- ◆ Thyroid hormones
- ◆ Vitamin D₃ metabolites

They can cross the plasma membrane.

Therefore:

Hormone → Intracellular receptor → Gene regulation → New proteins

[24/09/2026 7:50 AM] Imaanaa Harmee:

📖 LIFE CYCLE OF HORMONES

△ Think of every hormone as having a journey:

🔗 **SYNTHESIS → STORAGE → RELEASE → TRANSPORT → ACTION → METABOLISM → Excretion**

1. SYNTHESIS

△ Protein/peptide hormones

Synthesized through the:

Rough ER → Golgi apparatus

△ Steroid hormones

Synthesized mainly through:

Smooth ER + mitochondria
from cholesterol.

2. STORAGE

△ Peptide hormones

Stored in:
Secretory vesicles
until stimulation triggers release.

△ Steroid hormones

Generally not stored in large secretory vesicles.

They are synthesized when needed from cholesterol precursors.

3. RELEASE

△ Peptide hormones are released by:

Exocytosis

△ Steroid hormones generally diffuse out of the cell after synthesis.

4. TRANSPORT

Hormones circulate as:

Free hormone

or

● Protein-bound hormone

Binding proteins include:

Albumin + specific globulins

★ Why is protein binding important?

It:

◆ **Prolongs hormone half-life**

◆ Protects hormone from rapid clearance

◆ Provides a circulating reservoir

5. METABOLISM & EXCRETION

△ Hormones may be metabolized by:

☑ Liver + target tissues + other tissues

They are eliminated through:

◆ Urine

- ◆ Feces
- ◆ Other routes such as sweat

6 MECHANISMS OF HORMONE ACTION

△ Hormones produce cellular responses through three broad mechanisms.

⚡ 1. DIRECT MEMBRANE EFFECT

Hormone-receptor interaction can modify membrane function by:

- ◆ Opening ion channels
- ◆ Closing ion channels
- ◆ Changing carrier proteins
- ◆ Activating membrane pumps
- ◆ Inactivating membrane pumps

Result:

Rapid change in cellular function

📖 2. SECOND-MESSENGER SYSTEMS

💧 Commonly used by water-soluble hormones.

✨ Important second-messenger pathways:

cAMP

cGMP

IP₃ / DAG

Ca²⁺–Calmodulin

Tyrosine kinase pathways

📡 Signal Amplification

⚡ A small amount of hormone can produce a large cellular response.

[24/09/2026 7:50 AM] Imaanaa Harmee:

Male Reproductive system Physiology

Dr. Doc

[24/09/2026 7:50 AM] Imaanaa Harmee:

🧬 1. General Classification and Basics of the Reproductive System

🧬 The reproductive system is **divided into the male reproductive system and the female reproductive system.**

🌀 Its primary functions are:

- ◇ Perpetuation of the species through reproduction.
- ◇ Providing sexual pleasure.
- ◇ Gametogenesis (formation of haploid cells) and endocrine function (synthesis of steroid hormones) carried out by the gonads (testes in males, ovaries in females).

♀ *Comparison of Male vs. Female Gamete Development*

▽ Gametogenesis occurs through Meiosis I and II, converting a diploid ($2n$) stem cells into four haploid (n) gametes.

▽ However, male and female gamete development differ significantly:
[24/09/2026 7:50 AM] Imaanaa Harmee:

2. Gametogenesis and Spermatogenesis Overview

 *Male germ cells are formed during embryonic development but remain quiescent until puberty.*

 At puberty, hormonal signals from the hypothalamus and pituitary gland activate the seminiferous tubules, Sertoli cells, and Leydig cells.

 **At puberty, spermatogonia follow two paths:**

- ① Mitosis: Spermatogonia continuously undergo mitotic division **to retain a self-renewing pool of stem cells.**
- ② Meiosis: Spermatogonia enter meiotic division to produce functional gametes.

Three Main Stages of Sperm Production

▽ Sperm production proceeds through three distinct stages:

 **Spermatocytogenesis:**

☞ The initial stage where primary spermatogonia develop into secondary spermatocytes.

 **Spermatidogenesis:**

☞ The stage encompassing Meiosis I and Meiosis II, resulting in the formation of haploid spermatids.

 **Spermiogenesis:**

☞ The final maturation stage where round spermatids differentiate into motile spermatozoa during spermiation.

Spermiation & Early Transport

▽ Newly formed spermatozoa are initially immotile. Their flagella must develop motility, which is **held in check and not utilized until ejaculation**. Prior to ejaculation, movement of sperm through the male reproductive tract is achieved via peristalsis of the genital ducts.

3. Gender Determination and Embryonic Differentiation

Chromosomal Basis of Sex

 Human somatic cells contain 23 pairs of chromosomes (22 pairs of autosomes and 1 pair of sex chromosomes: XX for female, XY for male).

☺ **Syngamy (gamete fusion)** creates a diploid zygote through plasmogamy (fusion of cell membranes and cytosol) followed by karyogamy (fusion of nuclear genetic material).

The SRY Gene and Gonadal Differentiation

 Early in development, the human embryo exhibits gender bipotentiality.

 Around the 7th week of fetal development, the SRY (Sex-determining Region of the Y chromosome) gene becomes activated:

♂ Presence of SRY:

▽ Directs the bipotential gonads to descend and differentiate into testes, which subsequently secrete male-biased hormones.

♀ Absence of SRY:

▽ In the absence of the SRY gene on the X chromosome, the bipotential gonads naturally develop into ovaries, which then produce female-biased hormones.

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5. Differentiation of Internal and External Genitalia

 By the 7th week of embryonic development, the fetus possesses dual primitive genital tracts: the **Müllerian duct (female precursor)** and the **Wolffian duct (male precursor)**.

♂ Hormonal Control by Testosterone

① Internal Male Accessory Structures:

▽ Testosterone stimulates the development of the Wolffian duct into the epididymis, vas deferens, seminal vesicles, and prostate gland.

② External Male Genitalia:

▽ Testosterone induces the closure of the urogenital slit and formation of the penis and scrotum.

③ Testicular Descent:

▽ Testosterone is required for the embryonic descent of the testes into the scrotum.

④ Female Pathway (Default):

▽ In the absence of functioning fetal testes, the Wolffian duct regresses, the Müllerian duct differentiates into the fallopian tubes, uterus, and upper 2/3 of the vagina, and the urogenital slit remains open to form female external genitalia.

♂ 6. Functional Structures of the Male Reproductive System

♂ The male reproductive system consists of the primary gonads, genital tracts, accessory glands, and supporting structures.

I. Essential Structures:

The Gonads (Testes)

Functions:

① Spermatogenesis:

▽ Formation of functional spermatozoa.

② Endocrine Function:

▽ Production of testosterone by interstitial Leydig cells, inhibin, and minimal amounts of estrogen.

Blood Supply:

⊗ Arterial: Abdominal aorta gonadal/testicular arteries erectile tissues.

⊗ Venous: Testicular veins Inferior Vena Cava (IVC).

🧠 Innervation: **Pudendal nerve originating from the sacral plexus.**

II. Tubular Structures (Genital Tracts)

 The primary function of the genital tract is the transport, maturation, and storage of sperm:

① Epididymis:

▽ A heavily coiled duct (>6 meters in length) serving as the principal area for storage and functional maturation of spermatozoa.

② Vas Deferens:

▽ A muscular tube connecting the epididymis to the ejaculatory duct.

③ Ejaculatory Duct:

▽ Connects the vas deferens with the prostatic segment of the urethra.

④ Urethra:

▽ The terminal distal genital channel that allows passage and expulsion of semen out of the penis.

[24/09/2026 7:50 AM] Imaanaa Harmee:

♂ **6. Functional Structures of the Male Reproductive System**

♂ The male reproductive system consists of the primary gonads, genital tracts, accessory glands, and supporting structures.

🧬 **I. Essential Structures:**

★ **The Gonads (Testes)**

⚙️ Functions:

① Spermatogenesis:

▽ Formation of functional spermatozoa.

② Endocrine Function:

▽ Production of testosterone by interstitial Leydig cells, inhibin, and minimal amounts of estrogen.

Blood Supply:

🌀 Arterial:

★ Abdominal aorta ➞ gonadal/testicular arteries ➞ erectile tissues.

🌀 Venous:

➞ Testicular veins Inferior Vena Cava (IVC).

 Innervation:

★ **Pudendal nerve** originating from the sacral plexus.

II. Tubular Structures (Genital Tracts)

 The primary function of the genital tract is the transport, maturation, and storage of sperm:

① Epididymis:

▽ A heavily coiled duct (>6 meters in length) serving as the principal area for storage and functional maturation of spermatozoa.

② Vas Deferens:

▽ A muscular tube connecting the epididymis to the ejaculatory duct.

③ Ejaculatory Duct:

▽ Connects the vas deferens with the prostatic segment of the urethra.

④ Urethra:

▽ The terminal distal genital channel that allows passage and expulsion of semen out of the penis.

💧 III. Accessory Sex Glands (Seminal Plasma Production)

💧 The accessory glands secrete seminal plasma, which acts as a fluid medium, nutrient source, and protective buffer for sperm.

[24/09/2026 7:50 AM] Imaanaa Harmee: 💧 **Seminal Vesicles:**

▽ Produce 60% of seminal plasma volume. Secretes a thick, mucoid fluid rich in fructose (energy source), ascorbic acid, flavin, and prostaglandins (PGs).

💧 **Prostate Gland:**

▽ Produces 30% of seminal plasma volume. Secretes a thin, alkaline fluid that neutralizes the acidic environment of the male urethra and female vagina.

💧 **Bulbourethral (Cowper's)**

▽ Glands: Secretes an alkaline fluid similar to prostatic secretions to lubricate and neutralize the urethra prior to ejaculation.

IV. Supporting Structures

 Penis:

▽ Houses the urethra and contains three columns of erectile tissue—two corpora cavernosa and one corpus spongiosum—which facilitate erection.

 Scrotum:

▽ An external cutaneous sac that supports and regulates the temperature of the testes and epididymides.

Spermatic Cord:

▽ A protective vascular and neural bundle containing testicular blood vessels, autonomic nerves, and lymphatic channels.

V. Gross Anatomy of the Testis

 External Capsule:

▽ The testes are paired ovoid organs located in the scrotum, enclosed by a dense fibrous connective tissue capsule named the tunica albuginea testis.

 Testicular Parenchyma:

▽ Divided into 250 to 350 lobules separated by connective tissue septa (Septula testis) that extend inward from the mediastinum testis.

 Seminiferous Tubules:

▽ Each lobule contains one or more tightly coiled seminiferous tubules that loop and empty into the rete testis within the mediastinum, which then drains into the efferent ductules of the epididymis.

[24/09/2026 7:55 AM] Imaanaa Harmee:

FEMALE REPRODUCTIVE SYSTEM

1. Structural Organization of the Female Reproductive System

▽ The female reproductive system is divided into essential structures (gonads) and

accessory sex organs:

★ Essential Structures (Gonads)

🌀 Two Ovaries:

🌀 Responsible for oogenesis (gametogenic role) and endocrine function (secretion of Estrogen and Progesterone).

★ Accessory Sex Organs

🌀 Oviducts (Fallopian / Uterine Tubes)

🌀 Uterus

🌀 Vagina

🌀 External Genitalia

2. Anatomy & Cellular Organization of the Ovaries

▽ Anatomical Attachments

The ovaries are attached to the peritoneum via the mesovarian and broad ligaments.

▽ Functional Roles

Gametogenic potential:

★ Established early during fetal development.

🌀 Endocrine role: Not realized until puberty.

★ Structural Regions

1. Medullary Part:

▽ The central region containing blood vessels and connective tissues (elastin and collagen).

2. Cortical Part:

▽ The peripheral region containing actively proliferating ovarian follicles.

★ Cellular Differentiation

🌀 Mesenchymal Tissue:

🌀 Differentiates into interstitial tissue (hormone-secreting cells), serving as the primary source of ovarian hormones, associating with germinal elements, and providing a nutritive environment for oocytes.

★ Epithelial Tissue:

☞ Differentiates into granulosa cells.

3. Physiology of Accessory Sex Organs

- ✓ Oviduct / Fallopian Tube / ✓ Uterine Tube
- ✓ Serves as the normal site of fertilization.
- ✓ Transports ovum and sperm.
- ✓ Performs secretory functions.

★ Uterus

○ A hollow muscular organ in non-pregnant females consisting of three wall layers:

☒ Endometrium (with uterine glands),

☒ Myometrium, and

☒ Perimetrium.

☑ Functions as the site of pregnancy, supports the growth and development of the fetus, generates pressure during labor, and serves as the site of menstruation.

★ Vagina

☞ Lined with non-keratinized stratified epithelium.

☞ Vaginal glands secrete glycogen and lactate.

Harbors Lactobacillus acidophilus, which maintains an acidic pH within the vaginal wall.

☞ Functions as the female copulatory organ, the birth canal, and the passage for menstrual flow.

★ External Genitalia

▽ Includes the clitoris, labia minora, and labia majora.

[24/09/2026 7:55 AM] Imaanaa Harmee:

4. Oogenesis (Development of Ova)

▽ Oogenesis progresses through three major follicular stages:

★ Primordial follicles

★ Growing follicles (1° follicles $\rightarrow$ 2° follicles $\rightarrow$ 3° follicles)

★ Mature (Graafian) follicle $\rightarrow$ Ovum

▽ Oogenesis Sequence

Oogonia $\xrightarrow{\text{(Prenatal Mitosis)}}$ 1° Oocyte $\xrightarrow{\text{(1st Meiosis)}}$ 2° Oocyte $\xrightarrow{\text{(2nd Meiosis)}}$ Ovum

★ Prenatal & Follicular Population Dynamics

▽ Primitive ova (Oogonia) undergo active mitosis during prenatal development.

▽ Females have approximately 1 million primordial follicles at birth.
Primary Oocytes persist from birth to puberty, arrested in the Prophase of 1st meiosis ($2n$).

▽ By puberty, the count declines to approximately 400,000 follicles.

Meiotic Divisions & Ovulation

▽ Completion of the 1st meiotic division yields a Secondary Oocyte and a First Polar Body (which undergoes attrition).

▽ At ovulation, the secondary oocyte is arrested in Metaphase of meiosis II ($1n$).

▽ Completion of the second meiotic division yields the mature Ovum and a Second Polar Body (attrition).

5. Overview of the Female Sexual Cycle

✓ Also referred to as the menstrual cycle, involving cyclic structural and functional changes in female reproductive organs.

▽ Restricted exclusively to humans and large/great apes.

▽ Initiates at puberty and ends at menopause (the complete cessation of menstruation).

Menarche: The onset of the first menstrual flow, occurring at an average age of 12 years (range: 9–16 years).

🕒 Comprises two interconnected sub-cycles:

1. Ovarian Cycle:

📅 Follicular, Ovulatory, and Luteal phases.

2. Endometrial Cycle:

📖 Menstrual (bleeding), Proliferative, and Secretory phases.

6. The Ovarian Cycle

📖 Consists of three sequential phases regulated by pituitary gonadotropins (FSH and LH):

A. Follicular Phase (Days 1 to 13)

🔗 Driven by FSH and LH from the anterior pituitary gland.

🔗 Multiple primordial follicles enlarge at the cycle's onset; primary follicles produce fluid termed liquor folliculi, which accumulates in the antrum (rich in estrogen).

🔗 By Day 6, a single dominant mature follicle—the Graafian follicle—is established.

B. Ovulation Phase (Days 14 to 15)

🔗 The wall of the Graafian follicle ruptures, releasing the ovum, zona pellucida, and one surrounding layer of the corona radiata into the peritoneal cavity.

🔗 LH is the primary hormone of ovulation: 36 hours prior to ovulation, the anterior pituitary executes a massive LH surge, with peak LH levels reached 12 hours before ovulation.

★ Cascade

↑LH → ↑antral fluid volume → ↑progesterone secretion → ↑blood flow (BF) to Graafian follicles → ↑follicular swelling → ↑intrafollicular pressure → follicular rupture and ovum release

C. Luteal Phase (Days 15 to 25)

▽ Following rupture, the remaining Graafian follicle structure collapses.

🔗 Under LH stimulation, granulosa cells undergo hypertrophy and proliferation to transform into the corpus luteum.

🌀 **Function**

🔗 Secretes massive amounts of Progesterone alongside Estrogen, maintaining conditions necessary for pregnancy.

★ **Four Sequential Stages of the Corpus Luteum**

1. Stage of Vascularization (Corpus Hemorrhagicum)

☞ Vascularization of granulosa cells occurs.

2. Stage of Glandular Metamorphosis

☞ Follicular granulosa and theca cells transform into luteal cells capable of concentrating cholesterol as temporary glandular units.

3. Stage of Full Development

☞ Characterized by maximal output of progesterone and estrogen.

✦ If fertilization occurs: Human Chorionic Gonadotropin (hCG) preserves the corpus luteum, keeping it active for more than 10 days.

✦ If no fertilization occurs: The corpus luteum remains active for 10 days, after which it degenerates.

4. Stage of Involution

▽ Luteal cells are replaced by fibrous connective tissue, forming the corpus albicans.

[24/09/2026 7:55 AM] Imaanaa Harmee:

7. The Endometrial Cycle

▽ The endometrial wall undergoes three distinct phases driven directly by ovarian steroid hormones:

Bleeding / Menstrual Phase
Timeframe: 3 to 5 days

★ Hormonal Mechanism & Structural Features:

☞ Triggered by sudden withdrawal of sex hormones (estrogen & progesterone).

☞ Causes desquamation of the functional layer of the endometrium.

☞ Total blood loss averages 50 mL (normal range: 30–80 mL).

🌱 Proliferative Phase

Timeframe: Pre-ovulatory (mid-cycle)

★ Hormonal Mechanism & Structural Features:

☞ Driven primarily by Estrogen.

Causes growth and proliferation of endometrial stromal cells and glands.

☞ Promotes progressive elongation and growth of spiral arteries.

🕒 Secretory (Progestational) Phase

Timeframe: Post-ovulatory (luteal phase)

★ Hormonal Mechanism & Structural Features:

🕒 Driven chiefly by Progesterone (assisted by Estrogen).

🕒 Endometrial glands become tortuous and actively secretory.

🕒 Spiral arteries become increasingly coiled/spiral.

Total endometrial thickness reaches 6 mm.

[24/09/2026 7:55 AM] Imaanaa Harmee:

ANS

Dr.Doc

[24/09/2026 7:55 AM] Imaanaa Harmee:

Nerve Physiology

Dr. Doc

[24/09/2026 7:55 AM] Imaanaa Harmee: 🕒 **1. Excitable Tissue Fundamentals**

⚡ A. Definition of Excitability

▽ **Excitability** is the ability of a cell or tissue to detect a stimulus and respond by producing a change in its **membrane potential**.

▽ A stimulus can be **electrical, mechanical, chemical, or thermal**, depending on the type of cell and receptor involved.

▽ Excitable tissues can respond rapidly because changes in ion movement across the cell membrane quickly alter the electrical charge difference between the inside and outside of the cell.

▽ The membrane potential depends mainly on the unequal distribution of ions across the membrane and the membrane's selective permeability to those ions.

✓ **Excitability = the ability to respond to an appropriate stimulus by changing membrane potential.**

⚡ B. Excitable Tissues in the Human Body

▽ The two major excitable tissues are **nervous tissue and muscle tissue**.

✦ 1. Nervous Tissue

- ▽ Neurons are specialized to receive, process, and transmit electrical signals.
- ▽ When sufficiently stimulated, a neuron can generate an **action potential**, allowing information to travel along its membrane.
- ▽ These electrical signals are essential for sensation, communication between neurons, and control of muscles and glands.

✦ 2. Muscle Tissue

- ▽ Muscle cells are also excitable because their membrane potential can change in response to stimulation.
- ▽ Electrical excitation of muscle cells can trigger the processes that ultimately produce **muscle contraction**.
- ▽ Skeletal, cardiac, and smooth muscle have different patterns of electrical activity, but all depend on changes in membrane potential.

✓ **Nerve tissue uses excitability mainly for rapid communication, while muscle tissue uses excitability to initiate and regulate contraction.**

✦ C. Resting Membrane Potential (RMP)

- ▽ At rest, the inside of an excitable cell is normally **electrically negative relative to the outside**.
- ▽ This voltage difference across the cell membrane is called the **resting membrane potential (RMP)**.
- ▽ The RMP exists because ions are distributed unequally across the membrane and because the membrane has different permeabilities to different ions.
- ▽ In many neurons, the resting membrane potential is approximately **-70 mV**, although the exact value varies among cells.
- ▽ **Potassium (K^+)** plays an important role in establishing the RMP because resting membranes are generally more permeable to K^+ than to Na^+ .

✓ **The RMP provides the baseline electrical state from which an excitable cell can respond to stimulation.**

✦ D. From Stimulus to Action Potential

- ▽ When an excitable cell receives an adequate stimulus, ion channels in its membrane may open or close.
- ▽ This changes the movement of ions such as **Na⁺ and K⁺** across the membrane.
- ▽ The resulting change in membrane voltage may produce a **graded potential**, whose size depends on the stimulus and membrane response.
- ▽ If the membrane reaches the appropriate **threshold**, a rapid, regenerative electrical event called an **action potential** can occur.

[24/09/2026 7:55 AM] Imaanaa Harmee: ✦ E. Action Potential

- ▽ An **action potential** is a rapid, temporary change in membrane potential that allows electrical signals to travel along an excitable cell membrane.
- ▽ In a typical neuron, voltage-gated **Na⁺ channels** open during the early phase, allowing Na⁺ to enter the cell and causing **depolarization**.
- ▽ Na⁺ channels then become inactivated while voltage-gated **K⁺ channels** allow K⁺ to leave the cell.
- ▽ K⁺ leaving the cell produces **repolarization** and may briefly make the membrane more negative than its resting level, producing **hyperpolarization**.
- ▽ The membrane then returns toward its resting state as the voltage-gated channels reset and normal ion gradients are maintained.

✓ **Action potential = a rapid, all-or-none electrical signal triggered when an excitable membrane reaches threshold.**

✦ F. Overall Physiological Sequence

- ▽ **Stimulus** → changes ion-channel activity.
 - ▽ **Membrane potential changes** → depolarization or hyperpolarization occurs.
 - ▽ **Threshold reached** → an action potential is triggered.
 - ▽ **Action potential** → a rapid electrical signal travels along the excitable membrane.
 - ▽ **Return toward resting state** → the membrane potential is restored.
- ✓ **Key idea: Excitable tissues can generate rapid electrical responses because they**

maintain a resting membrane potential and possess ion channels that rapidly change membrane permeability when stimulated.

[24/09/2026 7:55 AM] Imaanaa Harmee: ☞ **2. Cellular Organization of the Nervous System**

▽ The nervous system is made up of **two major classes of cells: neurons and glial cells (glia)**.

▽ **Neurons** are specialized for receiving, generating, and transmitting electrical signals, while **glial cells** provide structural, metabolic, protective, and regulatory support for neurons.

✦ **A. Glial Cells (Support Cells)**

▽ Glial cells, or **neuroglia**, are the supporting cells of nervous tissue. They are very numerous and, in many CNS regions, may outnumber neurons.

▽ The term *glia* originally referred to the idea of a cellular “**glue**” that holds nervous tissue together. Unlike mature neurons, many glial cells retain the ability to **divide and multiply after birth**.

▽ Glial cells surround neuronal **cell bodies, axons, and dendrites** and create an environment that allows neurons to function properly.

▽ Although glial cells do not normally generate and conduct action potentials like neurons do, they are essential for **nutrition, protection, insulation, waste removal, and maintenance of the neuronal environment**.

✓ **Glia support neurons rather than serving as the primary signaling cells.**

✦ **1. Astrocytes — CNS**

▽ Astrocytes are star-shaped glial cells located throughout the **central nervous system (CNS)**.

▽ They help maintain the chemical environment around neurons and contribute to the movement of nutrients and other substances between blood vessels and nervous tissue.

▽ Astrocytes also contribute to the **blood-brain barrier (BBB)**, which helps regulate what substances can move from the bloodstream into brain tissue.

▽ The BBB protects neural tissue from many potentially harmful substances while still allowing essential substances to reach the brain.

✓ **Astrocytes = support, regulation, and contribution to the blood-brain barrier.**

✦ 2. Microglia — CNS

- ▽ Microglia are the main **immune-defense cells of the CNS**.
- ▽ They act similarly to macrophages by detecting, engulfing, and removing cellular debris, damaged cells, and certain invading microorganisms.
- ▽ They become particularly important when nervous tissue is injured or exposed to infection or inflammation.

✓ **Microglia = immune defense and cleanup in the CNS.**

✦ 3. Ependymal Cells — CNS

- ▽ Ependymal cells line the **ventricles of the brain** and parts of the central canal of the spinal cord.
- ▽ They are associated with the production, movement, and regulation of **cerebrospinal fluid (CSF)**, which surrounds and protects the brain and spinal cord.
- ▽ Their cilia can help move CSF through the ventricular system.

✓ **Ependymal cells = lining of CSF spaces and involvement in CSF movement.**

✦ 4. Oligodendrocytes — CNS

- ▽ Oligodendrocytes produce the **myelin sheath** around axons in the CNS.
- ▽ Myelin is a lipid-rich insulating material that electrically isolates portions of the axon.
- ▽ This insulation allows electrical impulses to travel much more rapidly along myelinated axons.

✓ **Oligodendrocytes = myelin formation in the CNS.**

✦ 5. Schwann Cells — PNS

- ▽ Schwann cells are glial cells of the **peripheral nervous system (PNS)**.
- ▽ They form the **myelin sheath around peripheral axons**.
- ▽ One Schwann cell typically myelinates one segment of a single axon, with gaps between myelinated segments called **nodes of Ranvier**.

▽ These gaps help allow rapid propagation of action potentials along myelinated axons.

✓ **Schwann cells = myelin formation in the PNS.**

[24/09/2026 7:55 AM] Imaanaa Harmee: ✦ 6. Satellite Cells — PNS

▽ Satellite cells are found mainly around the **cell bodies of neurons in peripheral ganglia.**

▽ They provide structural and metabolic support and help regulate the local chemical environment surrounding neuronal cell bodies.

✓ **Satellite cells = support and regulation around neuron cell bodies in PNS ganglia.**

✦ CNS vs PNS Glial Cells

▽ **CNS glia:** Astrocytes, microglia, ependymal cells, and oligodendrocytes.

▽ **PNS glia:** Schwann cells and satellite cells.

✓ **Easy distinction: Oligodendrocytes myelinate the CNS; Schwann cells myelinate the PNS.**

✦ B. Nerve Cells (Neurons)

▽ **Neurons** are the primary signaling cells of the nervous system.

▽ They are specialized to **receive information, integrate signals, generate action potentials, and communicate with other cells.**

▽ Neurons communicate with other neurons as well as with muscles and glands.

▽ A typical neuron can be organized into four major functional regions: **cell body, dendrites, axon, and presynaptic terminals.**

✦ 1. Cell Body (Soma)

▽ The cell body, or **soma**, is the main metabolic and maintenance center of the neuron.

▽ It contains the **nucleus** and most of the organelles required for protein synthesis, energy production, and cellular maintenance.

▽ The soma also integrates information received from different parts of the neuron.

✓ **Cell body = metabolic and maintenance center.**

✦ 2. Dendrites

- ▽ Dendrites are branching extensions that usually extend from the cell body.
- ▽ They are specialized mainly for **receiving signals** from other neurons or sensory receptors.
- ▽ Signals received by dendrites produce electrical changes that can be transmitted toward the cell body and axon.

✓ **Dendrites = major receptive region of the neuron.**

✦ 3. Axon

- ▽ The axon is a specialized neuronal process that conducts electrical signals **away from the cell body**.
- ▽ Its length can vary greatly, from very short processes to axons that extend considerable distances through the body.
- ▽ When the appropriate threshold is reached, an **action potential** can travel along the axon toward its terminals.

✓ **Axon = conducting region of the neuron.**

✦ 4. Presynaptic Terminals

- ▽ Presynaptic terminals are specialized endings at the end of an axon or its branches.
- ▽ They contain structures involved in releasing **neurotransmitters**, chemical messengers that communicate with the next cell.
- ▽ The junction where a neuron communicates with another cell is called a **synapse**.
- ▽ Neurotransmitter release allows the electrical signal arriving at the terminal to influence another neuron, muscle cell, or gland cell.

✓ **Presynaptic terminals = output region that communicates with the next cells.**

✦ Functional Organization of a Neuron

▽ In a simplified pathway, information generally moves through the neuron as:
Dendrites → Cell body → Axon → Presynaptic terminals → Next cell

▽ The dendrites primarily **receive** information, the cell body **maintains and integrates** neuronal activity, the axon **conducts** the action potential, and the presynaptic terminals **communicate the signal** to another cell.

✓ **The neuron is the primary signaling unit, while glial cells create and maintain the environment that allows neurons to function effectively.**

[24/09/2026 7:55 AM] Imaanaa Harmee: ☞ **3. Membrane Potential & Resting Membrane Potential (RMP)**

✦ **A. Membrane Potential Concepts**

▽ **Membrane potential** is the electrical voltage difference that exists across a cell membrane. It results from the separation of electrical charges between the **intracellular fluid (ICF)** and **extracellular fluid (ECF)**.

▽ This charge separation develops because ions such as **Na⁺, K⁺, Cl⁻, and negatively charged proteins** are not distributed equally between the ICF and ECF.

▽ The cell membrane is selectively permeable, meaning some ions can cross the membrane more easily than others. This selective movement of ions contributes to the development and maintenance of the membrane potential.

✦ **Polarized State**

▽ When the inside surface of a cell membrane is **more negative than the outside**, the cell is said to be **polarized**.

▽ At rest, neurons maintain a stable negative membrane potential called the **resting membrane potential (RMP)**.

▽ The RMP is often around **-70 mV in many neurons**, although the exact value varies among different excitable cells. Some physiological models use approximately **-90 mV**, particularly when emphasizing the strong influence of K⁺ permeability.

▽ A value of **-90 mV** means that the electrical potential inside the cell is approximately 90 mV more negative than the outside; it does **not** mean the inside is literally "90 times more negative."

✓ **Membrane potential = voltage difference across the cell membrane.**

✓ **Polarization = inside of the cell is electrically negative relative to the outside.**

✦ B. Factors Contributing to the Resting Membrane Potential

✦ 1. Differential Membrane Permeability

▽ The most important factor determining the resting membrane potential is the **different permeability of the membrane to different ions**.

▽ At rest, the membrane is much more permeable to **K⁺ than to Na⁺** because there are many more K⁺ leak channels available.

▽ K⁺ tends to move out of the cell down its concentration gradient. As positively charged K⁺ leaves, negative charge becomes relatively greater inside the cell.

▽ This creates an electrical force that eventually opposes further K⁺ movement, producing a stable electrical gradient.

▽ A small amount of **Na⁺ leaks into the cell**, making the RMP less negative than the equilibrium potential for K⁺ alone.

✓ **High resting K⁺ permeability is the major contributor to the negative RMP.**

✦ 2. Electrogenic Na⁺/K⁺ Pump

▽ The **Na⁺/K⁺-ATPase** is an active transport protein that uses ATP to maintain the Na⁺ and K⁺ concentration gradients.

▽ For each cycle, it transports:
3 Na⁺ out of the cell → 2 K⁺ into the cell

▽ Because more positive charge leaves than enters, the pump has a small **electrogenic effect**, making the inside of the cell more negative.

▽ More importantly, the pump continuously maintains the Na⁺ and K⁺ gradients that are necessary for the neuron to maintain its resting potential and generate action potentials.

✓ **Na⁺/K⁺ pump = maintains ion gradients and directly contributes a smaller electrogenic effect.**

[24/09/2026 7:55 AM] Imaanaa Harmee: ✦ 3. Intracellular Non-Diffusible Anions

▽ The cytoplasm contains large negatively charged molecules, especially **proteins and organic phosphates**, that cannot freely cross the cell membrane.

▽ These trapped negative charges contribute to the relatively negative electrical environment inside the cell.

▽ They also influence the distribution of diffusible ions across the membrane.

✓ **Non-diffusible intracellular proteins help maintain the negative charge inside the cell.**

✦ C. Quantitative Origin of the RMP

▽ The exact numerical value of the RMP depends on the ion concentrations, membrane permeability, and the activity of ion pumps.

▽ The **Nernst equation** can be used to calculate the equilibrium potential for an individual ion, such as K^+ .

▽ Because the resting membrane is much more permeable to K^+ than Na^+ , the K^+ equilibrium potential has a major influence on the overall RMP.

▽ The **Goldman-Hodgkin-Katz equation** considers the permeability and concentration gradients of multiple ions, providing a better estimate of the actual resting membrane potential.

✦ Approximate Quantitative Model

▽ In the model represented by your notes, the K^+ -dominated leak contribution is approximately **-86 mV**.

▽ Na^+ leakage shifts the predicted voltage in the positive direction by approximately **+8 mV** when considering the combined permeability of the major ions using the Goldman approach.

▽ The electrogenic action of the **Na^+/K^+ -ATPase** contributes approximately **-4 mV** in the simplified model.

▽ These values illustrate that the final RMP results from the **combined effects of ion permeability, ion concentration gradients, and active transport**, rather than from a single mechanism.

✓ **The K^+ leak pathway provides the dominant influence on the negative resting potential.**

✓ **The Na^+/K^+ pump maintains the concentration gradients and adds a smaller direct electrogenic effect.**

✓ **The exact RMP varies by cell type; approximately -70 mV is common for many neurons, while values near -90 mV can occur in cells with especially strong K^+ dominance.**

[24/09/2026 7:55 AM] Imaanaa Harmee: ☞ **4. Action Potentials, Graded Potentials & Propagation**

✦ A. Threshold & Stages of an Action Potential

▽ An **action potential (AP)** is a rapid, temporary change in the membrane potential of an excitable cell. It allows neurons to transmit electrical signals over long distances along their axons.

▽ An action potential begins when the membrane is sufficiently **depolarized to reach threshold**. In the model used in your notes, threshold is approximately **-65 mV**, requiring an initial depolarization of about **15–30 mV** from the resting level.

✦ 1. Resting Stage

▽ Before an action potential begins, the neuron is at its **resting membrane potential (RMP)**.

▽ The inside of the neuron is electrically negative compared with the outside, and voltage-gated Na^+ and K^+ channels are in their resting states.

✓ **Resting stage = membrane maintained at its normal negative RMP before stimulation.**

✦ 2. Depolarization — Rising Phase

▽ When a stimulus depolarizes the membrane to **threshold**, voltage-gated **Na^+ channels open rapidly**.

▽ Na^+ moves from the ECF into the neuron because of both its concentration gradient and electrical gradient.

▽ The influx of positive Na^+ causes the membrane to become progressively less negative and then positive.

▽ This produces a **positive-feedback cycle**: depolarization opens more Na^+ channels → more Na^+ enters → greater depolarization → still more Na^+ channels open.

▽ In large nerve fibers, the membrane potential can briefly rise above **0 mV**, producing an **overshoot**. In smaller fibers and many CNS neurons, the potential may approach 0 mV without a large overshoot.

✓ **Depolarization = rapid Na^+ entry makes the inside of the cell less negative and temporarily positive.**

✦ 3. Repolarization — Falling Phase

▽ Shortly after Na^+ channels open, they become **inactivated**, stopping the major inward Na^+ current.

▽ At the same time, voltage-gated **K^+ channels open more fully**.

▽ K^+ then moves out of the neuron into the ECF, carrying positive charge away from the cell.

▽ The membrane potential therefore moves back toward its negative resting value. This process is called **repolarization**.

✓ **Repolarization = Na^+ channel inactivation combined with K^+ efflux returns the membrane toward a negative potential.**

✦ 4. Hyperpolarization — Undershoot

▽ Voltage-gated K^+ channels do not close immediately. Because they remain open for a short time, additional K^+ leaves the cell.

▽ The membrane therefore becomes **more negative than its normal resting potential** for a brief period.

▽ This temporary phase is called the **undershoot** or **after-hyperpolarization**.

▽ As K^+ channels close and the normal resting ion distribution is maintained, the membrane returns to its resting state.

✓ **Hyperpolarization = brief membrane potential below the normal RMP.**

✦ B. Refractory Periods

▽ The **refractory period** is the period following the beginning of an action potential during which another action potential is either impossible or more difficult to generate.

✦ 1. Absolute Refractory Period (ARP)

▽ The absolute refractory period begins when voltage-gated Na^+ channels become activated and continues through the early part of repolarization.

▽ During this period, many Na^+ channels are **inactivated** and cannot immediately reopen.

▽ Therefore, another action potential **cannot be generated**, regardless of how strong the stimulus is.

▽ The ARP ensures that one action potential is completed before another can begin and helps limit the maximum firing frequency of the neuron.

✓ **ARP = no second action potential can occur, regardless of stimulus strength.**

✦ 2. Relative Refractory Period (RRP)

▽ The relative refractory period follows the absolute refractory period.

▽ During this period, some Na^+ channels have returned to a state in which they can participate in another action potential, but some K^+ channels are still open.

[24/09/2026 7:55 AM] Imaanaa Harmee: ▽ Because K^+ continues to leave the cell, the membrane may remain hyperpolarized.

▽ A **stronger-than-normal, or suprathreshold, stimulus** is therefore required to reach threshold again.

▽ An action potential generated during the RRP may have a **slower upstroke and reduced overshoot** compared with a normal action potential.

✓ **RRP = another action potential is possible, but it requires a stronger stimulus.**

✦ C. Key Properties of Action Potentials

✦ 1. Positive-Feedback Na^+ Channel Activation

▽ The rising phase of an action potential is driven by a self-reinforcing process.

▽ Initial depolarization opens some voltage-gated Na^+ channels.

▽ Na^+ enters the neuron and causes further depolarization.

▽ This additional depolarization opens even more Na^+ channels, producing still greater Na^+ entry.

▽ The process rapidly reaches a peak before Na^+ channel inactivation and K^+ channel activation dominate.

✓ **Na^+ entry during the rising phase creates positive feedback that rapidly amplifies depolarization.**

✦ 2. Threshold Dependency

▽ An action potential does not normally occur with every small membrane fluctuation.

▽ The membrane must reach a critical **threshold level** at which inward Na^+ current becomes sufficient to trigger the regenerative action potential.

▽ Once threshold is reached, the action potential proceeds through its characteristic sequence.

✓ **Below threshold → no full action potential.**

✓ **At threshold → regenerative action potential begins.**

✦ 3. All-or-None Law

▽ An action potential follows the **all-or-none principle**.

▽ If threshold is reached, a complete action potential is generated.

▽ If threshold is not reached, a full action potential does not occur.

▽ Increasing the intensity of a stimulus above threshold does **not** make an individual action potential larger. Stronger stimuli can instead increase the **frequency of action potentials**.

✓ **Action potential amplitude is essentially independent of stimulus intensity once threshold has been reached.**

✦ 4. Propagation

▽ Once generated, an action potential produces local electrical changes in adjacent sections of membrane.

▽ These local currents bring neighboring membrane to threshold, causing the action potential to regenerate along the axon.

▽ In an isolated axon, the disturbance can spread in both directions from the site where the action potential is initiated. In a normal neuron, however, physiological signaling is generally directed toward the axon terminals because the action potential normally begins near the axon initial segment and the refractory region behind the advancing impulse prevents backward re-excitation.

✓ **Propagation allows the action potential to travel along the axon without losing its amplitude.**

✦ D. Action Potential vs. Graded Potential

▽ **Graded potentials** are local changes in membrane potential whose size depends on the

strength of the stimulus.

▽ A stronger stimulus generally produces a larger graded potential, while a weaker stimulus produces a smaller one.

▽ Graded potentials can be **summated**, meaning several graded potentials can combine to produce a larger membrane change.

▽ They may be either **depolarizing or hyperpolarizing**.

▽ Graded potentials are **decremental**, meaning their amplitude decreases as they spread farther from their site of origin.

▽ They do not have the same absolute and relative refractory periods characteristic of action potentials.

▽ Graded potentials can result from sensory stimulation, neurotransmitter action, or spontaneous changes in membrane permeability.

▽ **Action potentials**, in contrast, are all-or-none events that occur after the membrane reaches threshold.

▽ Their amplitude remains relatively constant as they travel along the axon because the signal is **regenerated** at successive portions of the membrane.

▽ Action potentials cannot normally be summated in the same way as graded potentials because each AP has refractory periods.

[24/09/2026 7:55 AM] Imaanaa Harmee: ⚡ **Main Comparison**

▽ **Graded potential:** variable amplitude → can summate → decremental → local → may be depolarizing or hyperpolarizing.

▽ **Action potential:** all-or-none → cannot be summated as graded potentials are → non-decremental → travels along axon → follows a stereotyped sequence.

✓ **Graded potentials help determine whether threshold will be reached; action potentials transmit the signal over longer distances.**

⚡ E. Action Potential Conduction Modes

⚡ 1. Continuous Conduction

▽ **Continuous conduction** occurs in **unmyelinated axons**.

▽ Each adjacent section of the axonal membrane undergoes depolarization and repolarization in sequence.

▽ The action potential is therefore regenerated along essentially every neighboring section of membrane.

▽ Because the entire membrane must participate sequentially, continuous conduction is relatively **slow and energy-demanding** compared with saltatory conduction.

✓ **Continuous conduction = sequential propagation along the entire membrane of an unmyelinated axon.**

✦ 2. Saltatory Conduction

▽ **Saltatory conduction** occurs in **myelinated axons**.

▽ Myelin acts as an electrical insulator around the axon, while voltage-gated channels are concentrated at the gaps called **Nodes of Ranvier**.

▽ The action potential is regenerated mainly at these nodes, making the impulse appear to **“jump” from node to node**.

▽ The term *saltatory* comes from *saltare*, meaning **“to leap.”**

▽ Saltatory conduction allows myelinated axons to conduct signals much more rapidly than comparable unmyelinated axons.

✓ **Saltatory conduction = rapid node-to-node propagation through a myelinated axon.**

✦ F. Factors Influencing Conduction Velocity

✦ 1. Degree of Myelination

▽ Greater myelination generally increases conduction velocity.

▽ Myelin reduces current leakage across the membrane and allows the depolarizing current to travel rapidly between Nodes of Ranvier.

✓ **Myelinated fibers conduct faster than comparable unmyelinated fibers.**

✦ 2. Axon Diameter

▽ A larger axon diameter generally provides less internal resistance to the movement of electrical current.

▽ As a result, larger-diameter axons can conduct action potentials faster than smaller-diameter axons.

✓ **Larger axon diameter → generally faster conduction.**

✦ 3. Age

▽ Conduction velocity changes during development as the nervous system matures and myelination increases.

▽ In general, conduction is slower in very young children and may also decline with advanced age.

▽ The exact age of maximum conduction velocity depends on the neural pathway and developmental measure being studied, so a single age range should not be treated as universal.

✓ **Age affects conduction velocity mainly through nervous-system maturation, myelination, and later age-related changes.**

✦ 4. Axon Length

▽ Axon length does **not directly determine the conduction velocity** of an individual action potential.

▽ Instead, a longer axon means the signal has a greater distance to travel, so the **total travel time** can be longer even when conduction velocity is unchanged.

✓ **Axon length affects travel time, while myelination and diameter are major determinants of conduction velocity.**

[24/09/2026 7:55 AM] Imaanaa Harmee: ☞ 5. Neuronal & Nerve Fiber Classification

✦ A. Structural Classification of Neurons

▽ Neurons can be classified structurally according to the **number and arrangement of processes** extending from their cell body.

✦ 1. Multipolar Neurons

▽ Multipolar neurons have **one axon and multiple dendrites** extending from the cell body.

▽ They are the **most common structural type** of neuron and are found extensively throughout the CNS.

▽ Many motor neurons and interneurons are multipolar.

▽ Their numerous dendrites provide a large receptive surface for receiving signals from

many other neurons.

✓ **Multipolar = one axon + many dendrites.**

✦ 2. Bipolar Neurons

▽ Bipolar neurons have **one axon and one dendrite** extending from opposite sides of the cell body.

▽ They are less common than multipolar neurons and are particularly associated with specialized sensory systems.

▽ Bipolar neurons are found in the **retina, inner ear, and olfactory epithelium** of the nasal region.

▽ Their structure is well suited for transmitting information from specialized sensory receptors toward the CNS.

✓ **Bipolar = one dendrite + cell body + one axon.**

✦ 3. Unipolar Neurons

▽ Unipolar neurons have a **single process** extending from the cell body.

▽ In humans, the neurons commonly described in this category are more accurately called **pseudounipolar neurons**.

▽ A single process extends from the cell body and then divides into branches: one branch carries information from the peripheral receptor toward the cell body, while the other continues toward the CNS.

▽ Pseudounipolar sensory neurons are found in **dorsal root ganglia** and sensory ganglia associated with certain cranial nerves.

✓ **Unipolar/pseudounipolar = one main process associated with sensory neurons.**

✦ B. Functional Classification of Neurons

▽ Functional classification is based on the **direction in which information travels** relative to the CNS.

✦ 1. Sensory (Afferent) Neurons

▽ Sensory neurons carry information **from sensory receptors in the peripheral body toward the CNS.**

▽ They detect changes such as touch, pressure, temperature, pain, stretch, and chemical conditions.

▽ Their information enters the spinal cord or brain, where it can be processed and interpreted.

✓ **Afferent = information travels toward the CNS.**

✦ 2. Motor (Efferent) Neurons

▽ Motor neurons carry commands **away from the CNS toward peripheral effectors.**

▽ Their targets include **skeletal muscle, smooth muscle, cardiac muscle, and glands**, depending on the division of the nervous system involved.

▽ Motor output allows the nervous system to produce movement, regulate organ activity, and control glandular secretion.

✓ **Efferent = information travels away from the CNS.**

✦ 3. Association Neurons — Interneurons

▽ Interneurons are neurons that primarily connect neurons **within the CNS.**

▽ They receive and process information from sensory neurons and can influence motor neurons or other interneurons.

▽ They are essential for **integration, decision-making, reflexes, and complex neural processing.**

▽ For example, during a reflex, interneurons can connect incoming sensory information with the appropriate motor response.

✓ **Interneurons = integration and communication within the CNS.**

✦ C. Core Properties of Neurons

✦ 1. Excitability

▽ **Excitability** is the ability of a neuron to respond to an appropriate stimulus by producing changes in its membrane potential.

▽ When the membrane reaches the appropriate threshold, voltage-gated ion channels can generate an **action potential**.

✓ **Excitability = ability to respond electrically to a stimulus.**

[24/09/2026 7:55 AM] Imaanaa Harmee: ✦ 2. Conductivity

▽ **Conductivity** is the ability of a neuron to transmit an electrical signal along its membrane, particularly along its axon.

▽ Once an action potential is generated, it propagates along the axon toward the presynaptic terminals.

✓ **Conductivity = ability to transmit an electrical signal.**

✦ D. Classification of Nerve Fiber Types

▽ Nerve fibers can be classified according to characteristics such as **myelination, diameter, and conduction velocity**.

▽ In general, larger and more heavily myelinated fibers conduct impulses faster than smaller, less myelinated fibers.

✦ 1. Type A α Fibers

▽ Type A α fibers are **large-diameter, heavily myelinated fibers** and therefore have very rapid conduction.

▽ They are associated with **motor fibers supplying skeletal muscle** and with fast somatic sensory information from muscles, tendons, joints, and certain skin receptors.

▽ **Diameter:** approximately 10–20 μm .

▽ **Conduction velocity:** approximately 60–120 m/s.

✓ **A α = largest and fastest major nerve fibers.**

✦ 2. Type A β Fibers

▽ Type A β fibers are myelinated fibers that are smaller than A α fibers but still conduct signals rapidly.

▽ They are especially important for **sensory information such as touch, pressure, and vibration**.

▽ **Diameter:** approximately **5–10 μm** .

▽ **Conduction velocity:** approximately **30–60 m/s**.

✓ **A β = myelinated, relatively fast sensory fibers, especially for touch and pressure.**

✦ 3. Type A γ and A δ Fibers

▽ Type A γ **fibers** are myelinated fibers associated particularly with **motor control of muscle spindles**, helping regulate muscle spindle sensitivity.

▽ Type A δ **fibers** are smaller myelinated sensory fibers that carry relatively rapid **pain, temperature, and certain touch sensations**.

▽ **Diameter:** approximately **1–6 μm** , depending on the specific fiber.

▽ **Conduction velocity:** approximately **6–30 m/s**.

✓ **A γ = mainly muscle-spindle motor control.**

✓ **A δ = relatively fast pain and temperature sensation.**

✦ 4. Type B Fibers

▽ Type B **fibers** are **small, myelinated fibers**.

▽ They are classically associated with **preganglionic autonomic neurons**, which carry signals from the CNS toward autonomic ganglia.

▽ Their myelination allows them to conduct faster than unmyelinated Type C fibers.

▽ **Diameter:** approximately **2–6 μm** .

▽ **Conduction velocity:** approximately **6–30 m/s**.

✓ **Type B = myelinated preganglionic autonomic fibers.**

✦ 5. Type C Fibers

▽ Type C **fibers** are very small and **unmyelinated**, making them the slowest major nerve fiber type.

▽ They include **postganglionic autonomic fibers** and sensory fibers carrying slower forms of pain and temperature information.

▽ Because they lack myelin and have a small diameter, electrical signals travel much more slowly along them.

▽ **Diameter:** approximately **0.5–2 μm** .

▽ **Conduction velocity:** approximately **0.5–2 m/s**.

✓ **Type C = small, unmyelinated, slow-conducting fibers.**

✦ E. Relationship Between Fiber Structure and Conduction

▽ **Myelination** greatly increases conduction velocity because action potentials can propagate by **saltatory conduction** between Nodes of Ranvier.

▽ **Larger axon diameter** also increases conduction velocity because electrical current encounters less internal resistance.

▽ Therefore, the general pattern is:

Large diameter + myelination → faster conduction

Small diameter + little/no myelination → slower conduction

✓ **A α fibers are among the fastest, while Type C fibers are among the slowest.**

[24/09/2026 7:55 AM] Imaanaa Harmee: ☞ **6. Synaptic Transmission, Postsynaptic Potentials & Properties**

✦ A. Synapse Architecture & Signal Conduction

▽ A **synapse** is a specialized junction where a neuron communicates with another cell.

▽ The neuron sending the signal is called the **presynaptic neuron**, while the cell receiving the signal is the **postsynaptic cell**.

▽ The postsynaptic cell may be another neuron, a muscle cell, or a gland cell.

▽ In the autonomic nervous system, a **ganglion** is a collection of neuronal cell bodies in the PNS. It is not itself the synaptic junction, although synapses occur within autonomic ganglia.

✦ 1. Types of Synaptic Junctions

▽ **Neuroneuronal junction:** communication between one neuron and another neuron.

▽ **Neuromuscular junction:** communication between a motor neuron and a muscle fiber.

▽ **Neuroglandular junction:** communication between a neuron and a glandular cell.

✓ **Synapses allow neurons to communicate with neurons, muscles, and glands.**

✦ 2. Electrical and Chemical Components of Signal Transmission

▽ Within the presynaptic and postsynaptic neurons, signals can be transmitted electrically through changes in **membrane potential** and action potentials.

▽ At a typical **chemical synapse**, the electrical signal is converted into a chemical signal at the presynaptic terminal.

▽ This conversion of an electrical signal into a chemical signal is called **transduction** in the context of synaptic transmission.

▽ The chemical messenger, called a **neurotransmitter**, then crosses the synaptic cleft and produces an electrical or biochemical response in the postsynaptic cell.

✓ **Typical chemical synapse: electrical signal → chemical neurotransmitter → postsynaptic electrical response.**

✦ 3. Mechanism of Chemical Synaptic Transmission

▽ When an action potential reaches the **presynaptic axon terminal**, it depolarizes the terminal membrane.

▽ This depolarization opens **voltage-gated Ca^{2+} channels**.

▽ Ca^{2+} enters the presynaptic terminal because of its electrochemical gradient.

▽ The increased intracellular Ca^{2+} activates proteins involved in synaptic vesicle fusion with the presynaptic membrane.

▽ Synaptic vesicles then undergo **exocytosis**, releasing neurotransmitter molecules into the **synaptic cleft**.

▽ The neurotransmitter diffuses across the cleft and binds to specific **receptors on the postsynaptic membrane**.

▽ Receptor activation changes ion channel activity or activates intracellular signaling pathways, producing a postsynaptic response.

✓ **Key sequence:**

Action potential → Ca^{2+} entry → vesicle fusion → neurotransmitter release → receptor binding → postsynaptic response

✦ B. Postsynaptic Potentials & Summation

▽ A **postsynaptic potential** is a graded change in the membrane potential of the postsynaptic cell caused by neurotransmitter action.

▽ Depending on the receptor and ion channels involved, the response may make the postsynaptic neuron more likely or less likely to generate an action potential.

✦ 1. Excitatory Postsynaptic Potential (EPSP)

▽ An **EPSP** is a graded postsynaptic depolarization that generally moves the membrane potential **closer to threshold**.

▽ It commonly occurs when neurotransmitter binding opens channels that allow positively charged ions, particularly Na^+ , to enter the cell.

▽ The membrane becomes less negative, increasing the probability that the neuron will reach threshold at the axon initial segment.

▽ A single EPSP is usually too small to generate an action potential by itself.

✓ **EPSP = depolarizes the membrane and increases the likelihood of an action potential.**

[24/09/2026 7:55 AM] Imaanaa Harmee: ✦ 2. Inhibitory Postsynaptic Potential (IPSP)

▽ An **IPSP** is a graded postsynaptic change that generally makes the neuron **less likely to reach threshold**.

▽ It can result from opening channels that allow negative ions such as Cl^- to enter the cell or positive ions such as K^+ to leave the cell.

▽ These ion movements can hyperpolarize the membrane or stabilize it near a negative potential, making depolarization toward threshold more difficult.

✓ **IPSP = decreases the likelihood of an action potential.**

✦ 3. Temporal Summation

▽ **Temporal summation** occurs when the **same presynaptic neuron releases neurotransmitter repeatedly at short intervals**.

▽ If one EPSP occurs before the previous EPSP has completely disappeared, their effects can add together.

▽ Several rapidly occurring EPSPs may therefore produce enough depolarization to reach

threshold.

✓ **Temporal = repeated signals from one presynaptic neuron over time.**

✦ 4. Spatial Summation

▽ **Spatial summation** occurs when **multiple presynaptic neurons are active at approximately the same time.**

▽ Their individual EPSPs can combine at the postsynaptic neuron.

▽ If the combined depolarization reaches threshold, an action potential can be generated.

▽ IPSPs can also participate in summation and oppose excitatory inputs.

✓ **Spatial = signals from multiple presynaptic neurons combine at the same time.**

✦ C. Functional Properties of Synapses

✦ 1. Unidirectional Conduction

▽ Chemical synapses normally transmit information in **one direction.**

▽ The presynaptic neuron releases neurotransmitter, while the postsynaptic membrane contains the receptors that respond to it.

▽ Therefore, the signal normally travels:

Presynaptic neuron → synaptic cleft → postsynaptic cell

✓ **Chemical synapses are functionally unidirectional.**

✦ 2. Synaptic Delay

▽ Synaptic transmission takes a short amount of time because several steps must occur after the action potential reaches the terminal.

▽ These steps include Ca^{2+} entry, neurotransmitter release, diffusion across the cleft, receptor binding, and postsynaptic channel activation.

▽ A typical chemical synaptic delay is approximately **0.5–1.0 ms.**

✓ **Synaptic delay = brief time required for chemical transmission across a synapse.**

✦ 3. Synaptic Fatigue

▽ **Synaptic fatigue** refers to a temporary reduction in postsynaptic response during prolonged or repeated presynaptic stimulation.

▽ One important factor is depletion of readily releasable neurotransmitter vesicles.

▽ Other mechanisms, including changes in receptor responsiveness and ion concentrations, can also contribute.

▽ Synaptic fatigue can help limit excessive neuronal activity and may serve as a protective mechanism.

✓ **Fatigue = reduced synaptic effectiveness during prolonged repetitive stimulation.**

✦ 4. Synaptic After-Discharge

▽ **Synaptic after-discharge** occurs when the postsynaptic neuron continues to produce activity for a period after the original presynaptic stimulation has stopped.

▽ This can occur because neurotransmitter effects may persist and because activity can continue through interconnected neuronal circuits.

▽ Therefore, the postsynaptic response may outlast the original stimulus.

✓ **After-discharge = postsynaptic activity continues briefly after the initiating stimulus ends.**

✦ 5. Synaptic Potentiation (Facilitation)

▽ **Synaptic potentiation or facilitation** refers to an increase in the effectiveness of synaptic transmission following previous activity.

▽ Previous stimulation can make the postsynaptic response to a subsequent stimulus larger.

▽ In some forms of short-term facilitation, residual Ca^{2+} in the presynaptic terminal increases neurotransmitter release during subsequent action potentials.

✓ **Facilitation = previous activity can temporarily increase synaptic responsiveness.**

✦ D. Neurotransmitter Removal

▽ Neurotransmitters must be removed or inactivated after they have produced their effect.

▽ Rapid removal prevents continuous stimulation of postsynaptic receptors and allows the

synapse to respond appropriately to new signals.

[24/09/2026 7:55 AM] Imaanaa Harmee: ✦ 1. Enzymatic Degradation

▽ Specific enzymes break neurotransmitter molecules into inactive substances.

▽ This mechanism is particularly important for certain neurotransmitters, such as **acetylcholine**, which is rapidly broken down in the synaptic region.

✓ **Enzymatic degradation = neurotransmitter is chemically broken down.**

✦ 2. Diffusion

▽ Some neurotransmitter molecules simply **diffuse away from the synaptic cleft** into the surrounding extracellular fluid.

▽ Once the neurotransmitter concentration near the postsynaptic receptors decreases, its effect becomes weaker.

✓ **Diffusion = neurotransmitter moves away from the synaptic cleft.**

✦ 3. Reuptake

▽ **Reuptake** occurs when transporter proteins move neurotransmitter molecules back into the presynaptic neuron or, in some cases, into surrounding glial cells.

▽ The neurotransmitter can then be metabolized, recycled, or stored for later release.

✓ **Reuptake = neurotransmitter is transported back into cells for removal or recycling.**

✦ E. Overall Sequence of Synaptic Transmission

▽ The complete process can be understood as a series of linked events:

1. Action potential reaches the presynaptic terminal

2. Voltage-gated Ca^{2+} channels open

3. Ca^{2+} enters the presynaptic terminal

4. Synaptic vesicles release neurotransmitter

5. Neurotransmitter crosses the synaptic cleft

6. Neurotransmitter binds postsynaptic receptors

7. EPSP or IPSP develops

8. EPSPs and IPSPs are integrated

9. If threshold is reached, an action potential is generated

10. Neurotransmitter is removed by degradation, diffusion, or reuptake

✓ **The essential function of a synapse is to transfer information from one cell to another while allowing the nervous system to regulate whether the next neuron becomes more or less likely to fire.**

[24/09/2026 7:55 AM] Imaanaa Harmee:

Muscle Physiology

Dr. Doc

[24/09/2026 7:55 AM] Imaanaa Harmee:

🔗 **MUSCLE TISSUE**

- ▽ Structure
- ▽ Function
- ▽ Organization
- ▽ Contraction

🧬 1. Overview & Primary Functions of Muscle Tissue

★ Muscle tissue is one of the four primary tissue types in the body, functioning in close association with nerves, blood vessels, and connective tissue structures.

★ It represents a marvel of both biology and physics.

✦ **All muscle activities fall under four major functional categories:**

👤 **Movement Production**

▽ Drives the movement of body parts, blood through the circulatory system, lymph through lymphatic vessels, food through the GI tract, bile out of the gallbladder, urine through the urinary tract, semen through reproductive tracts, and the newborn through the birth canal.

👤 **Maintenance of Posture**

▽ *Continuous, involuntary contractions maintain upright posture*, e.g., neck muscles holding the head erect and leg muscles supporting standing.

💧 Thermogenesis

▽ Generates body heat through shivering, which consists of **involuntary contractions of skeletal muscle**.

🔗 Joint Stabilization

▽ Maintains firm tension on tendons crossing joints, thereby preserving joint integrity.
[24/09/2026 7:55 AM] Imaanaa Harmee:

⚡ 3. Universal Characteristics of Muscle Tissue

▽ Muscles tissue exhibits four cardinal physiological properties:

⚡ 1. Excitability

★ The capacity to receive and respond to stimuli.

Smooth Muscle Stimuli:

★ Neurotransmitters, hormones, mechanical stretch, pH, (PCO₂), or (PO₂).

Cardiac Muscle Stimuli:

▽ Neurotransmitters, hormones, stretch, and **intrinsic autorhythmicity**.

🔗 Skeletal Muscle Stimuli:

▽ **Exclusively neurotransmitters** released by motor neurons.(High yield for exam)

⚡ Response:

▽ Generation of an action potential (AP) propagating along the cell membrane.

★ 2. Contractility

▽ **The ability to shorten and thicken forcibly** when adequately stimulated, serving as **the hallmark of muscle tissue**.(High yield for exam)

↔ 3. Extensibility

▽ The ability to be stretched beyond resting length.

📦 4. Elasticity

🌀 **The ability to recoil** and return to original resting length and size following stretching or contraction.

🔑 4. Gross Anatomy & Connective Tissue Organization

^ A skeletal muscle organ is composed primarily of skeletal muscle tissue, integrated with nerves, blood vessels, and *dense irregular connective tissue* layers.

📌 Epimysium

☀ An outer layer of *dense irregular connective tissue* that envelops the entire muscle organ.

🌀 Fascicles

👉 **Bundles of elongated skeletal muscle fibers** enclosed by a **layer of *dense irregular connective tissue* called the perimysium.**

📌 Endomysium

⚡ A delicate layer of *loose connective tissue* surrounding each individual muscle fiber.

📏 Fiber Extent

☀ Individual skeletal muscle cells extend along the entire length of the muscle organ.

5. Neurovascular Supply & Muscle Attachments

Blood & Nerve Supply

🌀 Each skeletal muscle is **innervated by one nerve** (forming motor units), supplied by **one artery**, and drained by **one or more veins**.

🌿 These structures enter and exit through connective tissue sheaths and branch extensively throughout the tissue.

📍 Muscle Attachments

📌 Origin

★ The fixed, immovable attachment end.

📌 Insertion

★ The movable attachment end.

🔑 Direct Attachments

★ Less common; the **epimysium fuses directly** with the periosteum or perichondrium of

bone/cartilage.

Indirect Attachments

★ Typical arrangement; **connective tissue coverings extend beyond the muscle as a cordlike tendon or a broad, sheetlike aponeurosis that anchors into the periosteum or perichondrium.**

6. Microanatomy of Skeletal Muscle Fibers

Dimensions & Syncytium Structure

⚡ **Individual muscle cells (=muscle fibers)** reach diameters up to 100 µm and lengths up to 30 cm.

🧬 They are **multinucleated syncytia formed by the embryonic fusion of hundreds of precursor myoblasts.**

😊 Sarcolemma

The specialized plasma membrane consisting of a **true plasma membrane and an outer polysaccharide coat containing fine collagen fibrils.**

🌀 At cell ends, the outer coat fuses with tendon fibers, which collect into bundles to form muscle tendons.

⚡ Sarcoplasm

Cytoplasm rich in:

📄 Mitochondria

🍬 Glycogen granules — glucose store for energy

🏗️ Contractile myofibrils

➡ Transverse (T) Tubules

👉 *Deep invaginations of the sarcolemma that penetrate straight through the muscle cell.*

[24/09/2026 7:55 AM] Imaanaa Harmee:

🧬 7. Organelles & Myofilament Molecular Structure

🌀 Sarcoplasmic Reticulum (SR)

👉 The **muscle equivalent of smooth endoplasmic reticulum**, forming a loose membrane-bound network around every myofibril.

🛒 Its major role is to act as a **calcium storage depot.**

Myofibrils

▽ Long rod-like protein bundles extending the length of the cell, **composed of** repeating contractile filaments called **myofilaments**.

Thick Filaments — Myosin

🔗 Composed of approximately 300 myosin molecules.

🦶 **Each myosin protein resembles two golf clubs with shafts twisted together, terminating in double heads.**

Thin Filaments — Actin

🔗 Tropomyosin & Troponin

Double-helical strands composed of three distinct proteins:

Actin

🔗 Globular subunit beads (G-actin), each possessing a dedicated myosin binding site.

Tropomyosin

🔗 Filamentous protein loosely wrapped around the actin double helix, physically covering **myosin binding sites at rest**.

Troponin Complex

★ **Regulatory protein triad** (Troponin T, I, and C) bound to both actin and tropomyosin.
[24/09/2026 8:03 AM] Imaanaa Harmee: 🧑 FEMALE REPRODUCTIVE SYSTEM

1. Structural Organization of the Female Reproductive System

▽ The female reproductive system is divided into essential structures (gonads) and accessory sex organs:

★ Essential Structures (Gonads)

🌀 Two Ovaries:

🔗 Responsible for oogenesis (gametogenic role) and endocrine function (secretion of Estrogen and Progesterone).

★ Accessory Sex Organs

🌀 Oviducts (Fallopian / Uterine Tubes)

- 🌀 Uterus
- 🌀 Vagina
- 🌀 External Genitalia

2. Anatomy & Cellular Organization of the Ovaries

▽ Anatomical Attachments

The ovaries are attached to the peritoneum via the mesovarian and broad ligaments.

▽ Functional Roles

Gametogenic potential:

- ★ Established early during fetal development.

- 🔗 Endocrine role: Not realized until puberty.

★ Structural Regions

1. Medullary Part:

- ▽ The central region containing blood vessels and connective tissues (elastin and collagen).

2. Cortical Part:

- ▽ The peripheral region containing actively proliferating ovarian follicles.

★ Cellular Differentiation

🌀 Mesenchymal Tissue:

🔗 Differentiates into interstitial tissue (hormone-secreting cells), serving as the primary source of ovarian hormones, associating with germinal elements, and providing a nutritive environment for oocytes.

★ Epithelial Tissue:

🔗 Differentiates into granulosa cells.

3. Physiology of Accessory Sex Organs

- ✓ Oviduct / Fallopian Tube / ✓ Uterine Tube
- ✓ Serves as the normal site of fertilization.
- ✓ Transports ovum and sperm.
- ✓ Performs secretory functions.

★ Uterus

○ A hollow muscular organ in non-pregnant females consisting of three wall layers:

✧ Endometrium (with uterine glands),

✧ Myometrium, and

✧ Perimetrium.

☑ Functions as the site of pregnancy, supports the growth and development of the fetus, generates pressure during labor, and serves as the site of menstruation.

★ Vagina

☞ Lined with non-keratinized stratified epithelium.

☞ Vaginal glands secrete glycogen and lactate.

Harbors *Lactobacillus* *liquificious*, which maintains an acidic pH within the vaginal wall.

☞ Functions as the female copulatory organ, the birth canal, and the passage for menstrual flow.

★ External Genitalia

▽ Includes the clitoris, labia minora, and labia majora.

[24/09/2026 8:03 AM] Imaanaa Harmee: 4. Oogenesis (Development of Ova)

▽ Oogenesis progresses through three major follicular stages:

★ Primordial follicles

★ Growing follicles (1° follicles $\rightarrow$ 2° follicles $\rightarrow$ 3° follicles)

★ Mature (Graafian) follicle $\rightarrow$ Ovum

▽ Oogenesis Sequence

Oogonia $\xrightarrow{\text{(Prenatal Mitosis)}}$ 1° Oocyte $\xrightarrow{\text{(1st Meiosis)}}$ 2° Oocyte $\xrightarrow{\text{(2nd Meiosis)}}$ Ovum

★ Prenatal & Follicular Population Dynamics

▽ Primitive ova (Oogonia) undergo active mitosis during prenatal development.

▽ Females have approximately 1 million primordial follicles at birth.

Primary Oocytes persist from birth to puberty, arrested in the Prophase of 1st meiosis ($2n$).

▽ By puberty, the count declines to approximately 400,000 follicles.

Meiotic Divisions & Ovulation

▽ Completion of the 1st meiotic division yields a Secondary Oocyte and a First Polar Body (which undergoes attrition).

▽ At ovulation, the secondary oocyte is arrested in Metaphase of meiosis II (1n).

▽ Completion of the second meiotic division yields the mature Ovum and a Second Polar Body (attrition).

5. Overview of the Female Sexual Cycle

✓ Also referred to as the menstrual cycle, involving cyclic structural and functional changes in female reproductive organs.

▽ Restricted exclusively to humans and large/great apes.

▽ Initiates at puberty and ends at menopause (the complete cessation of menstruation).

Menarche: The onset of the first menstrual flow, occurring at an average age of 12 years (range: 9–16 years).

🔗 Comprises two interconnected sub-cycles:

1. Ovarian Cycle:

📅 Follicular, Ovulatory, and Luteal phases.

2. Endometrial Cycle:

📅 Menstrual (bleeding), Proliferative, and Secretory phases.

6. The Ovarian Cycle

📅 Consists of three sequential phases regulated by pituitary gonadotropins (FSH and LH):

A. Follicular Phase (Days 1 to 13)

🔗 Driven by FSH and LH from the anterior pituitary gland.

🔗 Multiple primordial follicles enlarge at the cycle's onset; primary follicles produce fluid termed liquor folliculi, which accumulates in the antrum (rich in estrogen).

🔗 By Day 6, a single dominant mature follicle—the Graafian follicle—is established.

B. Ovulation Phase (Days 14 to 15)

✂ The wall of the Graafian follicle ruptures, releasing the ovum, zona pellucida, and one surrounding layer of the corona radiata into the peritoneal cavity.

✂ LH is the primary hormone of ovulation: 36 hours prior to ovulation, the anterior pituitary executes a massive LH surge, with peak LH levels reached 12 hours before ovulation.

★ Cascade

↑LH → ↑antral fluid volume → ↑progesterone secretion → ↑blood flow (BF) to Graafian follicles → ↑follicular swelling → ↑intrafollicular pressure → follicular rupture and ovum release

C. Luteal Phase (Days 15 to 25)

▽ Following rupture, the remaining Graafian follicle structure collapses.

✂ Under LH stimulation, granulosa cells undergo hypertrophy and proliferation to transform into the corpus luteum.

🌀 Function

🌀 Secretes massive amounts of Progesterone alongside Estrogen, maintaining conditions necessary for pregnancy.

★ Four Sequential Stages of the Corpus Luteum

1. Stage of Vascularization (Corpus Hemorrhagicum)

🌀 Vascularization of granulosa cells occurs.

2. Stage of Glandular Metamorphosis

🌀 Follicular granulosa and theca cells transform into luteal cells capable of concentrating cholesterol as temporary glandular units.

3. Stage of Full Development

🌀 Characterized by maximal output of progesterone and estrogen.

✂ If fertilization occurs: Human Chorionic Gonadotropin (hCG) preserves the corpus luteum, keeping it active for more than 10 days.

✂ If no fertilization occurs: The corpus luteum remains active for 10 days, after which it degenerates.

4. Stage of Involution

▽ Luteal cells are replaced by fibrous connective tissue, forming the corpus albicans.

[24/09/2026 8:03 AM] Imaanaa Harmee: 7. The Endometrial Cycle

▽ The endometrial wall undergoes three distinct phases driven directly by ovarian steroid hormones:

Bleeding / Menstrual Phase

Timeframe: 3 to 5 days

★ Hormonal Mechanism & Structural Features:

⌚ Triggered by sudden withdrawal of sex hormones (estrogen & progesterone).

⌚ Causes desquamation of the functional layer of the endometrium.

⌚ Total blood loss averages 50 mL (normal range: 30–80 mL).

📅 Proliferative Phase

Timeframe: Pre-ovulatory (mid-cycle)

★ Hormonal Mechanism & Structural Features:

⌚ Driven primarily by Estrogen.

Causes growth and proliferation of endometrial stromal cells and glands.

⌚ Promotes progressive elongation and growth of spiral arteries.

📅 Secretory (Progestational) Phase

Timeframe: Post-ovulatory (luteal phase)

★ Hormonal Mechanism & Structural Features:

⌚ Driven chiefly by Progesterone (assisted by Estrogen).

⌚ Endometrial glands become tortuous and actively secretory.

⌚ Spiral arteries become increasingly coiled/spiral.

Total endometrial thickness reaches 6 mm.

[24/09/2026 8:03 AM] Imaanaa Harmee: ANS

Dr.Doc